



BASIC MATHEMATICS

Student's book



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Experimental version

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FOREWORD

Dear student,

The Rwanda Basic Education Board (REB) is honoured to present this Basic Mathematics textbook for Senior Four (S4). This textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC). It promotes learner-centred teaching and learning practices that align with Rwanda's vision of empowering students to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances towards building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding students to connect classroom knowledge with real-life experiences, thereby fostering both individual growth and national development.

This student's book serves as a comprehensive resource for the teaching and learning of Basic Mathematics at Senior Four in the following pathways: Languages, Art and Humanities. The subject is an integral part of the CBC, aimed at equipping students with essential knowledge, skills, values, and attitudes necessary to become responsible and competent citizens. Through this subject, you are encouraged to develop problem-solving skills and contribute positively to your communities.

In this regard, the only way to learn mathematics effectively is by doing mathematics through active problem-solving rather than passive reading. You are expected to work through problems, attempt proofs, and practise regularly, rather than only reviewing theorems or looking up answers.

In its pursuit of quality education, the Government of Rwanda emphasizes the importance of using this kind of teaching and learning materials aligned with the curriculum.

I sincerely thank everyone who contributed to the development and refinement of this textbook. Special recognition goes to the REB team that coordinated the entire process. We also thank all partners, teachers, lecturers, and individuals who were involved from the beginning to the end.

Each feedback was highly valued and will contribute to improving the learning outcomes of our students.



Dr. MBARUSHIMANA Nelson

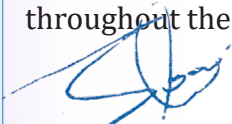
Director General/REB

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I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Basic Mathematics students' book for Senior Four (S4). This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff, subject experts, university lecturers and teachers for their valuable work in the adaptation of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.



MURUNGI Joan

Head of CTLR department/REB

Table of Contents

FOREWORD	iii
ACKNOWLEDGEMENT	iv
UNIT 1: ARITHMETIC.....	1
Introduction	2
1.1 Negative numbers and related problems.....	3
1.2 Decimals and related problems	5
1.3 Fractions and related problems.....	8
1.4 Ratios and related problems	11
1.5 Percentages and related problems	13
1.6 Absolute value	15
1.7 Problems involving absolute value	18
1.8 Concept and properties of powers.....	20
1.9 Problems involving powers.....	23
1.10 Concept and properties of radicals	25
1.11 Rationalization of denominators	27
1.12 Problems involving radicals	29
1.13 Concept and properties of decimal logarithms	31
1.14 Simple equations involving decimal logarithms	34
1.15 Simple interest and related problems	38
1.16 Compound interest.....	40
1.17 Problems related to compound interest.....	41
1.18 Final value of investment	43
1.19 Problems related to final value of investment	48
1.20 Arc of elasticity for demand	49
End of unit assessment	53
UNIT 2: EQUATIONS AND INEQUALITIES.....	54
Introduction	55
2.1 Linear equations in one unknown	56
2.2 Real life problems involving linear equations.....	57
2.3 Meaning of inequalities.....	59
2.4 Concept and properties of intervals.....	61
2.5 Problems involving intervals.....	63
2.6 Inequalities involving products.....	65
2.7 Inequalities involving quotients	67
2.8 Real life problems involving linear inequalities.....	69
2.9 Solving algebraically simultaneous linear equations in two unknowns (Solving by equating two same variables)	70

2.10 Solving algebraically simultaneous linear equations in two unknowns (solving by row operations or elimination method).....	72
2.11 Solving graphically simultaneous linear equations in two unknowns.....	74
2.12 Solving algebraically and graphically simultaneous linear inequalities in two unknowns.....	76
2.13 Solving quadratic equations using factorization.....	79
2.14 Concept of discriminant.....	81
2.15 Solving quadratic equations using discriminant.....	81
2.16 Applications of linear equations: Problems about supply, demand and finance	83
2.17 Revision of unit content.....	92
End of unit assessment	93
End of unit assessment	94
UNIT 3: DESCRIPTIVE STATISTICS.....	95
Introduction	96
3.1 Definition of terms related to statistics and type of data.....	96
3.2 Presentation / organization of data using frequency distribution table.....	99
3.3 Presentation of data using bar chart.....	106
3.4 Presentation of data using pie chart	108
3.5 Presentation of data using histogram and polygon.....	111
3.6 Interpretation of a statistical graph	118
3.7 Measures of central tendency for ungrouped data: Mode and mean	120
3.8 Median as a measure of central tendency for ungrouped data	123
3.9 Mode as the measure of central tendency for grouped data	124
3.10 Mean as the measure of central tendency for grouped data.....	126
3.11 Median as the measure of central tendency for grouped data.....	127
3.12 Quartiles as Measures of dispersion	129
3.13 Variance as the measure of dispersion.....	131
3.14 Standard deviation.....	133
3.15 Coefficient of variation as the measure of dispersion.....	135
3.16 Application: Collection, organization, presentation and interpretation of data and drawing conclusions.....	136
3.17 Questions for revision of unit content.....	138
End of unit assessment	140
BIBLIOGRAPHY.....	141

UNIT 1

ARITHMETIC



Key Unit Competence:

Use arithmetic operations in solving simple real-life problems.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Identify the properties of absolute value, powers, radicals, and logarithms.
- Differentiate simple from compound interest.
- Explain the demand and supply functions.

Skills

2

- Perform operations (+, −, ×, ÷) with fractions, decimals, negative numbers, and radicals.
- Convert between fractions, decimals, and percentages.
- Solve real-life problems involving ratios, percentages, and absolute value.
- Apply properties of powers and radicals to simplify expressions.
- Rationalize the denominator of a radical expression.
- Solve simple equations involving decimal logarithms.
- Calculate simple interest, compound interest, and the final value of an investment.

Attitudes and values

3

- Appreciate the role of arithmetic in daily life and financial planning.
- Value accuracy and precision in numerical calculations of final value of an investment.
- Demonstrate critical thinking and confidence when interpreting financial concepts like interest and elasticity.

Introduction

This unit introduces the fundamental ideas of arithmetic operations and numerical relationships, which form an important foundation for many topics in Mathematics and its applications in real-life situations. The purpose of this unit is to help you understand how numbers can be manipulated, compared, and applied to solve practical problems involving quantities, financial situations, and economic relationships. These skills are essential for developing analytical thinking, problem-solving abilities, and quantitative reasoning, which will be useful in higher-level mathematics and everyday decision-making. In this unit, you will study: Arithmetic operations and their applications in real-life contexts.

The key words to be explained in this unit are the following: fraction, decimal, percentage, ratio, negative number, absolute value, power (exponent), radical, rationalization, logarithm, simple interest, compound interest, principal, rate, time, final investment value, elasticity of demand, and arc of elasticity.

Introductory activity



Imagine you are helping a small business or cooperative in your community. They sell fruits, vegetables, and school supplies. They also invest some money in a bank account to earn interest. Your task is to help them make the best financial and operational decisions using mathematics.

The table below shows items:

Items	Quantity	Price per item (Frw)
Apples	12	300
Bananas	25	150
Exercise books	20	200
Pieces of chalk	0	50

Questions:

- a)
 - i. The shop sells half of apples today. How many apples remain?
 - ii. If all 12 apples are sold and 25 bananas are sold, what is the ratio of apples to bananas sold?
 - iii. If the price of exercise books increases by 10%, what is the new price?
 - iv. If the cooperative buys a bundle of 16 apples arranged in a squared array, how many apples are on each side? (Use square roots)
- b) The cooperative invests 50,000 Frw in a bank at 5% simple interest per year. How much will they have after 3 years?

1.1 Negative numbers and related problems

Activity 1.1



“Negative numbers are used to represent situations where there is a loss, debt, or movement below a reference point. For example, if the temperature drops below zero, or if you owe money, or if a submarine goes below sea level, we use negative numbers.”

1. Identify and list four other real-life situations from your environment where negative numbers can be used.
2. Solve the following problems:
 - a) If the temperature was 5°C in the morning and rose by 8°C by noon, what is the new temperature at that moment?
 - b) The shop had a debt of 12,000 Frw. After paying 5,000 Frw, what is the remaining debt?
 - c) The elevator goes down 3 floors from ground level (0), then goes up 2 floors. What floor is it on now?

Content Summary

i. Negative numbers

A negative number is a number that is less than zero. It is usually written with a minus ($-$) sign in front. Negative numbers are used to represent **loss, debt, temperature below zero, depth below sea level, or any quantity that decreases from a reference point.**

ii. Visual representation

Using a number line:

- Zero is the middle point.
- Positive numbers are to the right of zero (0).



- Negative numbers are to the left of zero (0).

iii. Notation

Negative numbers are written with a minus sign: -1 , -5 , -12 .

Note: Positive numbers are written without a sign or with a plus sign: $+1$, $+5$, $+12$, or without the sign: 1 , 5 , 12 .

iv. Applications for negative numbers

- a) **Temperature:** Temperatures below zero are negative.

Example:

If the temperature in Kigali is -2°C , it means it is 2 degrees below zero.

b) Finance and debts: Negative numbers represent money owed or a deficit.

Example:

If a person has $-5,000$ FRW in their bank account, it means they are in debt by 5,000Frw.

c) Elevation and depth: Heights below a reference level (like sea level) are negative.

Example:

The Dead Sea is approximately -430 m below sea level.

d) Games and scores: Negative numbers can represent points lost in a game.

Example:

If a player loses 10 points, the score is -10 .

v. Operations with negative numbers

Operation	Example	Explanation
Addition	$-5 + 3 = -2$	Moving up from a negative number
Subtraction	$4 - 7 = -3$	Decrease past zero
Multiplication	$-3 \times 2 = -6$	A negative number multiplied by a positive number is negative
Division:	$-12 \div 3 = -4$	A negative number divided by a positive number is negative

Application activity 1.1

- Identify five real-life situations where negative numbers are used. For each situation:
 - Name the situation (e.g., debt, temperature below zero).
 - Describe the meaning of the negative number in that context.
- List the first ten negative integers.
- Consider the following sets of negative numbers based on real-life applications:
 - $A = \{-1, -2, -3, -4\}$ (loss in a game)
 - $B = \{-2, -3, -5, -7\}$ (bank debts)
 - $C = \{-1, -2, -5, -8\}$ (temperature drops)

iv. $D = \{-3, -4, -5, -8\}$ (stock decreases)

Questions

- Represent these sets using Venn diagrams.
- Identify which negative numbers appear in more than one set.
- Discuss what it means for a negative number to belong to multiple real-life contexts.

1.2 Decimals and related problems

Activity 1.2



Refer to the meaning of decimals and fractions learnt in previous years,

- Calculate $50 \div 100$ and write it in the form of a fraction and then as a decimal number.
- Express $\frac{1}{4}$ and $\frac{22}{7}$ in the form of decimal numbers.
- A student goes to a school shop to buy some learning materials. A mathematics notebook costs 1000 Frw, a pen costs 200 Frw, and a ruler costs 100 Frw. The student buys 2 notebooks, 3 pens, and 1 ruler. Calculate the total amount of money the student spends. The shopkeeper later decides to reduce the total bill by 0.5 of a thousand francs as a small reduction. Calculate the final amount the student pays after the reduction.
- Do you sometimes use decimal numbers in your life? Explain your answer.

Content Summary

i. Definition of decimal numbers

A **decimal number** is a number expressed in the base-ten place value system that includes a decimal point separating the whole number part from the fractional part.

Decimals represent fractions whose denominators are powers of ten, such as (10, 100, 1000), and so on.

Example:

Decimal numbers: 2.25; 0.46; 0.0001; etc.

Conversion between decimals and fractions

a) Convert decimals to fractions

Example 1:

Convert 0.4 to fraction.

Solution

- Identify the place value: 0.4 means four-tenths.
- Write as a fraction: $\frac{4}{10}$

Example 2:

$$\text{i. } 0.5 = \frac{5}{10} \quad \text{ii. } 0.75 = \frac{75}{100} \quad \text{iii. } 0.375 = 0 + \frac{3}{10} + \frac{7}{100} + \frac{5}{1000} = \frac{375}{1000}$$

b) Convert fractions to decimals

Fractions are converted to decimals by division.

Example 1:

Convert $\frac{3}{5}$ to decimal form.

Solution

Divide the numerator by the denominator: $3 \div 5 = 0.6$

Therefore, the decimal form of $\frac{3}{5} = 0.6$

Example 2:

Convert $\frac{7}{8}$ to decimal form.

Solution

Perform division: $7 \div 8 = 0.875$

Therefore, the decimal form of $\frac{7}{8} = 0.875$

Example 3:

Convert $\frac{9}{4}$ to decimal form.

Solution

Divide: $9 \div 4 = 2.25$

Therefore, the decimal form of $\frac{9}{4} = 2.25$

ii. Real-life situations where decimals are used

Decimals are widely used in scientific measurements, financial calculations, statistics, and everyday problem-solving where values are not whole numbers.

Example 4:

- **Measurement:** A table length of 1.75 m means 1 meter and 75 hundredths of a meter.
- **Distance:** 2.5 km means 2 kilometers and a half a kilometer.
- **Time:** 1.5 hours means 1 hour and a half.
- **Mass:** 3.25 kg means 3 kilograms and 25 hundredths of a kilogram.

The key idea is that decimals represent quantities between whole numbers.

Example 5

A water tank contains 18.75 liters of water. During a laboratory experiment, a student uses 2.4 liters of water for each trial. After 5 trials, the student measures the remaining water in the tank.

- a) Calculate the total amount of water used in the 5 trials.
- b) Determine the amount of water remaining in the tank.

Solution

Step 1: Calculate the total water used

Each trial uses 2.4 liters, and there are 5 trials.

$$2.4 \times 5 = 12.0$$

So, the total water used is 12.0 liters.

Step 2: Calculate the remaining water

Initial amount of water = 18.75 liters

Water used = 12.0 liters

So, 6.75 liters of water remain in the tank.

Application activity 1.2



1. Evaluate the following:
 - a) $1.345 + 0.00041 + 0.20023 =$
 - b) $93.954 \div 2.4 =$
2. A science laboratory has three types of chemical solutions stored in separate tanks: Tank A contains 12.875 liters, Tank B contains 8.462 liters, and Tank C contains 15.309 liters. The lab uses the solutions for different experiments. In the first experiment, 3.578 liters were taken from Tank A and 2.345 liters from Tank B. In the second experiment, 4.125 liters are taken from Tank B and 5.874 liters from Tank C. After completing the experiments, the remaining solutions in all tanks are combined and mixed for storage.

- i. Calculate the total volume of solutions used in the first experiment.
- ii. Calculate the total volume of solutions used in the second experiment.
- iii. Determine the remaining volume in each tank after the experiments.

Homework:

- iv. Find the total volume of the remaining solution that is mixed.
- v. If the remaining solution is poured into 2.75-litre bottles, how many full bottles can be filled?

1.3 Fractions and related problems

Activity 1.3



1. Sam had 120 teddy bears in his toy store. He sold $\frac{2}{3}$ of them at 100 Rwandan francs each. How much did he receive?
2. Considering that the denominator is different from zero, work out the following:

$$\frac{1}{x+1} - \frac{1}{2x+2}$$
3. Do you sometimes use fractions in your life? Explain your answer.

Content Summary

Consider the expressions, $\frac{x}{4} + \frac{3y}{x}, \frac{5}{x+4}$ in each of these the numerator or the denominator or both contain a variable or variables. These are examples of algebraic fractions. Since the letter used in these fractions stand for real numbers, we deal with algebraic fractions in the same way as we do with fractions in arithmetic. Fractions can be simplified, and operations on fractions, such as addition, subtraction, multiplication, and division, can be applied in simple arithmetic for solving related problems. The dividend (upper number) is called the numerator, and the divisor (lower number) is called the denominator. From the operations of fractions, different fractions can be found by taking Lowest Common Multiple (L.C.M.) and then adding all the fractions.

Adding fractions

- To add fractions with the common denominator, there is a simple rule $\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$ where $(b \neq 0)$ is a common denominator.
- When fractions have different denominators, find a common denominator to ease the addition. That is to say $\frac{a}{b} + \frac{c}{d} = \frac{ad+cb}{bd}$ where $(b \neq 0, d \neq 0)$ and bd becomes a common denominator.

Example:

$$1. \frac{x}{4} + \frac{5}{4} = \frac{x+5}{4}$$

$$2. \frac{x}{2} + \frac{y}{5} = \frac{5x}{(2)(5)} + \frac{2y}{(2)(5)} = \frac{5x}{10} + \frac{2y}{10} = \frac{5x+2y}{10}$$

i. Subtracting fractions

Subtracting fractions are very similar to addition of fractions, except that the sign changes.

Example:

Considering that the denominator is different from zero,

$$\frac{x+2}{x} - \frac{x}{x-2} = \frac{(x+2)(x-2) - (x)(x)}{x(x-2)} = \frac{-4}{x^2-2x} \text{ with } x \neq 0 \text{ and } x \neq 2$$

ii. Multiplying fractions

Multiplying fractions is the easiest one of all, just multiply the numerators together, and the denominators together

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}, \text{ where } (b \neq 0, d \neq 0)$$

Example:

Considering that the denominator is different from zero,

$$\frac{3x}{x-2} \times \frac{x}{3} = \frac{(3x)(x)}{3(x-2)} = \frac{3x^2}{3(x-2)} = \frac{x^2}{x-2} \quad (x \neq 2)$$

iii. Dividing fractions

To divide fractions, first flip the fraction we want to divide, then use the same method as for multiplication: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$ where $(b \neq 0, c \neq 0, d \neq 0)$

Example:

Considering that the denominator is different from zero, we have:

$$\frac{3y^2}{x+1} \div \frac{y}{2} = \frac{3y^2}{x+1} \times \frac{2}{y} = \frac{(3y^2)(2)}{(x+1)(y)} = \frac{6y^2}{(x+1)(y)} = \frac{6y}{x+1} \quad (x \neq -1, y \neq 0)$$

Rational fractions

We know that $\frac{p}{q}$ for $q \neq 0$ is called a rational number. Similarly, the quotient of two polynomials $N(x)$ and $D(x)$: $\frac{N(x)}{D(x)}$ for $D(x) \neq 0$ with no common factors. There are two types of rational fractions: proper or Improper fractions.

Proper fraction: A rational fraction $\frac{N(x)}{D(x)}$ is called a proper fraction if the degree of numerator is less than the degree of Denominator $D(x)$.

Example:

Considering that the denominator is different from zero, $\frac{6x + 27}{3x^2 - 9x}$ is a proper fraction.

Improper Fraction: A rational fraction $\frac{N(x)}{D(x)}$ is called improper fraction if the degree of numerator is greater than or equal to the degree of Denominator $D(x)$.

Example:

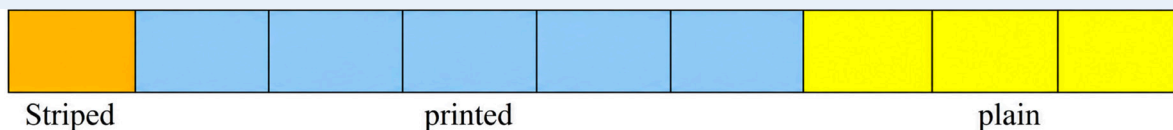
$$\frac{6x^3 - 5x^2 - 3x - 10}{x^2 + 1}$$

Problems related to fractions

Example:

1. One-ninth of the shirts sold at Peter's shop are striped, $\frac{5}{9}$ are printed and the rest of the shirts are plain colored shirts. If Peter's shop has 81 plain color shirts, how many more printed shirts than plain color shirts do the shop have?

Solution



3 units equals 81

1 unit $81 \div 3 = 27$

Printed shirts have 2 more parts than plain shirts.

2 units $= 27 \times 2 = 54$

Peter's shop has 54 more printed color shirts than plain shirts.

2. Oscar sold 2 glasses of milk for every 5 sodas he sold. If he sold 10 glasses of milk, how many sodas did he sell?

Solution

Let x be the number of Sodas that Oscar will sell.

Set up a proportion of milk with soda as $\frac{\text{milk}}{\text{soda}}, \frac{2}{5} = \frac{10}{x} \Rightarrow 2x = 50, x = \frac{50}{2} = 25$

He will sell 25 sodas.

Application activity 1.3



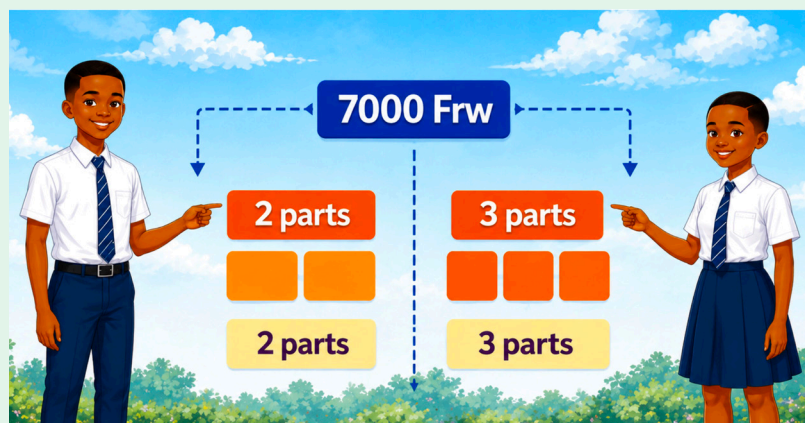
- Given that $\frac{1}{5}$ of the mangoes are sold at the market. $\frac{1}{3}$ of the remaining mangoes are given to friends. The rest are kept at home. If the farmer kept 40 mangoes at home, how many did he originally have?
- A juice shop mixes apple juice and orange juice in a ratio of $\frac{1}{2}$. If the shop uses 6 liters of apple juice, how many liters of orange juice are needed?
- Simplify and classify the following rational fractions as proper or improper:
- (a) $\frac{3x+2}{5x+7}$ (b) $\frac{7x^2+2x+1}{3x^2+4}$

1.4 Ratios and related problems

Activity 1.4



Suppose that two brothers from your village received 7000Frw from their grand brother who lives in the City. The condition to share this amount of money is that for every 2Frw that the younger brother gets, the other one gets 3Frw.



The two brothers have come to you for help after they disagreed on how to share that money.

- In what ratio would you share the money between them?
- Tell your partner how you would share the money and how much each would get.

Content Summary

The mathematical term '**ratio**' defines the relationship between two numbers of the same kind. The relationship between these numbers is expressed in the form "**a to b**" or more commonly in the form $a : b$.

A ratio is used to represent how much of one object or value there is in relation to another object or value. The two numbers in a ratio can only be compared when they have the same unit.

Example:

If there are 10 apples and 5 oranges in a bowl, then the ratio of apples to oranges would be 10 to 5 or 10: 5. This is equivalent to 2:1. In contrast, the ratio of oranges to apples would be 1:2.

Ratio occurs in many situations such as in business where people compare profit to loss, in sports where compare wins to losses etc.

A comparison of two or more numbers is a **ratio**. You can write a 2-term ratio in fractional form when the second term is non-zero. Thus $a : b = \frac{a}{b}$ where $b \neq 0$ and we read a is to b .

When you equate two ratios you are writing a **proportion**. A proportion is a relationship between four numbers or quantities in which the ratio of the first pair equals the ratio of the second pair, and is written as $a : b = c : d$ and is read as "**a is to b while c is to d**"

The proportion $a : b = c : d$ is written as $\frac{a}{b} = \frac{c}{d}$ where a and d are known as **extremes** of the proportion. While b and c are known as **means**. The product of the extremes equals the product of the means. If $\frac{a}{b} = \frac{c}{d}$, $b \neq 0$ and $d \neq 0$, $a \times d = b \times c$

Example 1:

Share 38 400 Frw between Linda and Jean in the ratio 5:7, respectively.

Solution

38 400 Frw is to be shared in the ratio. It is split into 12 equal parts. It means $5 + 7 = 12$ equal parts.

The amount of money Linda receives is $\frac{5}{12} \times 38400 \text{ Frw} = 16000 \text{ Frw}$

The amount of money Jean receives is $\frac{7}{12} \times 38400 \text{ Frw} = 22400 \text{ Frw}$

Example 2:

The working capital of "X" Company Ltd has deteriorated in recent years and now stands as under current assets of 980 000Frw to the current liabilities of 700 000Frw. Compute the current ratio.

Solution

The current ratio is given by $\frac{\text{current assets}}{\text{current liabilities}} = \frac{980000 \text{ Frw}}{700000 \text{ Frw}}$

Taking 98: 70 (after dividing by 10000Frw at both numerator and denominator)

Find the Greatest Common Divisor (GCD) of 98 and 70.

The GCD = 14.

Divide both numbers by 14:

$$\frac{98}{14} : \frac{70}{14} \text{ or } 7 : 5$$

Final ratio is 7 : 5

Application activity 1.4



1. John weighs 56.7 kilograms. If he is going to reduce his weight in the ratio 7: 6, find his new weight.
2. The average age of three boys is 25 years, and their ages are in proportion 3: 5: 7. Find the age of the youngest boy.

1.5 Percentages and related problems

Activity 1.5



Refer to the meaning of decimals and percentage learnt in previous years and

1. Calculate 60:100 and write it in the form of:
 - a) a fraction,
 - b) a percentage and
 - c) a decimal number.
2. Express $\frac{1}{3}$ and $\frac{22}{7}$ in the form of decimal numbers. Is it possible to express this number in the form of percentage? is the percentage obtained an exact number?
3. 24 students in a class took English test. If 18 students passed the test, what percentage of the students did not pass?
4. Are percentages used by bank managers? Explain your answer.

Content Summary

Percentage is defined as the proportion, rate or ratio expressed with a denominator of 100.

Example:

$$\frac{3}{100}, \frac{25}{100}, \text{ etc.}$$

Fraction can be expressed in the form of percentage as $\frac{1}{4} \times 100 = 25\%$.

Decimal format is often required for a value that is usually specified as a percentage in everyday usage. For example, interest rates are usually specified as percentages. A percentage format is just another way of specifying a decimal fraction, $62\% = \frac{62}{100} = 0.62$ and so, percentages can easily be converted into decimal fractions by dividing by 100. Because some fractions cannot be expressed exactly in decimals, one may need to 'round off the answer for convenience. In many of the economic problems (of various books) there is not much point in taking answers beyond two decimal places (2dp). Where this is done then we denote 'to 2 dp' normally put after the answer.

Example:

$$\frac{1}{7} \text{ as a percentage is } 14.29\% \text{ (to 2 dp)}.$$

In solving word problems involving percentage, 3 steps can help you:

1. Make sure you understand the question
2. Sort out the information to make a basic percentage problem
3. Apply the operations to find out what is asked.

Example:

A town council imposes different taxes on different fixed assets as follows: commercial property 25% per year, residential property 15% per year, industrial property 20% per year.

An investor owns a residential building on a plot all valued at 80 000 000Frw an industrial plot worth 75 000 000Frw and a commercial premise worth 12 500 000Frw. How much tax does the investor pay annually?

Solution

$$\text{Commercial: } \frac{25}{100} \times 12,500,000 \text{ Frw} = 3,125,000 \text{ Frw}$$

$$\text{Residential: } \frac{15}{100} \times 80,000,000 \text{ Frw} = 12,000,000 \text{ Frw}$$

Industrial: $\frac{20}{100} \times 75,000,000 \text{ Frw} = 15,000,000 \text{ Frw}$

Total tax the investor can pay annually:

$$32,125,000 \text{ Frw} + 12,000,000 \text{ Frw} + 15,000,000 \text{ Frw} = 30,125,000 \text{ Frw}$$

Application activity 1.5



1. In the middle of the first term, the school organized the test, and the mathematics teacher prepared 20 questions for both section A and B. Peter got 80% of correct answers. How many questions did Peter miss?
2. A student earned a grade of 80% on mathematics test that has 20 questions. How many questions did the student answer correctly? And what percentage of that is not answered correctly?
3. John took a mathematics test and got 35 correct answers and 10 incorrect answers. What was the percentage of correct answers?
4. As an S4 student, is it necessary to know how to determine the percentage? Explain it with supportive examples.

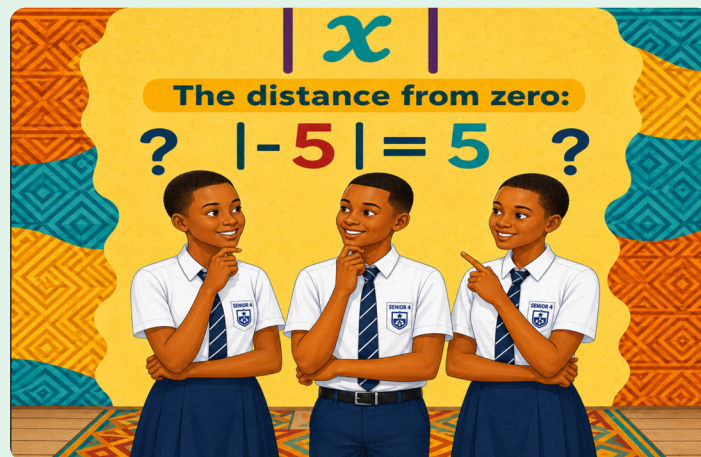
1.6 Absolute value

1.6.1. Concept and properties

Activity 1.6.1



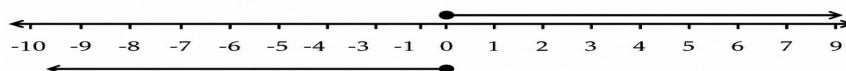
1. Draw a number line and state the number of units found between:
 - a) 0 and -8
 - b) 0 and 8
 - c) 0 and $\frac{1}{2}$
 - d) 4 and 17
2. Do you think that a distance can be expressed by a negative number? What is the distance from 0 to -5?



Content Summary

Absolute value of a number x is the distance (number of units) of that number from the origin (zero point) on a number line. The symbol $| \quad |$ is used to denote the absolute value.

7 is at 7 units from zero, thus the absolute value of 7 is 7 or $|7| = 7$. Also -7 is at 7 units from zero, thus the absolute value of -7 is 7 or $|-7| = 7$. So $|-7| = |7| = 7$ since -7 and 7 are on equal distance from zero on number line.



Note:

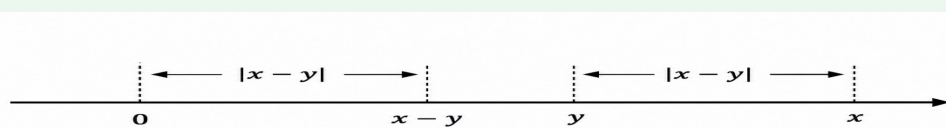
- The absolute value of zero is zero
- The absolute value of a non-zero real number is a positive real number.
- Given that $|x| = k$ where k is a positive real number or zero, then $x = -k$ or $x = k$.

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases},$$

- Geometrically, it represents the non-negative distance from x to 0 on the real line.

More generally, $|x - y|$ represents the nonnegative distance between the point x and y on the real line, since this distance is the same as that from the point to 0. (See the following figure)

$$|x - y| = \begin{cases} x - y, & \text{if } x - y > 0 \text{ or } x \geq y \\ -(x - y), & \text{if } x - y < 0 \text{ or } x \leq y \end{cases}$$



Example:

Find x in the following:

- a) $|x| = 5$ b) $|x| + 5 = 1$ c) $|x - 4| = 10$ d) $|x - 4| < 10$

Solution

a) For $|x| = 5$, we have $x = -5$ or $x = 5$.

b) For $|x| + 5 = 1$, we can write it as $|x| = 1 - 5$ or $|x| = -4$, which is impossible in the set of real numbers

There is no value of x satisfying the given equation since the absolute value of x must be a positive real number.

c) For $|x - 4| = 10$, we have

$$\begin{cases} x - 4 = -10 \\ x - 4 = 10 \end{cases} \quad \text{or} \quad \begin{cases} x = -10 + 4 \\ x = 10 + 4 \end{cases}$$

$$x = -6 \text{ or } x = 14.$$

This means that if $|x - 4| = \begin{cases} x - 4, & \text{if } x - 4 > 0 \\ -(x - 4) & \text{if } x - 4 < 0 \end{cases}$

1.6.2 Properties of the absolute Value

Activity 1.6.2



Evaluate and compare the following:

1. $|3|$ and $|-3|$
2. $|3 \times 5|$ and $|3| \times |5|$
3. $|(-8) + 5|$ and $|-8| + |5|$

Content Summary

From answers you provided for the activity, you can deduce the following properties:

1. Opposite numbers have equal absolute value. $|a| = |-a|$

Example:

$$|5| = |-5| = 5$$

2. The absolute value of a product is equal to the product of the absolute values of the factors. $|ab| = |a||b|$

Example:

$$|4(-6)| = |4||-6|$$

$$|4(-6)| = |-24| = 24$$

$$|4||-6| = 4 \times 6 = 24$$

3. The absolute value of a quotient is equal to the quotient of the absolute values. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

Example:

$$\left|\frac{-8}{2}\right| = \frac{|-8|}{|2|} = \frac{8}{2} = 4 \text{ or } \left|\frac{-8}{2}\right| = |-4| = 4$$

4. The absolute value of a sum is less than or equal to the sum of the absolute values of the addends. $|a+b| \leq |a|+|b|$

Example:

$$|-3+2| \leq |-3|+|2|$$

$$|-1| \leq 3+2$$

$$1 \leq 5$$

Application activity 1.6



1. Find the value(s) of x :

a) $|x| = 6$

b) $|x+1| = 0$

c) $|x-3|-4 = 2$

d) $|2x+1| = 4$

e) $|x-3|+3 = 5$

2. Simplify

a) $|-5|$

b) $|-4||-5|$

c) $|-7|+|4|$

d) $-|4 \times 6|$

e) $-|-6+8|$

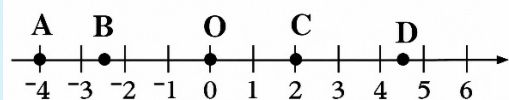
f) $\left|\frac{-18}{6}\right|$

g) $\left|\frac{-4ab}{2c}\right|$

3. Let a and b be the coordinates of points A and B, respectively, on a number line.

The distance between A and B, denoted by $d(A, B)$ is defined by $d(A, B) = |b - a|$.

Refer to the figure below and determine the distance $d(A, B)$, $d(C, B)$ and $d(O, A)$.



1.7 Problems involving absolute value

Activity 1.7



Using absolute value, solve the following equations

a) $|4x-7| = 3$

b) $|2-4x| = 1$

c) $|1+3u| = -6u$

d) $|12z+4| = |4z-6|$

Content Summary

To solve problems involving absolute value, use the definition of absolute value and then solve. After applying the definition to the given equation, you will have two equations to solve.

In fact, when solving absolute value equations, you will usually get two solutions. That is important to keep in mind. Remember that

- If x is positive, $|x| = x$
- If x is negative, $|x| = -x$

Example:

Solve for x when $|x - 4| = 7$

Before we apply the definition, let's make a useful substitution

Let $y = x - 4$, so $|x - 4| = 7$ becomes $|y| = 7$. You must understand this step.

Now, let's apply the definition to $|y| = 7$. Again, you will have two equations to solve

Once again, when solving absolute value equations, you will usually get two solutions.

If y is positive, $|y| = y$, so the first equation to solve is $y = 7$. You must substitute $x - 4$ for y .

After substitution, $y = 7$ becomes $x - 4 = 7$

$$x - 4 + 4 = 7 + 4 \text{ or } x = 11$$

If y is negative, $|y| = -y$, so the second equation to solve is $-y = 7$ or $y = -7$

Substituting $x - 4$ for y , we get $x - 4 = -7$ or $x - 4 + 4 = -7 + 4$ or $x = -3$

The solutions are $x = -3$ and $x = 11$

Example:

Solve for x when $|3x + 3| = 15$

Before we apply the definition, let's make a useful substitution.

Let $y = 3x + 3$, so $|3x + 3| = 15$ becomes $|y| = 15$

Now, let's apply the definition to $|y| = 15$

Lastly, when solving absolute value equations, we usually get two solutions. If y is positive, $|y| = y$, so the first equation to solve is $y = 15$. We substitute $3x + 3$ for y .

After substitution $y = 15$ becomes $3x + 3 = 15$ or $3x + 3 - 3 = 15 - 3$ or $3x = 12$ or $x = \frac{12}{3}$ or $x = 4$.

If y is negative, $|y| = -y$, so the second equation to solve is $-y = 15$ to $y = -15$.

After substitution, becomes $3x + 3 = -15$ or $3x = -18$ or $x = -6$.

The solutions are $x = 4$ and $x = -6$.

Application activity 1.7



1. Solve the following equations:

a) $|2x - 3| = 7$ b) $|4 - 5x| = 16$ c) $|x - 8| = 3$

2. You have money in your wallet, but you don't know the exact amount. When a friend asks you to lend him/her 5000 *Frw*, you say that if you do, from the amount you have, you remain with or miss 1500 *Frw*. Use an absolute value equation to find minimum and maximum amount of money in your pocket?

1.8 Concept and properties of powers

Activity 1.8



Knowing that $4 = 2 \times 2 = 2^2$, examine each expression and rewrite it in power form:

a) $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$

b) $3 \times 3 \times 3 \times 3 \times t \times t \times s \times s \times s \times s \times s$

c) $(x + 2)(x + 2)(x + 2)$

Content Summary

We call n^{th} power of a real number a that we denote a^n , the product of n factors of a . that is

$$a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ factors}} \quad \begin{cases} n \text{ is an exponent} \\ a \text{ is the base} \end{cases}$$

Example:

i. $2^4 = \underbrace{2 \cdot 2 \cdot 2 \cdot 2}_{4 \text{ factors}} = 16$ ii. $3^3 = \underbrace{3 \cdot 3 \cdot 3}_{3 \text{ factors}} = 27$

Notice

- $a^1 = a$
- $a^0 = 1, a \neq 0$
- If $a = 0, a^0$ is not defined

Properties of powers

Let $a, b \in \mathbb{R}$ and $m, n \in \mathbb{Z}$

a) $a^m \cdot a^n = a^{m+n}$

In fact, $a^m \cdot a^n = \underbrace{a \cdot a \cdot a \cdots a}_{m \text{ factors}} \times \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ factors}} = \underbrace{a \cdot a \cdot a \cdots a}_{m+n \text{ factors}} = a^{m+n}$

b) $(a^m)^n = a^{m \times n}$

In fact, $(a^m)^n = \underbrace{a \cdot a \cdot a \cdots a}_{m \text{ factors}} \times \underbrace{a \cdot a \cdot a \cdots a}_{m \text{ factors}} \cdots \underbrace{a \cdot a \cdot a \cdots a}_{m \text{ factors}} = a^{m \times n}$

c) $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

$$\text{In fact, } \left(\frac{a}{b}\right)^m = \underbrace{\frac{a}{b} \cdot \frac{a}{b} \cdot \frac{a}{b} \cdots \frac{a}{b}}_{m \text{ factors}} = \frac{\overbrace{a \cdot a \cdot a \cdots a}^{m \text{ factors}}}{\underbrace{b \cdot b \cdot b \cdots b}_{m \text{ factors}}} = \frac{a^m}{b^m}$$

d) $\frac{1}{b^m} = b^{-m}$

In fact, $\frac{1}{b^m} = \frac{1^m}{b^m} = \left(\frac{1}{b}\right)^m = (b^{-1})^m = b^{-m}$

e) $\frac{a^m}{a^n} = a^{m-n}$

In fact, $\frac{a^m}{a^n} = a^m \frac{1}{a^n} = a^m a^{-n} = a^{m-n}$

f) $(ab)^m = a^m b^m$

In fact, $(ab)^m = \underbrace{ab \cdot ab \cdots ab}_{m \text{ factors}} = \underbrace{a \cdot a \cdots a}_{m \text{ factors}} \times \underbrace{b \cdot b \cdots b}_{m \text{ factors}} = a^m b^m$

These properties help us to simplify some powers.

There is no general way to simplify the sum of powers, even when the powers have the same base. For instance, $2^5 + 2^3 = 32 + 8 = 40$, and 40 is not an integer power of 2. But some products or ratios of powers can be simplified using repeated multiplication models of $\frac{a^m}{a^n}$.

Example 1:

a) $2^4 \cdot 2^3 \cdot 4 = 2^4 \cdot 2^3 \cdot 2^2 = 512$

b) $a^4.b^3.a^5.b^8 = a^4.a^5.b^3.b^8 = a^9.b^{11}$

$a^9 b^{11}$ cannot be simplified further because the bases are different.

$$\text{c) } \frac{y^9}{y^2} = y^{9-2} = y^7$$

Example 2

Expressions used for powers

The Third Power: “Cubed”

Directly following a square, the third power (x^3) has its own dedicated geometric name.

- **Expression:** “To the power of three” or “Cubed”
- **Example:** 5^3 is read as “five cubed” or “five to the power of three.”

- **Geometric Origin:** Just as a square represents a two-dimensional area, a cube represents a three-dimensional volume.

For any exponent greater than 3, we drop the specific geometric nicknames and use a standard, ordinal phrasing.

Power	Mathematical Notation	Common Verbal Expression	Alternative Expression
4 th	x^4	"x to the (four) th power"	"x to the power of four"
5 th	x^5	"x to the (five) th power"	"x to the power of five"
6 th	x^6	"x to the (six) th power"	"x to the power of six"
n th	x^n	"x to the n th power"	"x to the power of n"

Example 3:

On **Day 1**, a school environment club successfully collects and recycles **3 old smartphones**. The club sets a community goal to triple the number of old phones collected on each subsequent day compared to the day before, utilizing social media campaigns to mobilize more students.

1. **Write an exponential expression** using powers to represent the number of smartphones collected on the **6th day** of the challenge.
2. **Calculate the total number** of smartphones collected on that 6th day.
3. **Cross-Cutting Reflection:** If each recycled smartphone prevents approximately 0.5 kg of hazardous e-waste from entering local landfills, how much waste is prevented on the 6th day alone?

Solution

The progression follows the pattern 3^d , where 3 is the base (representing the tripling effect), and d is the exponent representing the day.

- Day 1: $3^1 = 3$
- Day 2: $3 \times 3 = 3^2 = 9$
- Day 6: 3^6

Calculation: $3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 729$ smartphones

Environmental Impact: $729 \times 0.5 \text{ kg} = 364.5 \text{ kg}$ of e-waste prevented.

Application activity 1.8



1. Simplify

a) $x^3 x^2$ b) $(xy^3)^2 + 4x^2 y^6$ c) $\frac{6xy^2}{3xy}$ d) $\frac{ab}{a^3} - \frac{a^3 b^2}{a^5 b}$ e) $\frac{yx}{4xy}$

2. Referring to your real-life experience, where are powers used?

1.9 Problems involving powers

Activity 1.9



A small city has a population P_0 of 50,000 and is growing at a rate r of 2% per year due to a new tech hub. Using the formula $P = P_0(1+r)^t$. What will the population be in a time t of 10 years?

Content Summary

To solve problems involving powers, you can use this simple 5-step rule:

1. Understand the problem fully as described in the statement of the problem;
2. Write the formula as suggested or required by the given problem;
3. Substitute numbers in the formula;
4. Calculate powers step by step;
5. Interpret the result.

This approach reduces fear of formulas.

Example 1:

Suppose a slogan is repeated according to the model: $S = (5^2)^n$ where n is the number of campaign events.

- a) Write the expression for 3 events.
- b) Simplify the power.
- c) Express the result as a single power.
- d) Compare the exponents.
- e) What rule do you use? Can you explain it?

Solution

- a) For 3 events, we have, $S = (5^2)^3$
- b) Power can be simplified as $S = 25^3$
- c) As a single power, $S =$
- d) Comparing the exponents, the inner exponent was 2 and the outer exponent was 3. In the simplified single power, we have $S = 5^6$. The new exponent 6 is the product of the original exponents: $6 = 2 \times 3$.
- e) The rule used is to raise the power to another power: $(a^m)^n = a^{n \times m}$. That is, raising power to another power implies multiplying the exponents.

Example 2:

A transport model for field trips is: $T = (4 \times 5)^n$ where

4 = buses

5 = students per bus group.

- Write the model when $n=3$.
- Evaluate $T = (4 \times 5)^n$.
- Compute 4^3 and 5^3 .
- Multiply the results.
- What property of powers is illustrated?

Solution

- When $n = 3$, the model is $T = (4 \times 5)^3$
- $(4 \times 5)^3 = 20^3 = 20 \times 20 \times 20 = 8,000$
- $4^3 = 4 \times 4 \times 4 = 64$ while $5^3 = 5 \times 5 \times 5 = 125$
- Multiplying 4^3 and 5^3 gives $64 \times 125 = 8,000$
- The property of powers illustrated here is the power of a product: $(ab)^n = a^n b^n$.

Application activity 1.9



- The number of football matches is modeled by: $M = (2^3 \times 2^2)^n$
 - Simplify inside the bracket.
 - Let $n=2$. Compute the value of M .
 - Apply the power rule and write the final answer.
- A geographical map scaling model is: $M = \left(\frac{3}{5}\right)^n$ where n represents repeated scaling.
 - Write the expression for $n=2$
 - Expand the power.
 - Multiply numerators and denominators separately.
 - Write the simplified fraction.
 - Explain the rule(s) used.

1.10 Concept and properties of radicals

Activity 1.10



1. Evaluate the following powers using a calculator

a) $\sqrt{81}$ b) $(216)^{\frac{1}{3}}$ c) $(-27)^{\frac{1}{3}}$ d) $\sqrt[4]{16}$

2. Using examples, explain whether there is a difference between the square and the square root of a number. How do you calculate the square root of a given number?

Content Summary

A radical, or root, is the mathematical opposite of an exponent, in the same sense that addition is the opposite of subtraction. The smallest radical is the square root, represented with the symbol $\sqrt{}$. The next radical is the cube root, represented by the symbol $\sqrt[3]{}$. The small number in front of the radical is its index.

The n^{th} root of a real number is $\frac{1}{n}$ power of that real number. It is noted by

$$\sqrt[n]{b}, b \in \mathbb{R}, n \in \mathbb{N} - \{1\}$$

For $b \geq 0$ when n is even (or for any $b \in \mathbb{R}$ when n is odd), $\forall a, b \in \mathbb{R}, \sqrt[n]{b} = a \Leftrightarrow a = b^{\frac{1}{n}} \Rightarrow a^n = b$

Terminology

In the expression $\sqrt[n]{b}$

- n is called the **index** (or **order**) of the radical.
- b is called the **radicand**.
- $\sqrt{}$ is called the **radical sign**.
- $\sqrt[n]{b}$ is called the **radical expression** (or n^{th} root of b).

Example:

a) $\sqrt[3]{27} = (27)^{\frac{1}{3}} = (3^3)^{\frac{1}{3}} = 3^{3 \times \frac{1}{3}} = 3$

b) $\sqrt[4]{16} = (16)^{\frac{1}{4}} = (2^4)^{\frac{1}{4}} = 2$

If $n = 2$, we say square root and is written as $\sqrt{b}$. Here b must be a positive real number or zero.

If $n = 3$, we say cube root noted $\sqrt[3]{b}$. Here b can be any real number.

If $n = 4$, we say 4th root noted $\sqrt[4]{b}$. Here b must be a positive real number or zero.

Generally, for index n , we say n^{th} root noted $\sqrt[n]{b}$. Here if n is even, b must be a positive real number or zero and if n is odd b can be any real number.

Example:

$\sqrt{-9}$ is not defined in $\mathbb{R}$ because the index in radical is even but $\sqrt[3]{-27} = (-27)^{\frac{1}{3}} = [(-3)^3]^{\frac{1}{3}} = -3$

Properties of radicals

$$\forall n \in \mathbb{N} - \{1\}, m \in \mathbb{Z}$$

$$\text{a) } \sqrt[n]{a^m} = a^{\frac{m}{n}}$$

$$\text{In fact, } \sqrt[n]{ab} = (ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}} = \sqrt[n]{a} \sqrt[n]{b}$$

$$\text{b) } \sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$$

$$\text{In fact, } \sqrt[n]{ab} = (ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}} = \sqrt[n]{a} \sqrt[n]{b}$$

$$\text{c) } \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\text{In fact, } \sqrt[n]{\frac{a}{b}} = \left(\frac{a}{b}\right)^{\frac{1}{n}} = \frac{a^{\frac{1}{n}}}{b^{\frac{1}{n}}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\text{d) } \sqrt[n]{\sqrt[m]{a}} = \sqrt[nm]{a} = a^{\frac{1}{nm}}$$

$$\text{In fact } \sqrt[n]{\sqrt[m]{a}} = \left(a^{\frac{1}{m}}\right)^{\frac{1}{n}} = a^{\frac{1}{m} \times \frac{1}{n}} = a^{\frac{1}{mn}} = \sqrt[nm]{a}$$

Example:

Simplify

$$\text{a) } \sqrt{46656}$$

$$\text{b) } \sqrt[3]{\sqrt{64}}$$

$$\text{c) } \sqrt[3]{ab} \times \sqrt[3]{a^2b^2}$$

$$\text{d) } \sqrt{\frac{36}{81}}$$

Solution

$$\text{a) } \sqrt{46656} = \sqrt{6^6} = 6^3 = 216$$

$$\text{b) } \sqrt[3]{\sqrt{64}} = \sqrt[6]{64} = \sqrt[6]{2^6} = 2$$

$$\text{c) } \sqrt[3]{ab} \times \sqrt[3]{a^2b^2} = \sqrt[3]{a^3b^3} = \sqrt[3]{(ab)^3} = ab$$

$$\text{d) } \sqrt{\frac{36}{81}} = \frac{\sqrt{36}}{\sqrt{81}} = \frac{6}{9} = \frac{2}{3}$$

Addition and subtraction

When adding or subtracting the radicals we may need to simplify if we have similar radicals. Similar radicals are the radicals with the same indices and same bases.

Example:

$$\text{a) } \sqrt{2} + \sqrt{8} = \sqrt{2} + \sqrt{2 \times 4} = \sqrt{2} + \sqrt{2} \times \sqrt{4} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$$

$$\text{b) } \sqrt{3} - \sqrt{27} = \sqrt{3} - \sqrt{3 \times 9} = \sqrt{3} - \sqrt{3} \times \sqrt{9} = \sqrt{3} + 3\sqrt{9} = -2\sqrt{3}$$

Application activity 1.10



1. Simplify the following expressions:

a) $\sqrt{a} \times \sqrt{ab^3} \times \sqrt{bc^2}$	b) $\sqrt[3]{abc} \times \sqrt[3]{a^2b^2c^2}$	c) $\sqrt[3]{\frac{8}{27}}$
d) $\sqrt[4]{x^8}$	e) $\sqrt{\frac{x^3y^4}{4x}}$	f) $\sqrt{18} + \sqrt{2}$
g) $\sqrt{20} + \sqrt{5}$	h) $\sqrt{2} \times \sqrt{3}$	i) $\frac{\sqrt{6}}{\sqrt{2}}$

2. Compute the following radicals

a) $\sqrt{12} - \sqrt{3}$	b) $4\sqrt{3} - \sqrt{12}$	c) $5\sqrt{7} - \sqrt{28}$
d) $\sqrt{18} \times \sqrt{8}$	e) $\sqrt{45} + \sqrt{80} + \sqrt{180}$	f) $\sqrt{108} - \sqrt{48}$

1.11 Rationalization of denominators

Activity 1.11



Make the denominator of each of the following a rational number:

1. $\frac{1}{\sqrt{2}}$	2. $\frac{2-\sqrt{3}}{2\sqrt{5}}$	3. $\frac{2}{1-\sqrt{6}}$	4. $\frac{\sqrt{2}+\sqrt{3}}{\sqrt{3}+\sqrt{5}}$
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Content Summary

Rationalizing a denominator is to convert a fraction with an irrational denominator to a fraction with rational denominator. To do this,

1. If the denominator involves radicals, we multiply the numerator and denominator by the conjugate of the denominator when this is made by two terms.

Some examples: the conjugate of $a + \sqrt{b}$ is $a - \sqrt{b}$ and the conjugate of $a - \sqrt{b}$ is $a + \sqrt{b}$.

2. When the denominator is made by one term, we multiply the numerator and denominator by the same radical. For example, use $\sqrt{a}$ if the denominator is $\sqrt{a}$; you can use $\sqrt{b}$ when the denominator is $a\sqrt{b}$.

Remember that $(a+b)(a-b) = a^2 - b^2$

Example 1:

In a hilly district prone to heavy seasonal rainfall, a civil engineer is designing a concrete drainage channel to divert rainwater safely away from a residential area and prevent soil erosion.

The engineer determines that the optimal cross-sectional velocity of the water flow requires a hydraulic radius (R) calculated by the formula: $R = \frac{6}{\sqrt{3}}$ meters

To easily estimate material costs and communicate the exact measurements to the construction site workers without relying on decimal approximations, the dimension needs to be written with a rational denominator.

- Rationalize the denominator** of the expression $\frac{6}{\sqrt{3}}$ to find the simplified, exact value of the hydraulic radius.
- Calculate the final measurement** as a simplified radical expression.
- Disaster Management Reflection:** Why is it crucial for community safety to ensure that infrastructure calculations like drainage capacity are precise and easy to interpret during construction?

Solution

- To eliminate the radical from the denominator, multiply both the numerator and the denominator by $\sqrt{3}$: $\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$
- Simplification Steps: $\frac{6\sqrt{3}}{\sqrt{9}} = \frac{6\sqrt{3}}{3}$ then, the simplified answer is.
- Why This Context Matters (Pedagogical Note)
 - Real-World Necessity: In practical fields like construction and engineering, expressions are often rationalized because it is significantly easier to divide a number by a whole integer (like 3) than it is to divide by an infinite, non-repeating decimal (like $\sqrt{3} \approx 1.732$).
 - Cross-cutting them: This problem seamlessly integrates Disaster Management and Sustainability by showing us how algebraic precision directly impacts a community's resilience against climate-related challenges like heavy flooding.

Example 2:

$$\begin{aligned} \text{a) } \frac{1}{1+\sqrt{2}} &= \frac{1-\sqrt{2}}{(1+\sqrt{2})(1-\sqrt{2})} = \frac{1-\sqrt{2}}{1-2} = \frac{1-\sqrt{2}}{-1} = \sqrt{2}-1 \\ \text{b) } \frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}} &= \frac{\sqrt{2}(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})(\sqrt{5}+\sqrt{3})} = \frac{\sqrt{10}+\sqrt{6}}{5-3} = \frac{\sqrt{10}+\sqrt{6}}{2} \\ \text{c) } \frac{\sqrt{3}+\sqrt{7}}{4\sqrt{2}} &= \frac{(\sqrt{3}+\sqrt{7})\sqrt{2}}{(4\sqrt{2})\sqrt{2}} = \frac{\sqrt{6}+\sqrt{14}}{8} \end{aligned}$$

Application activity 1.11



1. Rationalize the denominator: a) $\frac{2\sqrt{2}}{4+3\sqrt{3}}$ b) $\frac{a-\sqrt{b}}{\sqrt{d}}$ c) $\frac{3\sqrt{3}+2\sqrt{2}}{1+2\sqrt{2}}$
2. Find the value of a and b if $\frac{\sqrt{7}-2}{\sqrt{7}+2} = a\sqrt{7} + b$

1.12 Problems involving radicals

Activity 1.12



Jane is going to Nene's house to prepare a school project. Instead of walking two perpendicular streets to his classmates' house, Jane will take a short-cut diagonal path through the city plaza. Jane is 15 meters away from Nene's street. The distance from the intersection of the two streets to Nene's house is 8 meters.

1. How will you illustrate the problem?
2. How far will Jane travel along the shortcut?
3. How many meters will she save by taking the short cut rather than walking along the sidewalks?
4. If one of the distances increases or decreases, what might happen to the distance of the short cut? Justify your answer.
5. What mathematical concepts did you use?

Content Summary

Basic word problems involving radical equations are word problems where the radical equation is given along with some of the known information that we can plug into the equation before we simplify/solve for what we are being asked to find.

Since the equation is already given in a basic word problem, the tricky part of solving this type of problem relates on how to solve an equation involving a radical.

Example 2:

A local agricultural cooperative receives a grant to develop a modern, high-yield vegetable farm on a square plot of land. The total area of the plot is exactly 10 hectares.

To maximize solar exposure for the crops, the cooperative decides to split the plot into parallel rectangular cultivation strips. The width (W) of the primary irrigation zone is determined by dividing the total area by the ideal length of the modern sprinkler system, resulting in the following calculation: $W = \frac{10}{\sqrt{5}}$ decameters. To accurately budget for the perimeter fencing materials and ensure the cooperative members can easily measure the boundaries using standard tools, the width needs to be expressed with a rational denominator.

- Rationalize the denominator of the expression $\frac{10}{\sqrt{5}}$.
- Simplify the expression to find the exact, standard measurement of the width.
- Financial Education Reflection: Imagine the cooperative needs to buy fencing material that costs 5,000 Frw per decameter. How does transforming the measurement from an unrationalized fraction ($\frac{10}{\sqrt{5}}$) to a simplified form help the cooperative avoid mathematical errors and budget accurately for their expenses?

Solution

Multiply the numerator and the denominator by $\sqrt{5}$ to eliminate the radical from the bottom:

$$\frac{10}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{10\sqrt{5}}{5}. \text{ The simplified answer is } 2\sqrt{5} \text{ decameters.}$$

Financial Literacy Integration

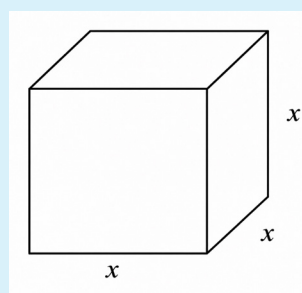
This scenario helps students see that keeping radicals in the denominator makes mental math, scaling, and financial budgeting unnecessarily complex. Simplifying the expression allows the cooperative to estimate values far more efficiently (e.g., knowing $\sqrt{5} \approx 2.24$, the width is roughly 4.48 decameters, making cost estimation straightforward).

Application activity 1.12

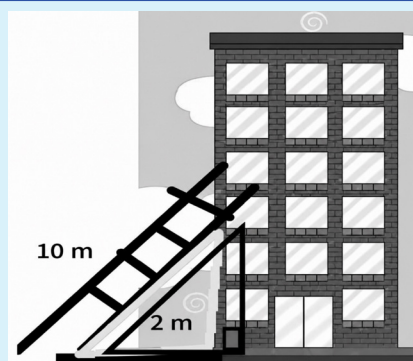


1. On a clear day, the distance d (in meters) that can be seen from the top of a tall building of height h (in meter) can be approximated by $d = 1.2\sqrt{h}$. Approximate the distance that can be seen from Kigali Tower which is 30m tall.

2. A cube has a total surface area of 96 square cm. Find the volume of that cube.



3. To save a woman from a burning building, the firemen placed a 10-meter ladder against the window. The foot of the ladder was 2 meters from the wall as illustrated on the right figure. Approximately, how far up the wall is the ladder touching the wall?



1.13 Concept and properties of decimal logarithms

Activity 1.13



- What is the real number at which 10 must be raised to obtain:
a) 1 b) 10 c) 100 d) 1000 e) 10000 f) 100000
- Explain your classmate how you can find the number x if $x^3 = 64$.

Content Summary

The decimal logarithm of a positive real number x is defined to be a real number y for which 10 must be raised to obtain x . We write $\forall x > 0, y = \log x$ as $x = 10^y$

$\log x$ is the same as $\log_{10} x$ and is defined for all positive real numbers only. 10 is the base of this logarithm. In general notation we do not write this base for decimal logarithm.

In the notation $y = \log x$, x is said to be the **antilogarithm** of y .

Example 1:

$$\log(100) = ?$$

We are required to find the power to which 10 must be raised to obtain 100

$$\text{So } \log(100) = 2 \text{ as } 100 = 10^2$$

$$y \quad \log x \text{ means } 10^y = x$$

Be careful! $\log 2x + 1 \neq \log(2x + 1)$
 $\log 2x + 1 = (\log 2x) + 1$

Properties

$$\forall a, b \in]0, \infty[$$

a) $\log a = \log b \Leftrightarrow a = b$. This is known as one to one property

b) $\log ab = \log a + \log b$

c) $\log \frac{1}{b} = -\log b$

d) $\log \frac{a}{b} = \log a - \log b$

e) $\log a^n = n \log a$

f) $\log \sqrt{a} = \log a^{\frac{1}{2}} = \frac{1}{2} \log a$

g) $\log \sqrt[m]{a^n} = \log a^{\frac{n}{m}} = \frac{n}{m} \log a$

h) $\text{co} \log x = \log \left(\frac{1}{x} \right) = -\log x$

Example 2:

Calculate in terms of $\log a$, $\log b$ and $\log c$

a) $\log a^2 b^2$

b) $\log \frac{ab}{c}$

c) $\log \frac{ab}{\sqrt{c}}$

Solution

a) $\log a^2 b^2 = \log (ab)^2 = 2 \log ab = 2(\log a + \log b)$

b) $\log \frac{ab}{c} = \log ab - \log c = \log a + \log b - \log c$

c) $\log \frac{ab}{\sqrt{c}} = \log ab - \log \sqrt{c} = \log a + \log b - \log (c)^{\frac{1}{2}} = \log a + \log b - \frac{1}{2} \log c$

Example 3:

Given that $\log 2 = 0.30$, $\log 3 = 0.48$ and $\log 5 = 0.69$.

Calculate: a) $\log 6$ b) $\log 0.9$

Solution

a) $\log 6 = \log (2 \times 3) = \log 2 + \log 3 = 0.30 + 0.48 = 0.78$

$$\log 6 = \log(2 \times 3) = 0.78$$

b) $\log 0.9 = \log \left(\frac{9}{10} \right) = \log 9 - \log 10 = \log 3^2 - \log (2 \times 5) = 2 \log 3 - \log 2 - \log 5$

$$= 2(0.48) - 0.30 - 0.70 = -0.04$$

Co-logarithm

Co-logarithm, sometimes shortened to **colog**, of a number is the logarithm of the reciprocal of that number, equal to the negative of the logarithm of the number itself,

$$\text{colog } x = \log \left(\frac{1}{x} \right) = -\log x$$

Example 4:

$$\text{colog } 200 = -\log 200 = -2.3010$$

Change of base formula

If ($u > 0$) and if a and b are positive real numbers different from 1, $\log_b u = \frac{\log_a u}{\log_a b}$

This means that if you have a logarithm in any other base, you can convert it into the decimal

logarithm in the following way where $a = 10$: $\log_b u = \frac{\log_{10} u}{\log_{10} b} = \frac{\log u}{\log b}$

This is for example: $\log_2 5 = \frac{\log_{10} 5}{\log_{10} 2} \approx 2.322$

There is another special logarithm called natural logarithm which has the base a number $e \approx 2.71828$. This logarithm is written as: $\log_e x = \ln x$.

Example 5:

Following a heavy rainy season, a community water management committee needs to test the acidity of a local drinking water reservoir to ensure it remains safe for consumption.

The acidity of water is measured on the pH scale, which is defined mathematically using a decimal logarithm: $pH = -\log_{10}[H^+]$

Where $[H^+]$ represents the concentration of hydrogen ions in moles per liter.

A laboratory technician tests two different water samples from the reservoir network:

- Sample A (Untreated Water): Has a hydrogen ion concentration of $[H^+] = 10^{-5}$ mol/L.
- Sample B (Treated Water): Has a hydrogen ion concentration that is exactly 100 times less acidic than Sample A (meaning its hydrogen ion concentration is divided by 100).

1. Express the pH of Sample B as a logarithmic expansion using the quotient property of

$$\text{logarithms: } \log_{10} \left(\frac{M}{N} \right) = \log_{10} M - \log_{10} N$$

2. Calculate the exact pH level of both Sample A and Sample B using the power property ($\log_{10} x^k = k \log_{10} x$) and the fact that $\log_{10} 10 = 1$.
3. Health and Value Reflection: Safe drinking water typically requires a pH between 6.5 and 8.5. Based on your calculations, which sample is safe for the community to drink, and why is regular scientific monitoring a vital civic responsibility?

Solution

- Find the concentration of Sample B: Concentration of Sample B = $\frac{10^{-5}}{100} = \frac{10^{-5}}{10^2}$
- Apply Logarithm Properties for Sample A

$$pH_A = -\log_{10}(10^{-5}) \text{ Applying the power property:}$$

$$pH_A = -(-5) \times \log_{10} 10 = 5 \times 1 = 5 \text{ (Sample A is acidic and unsafe).}$$

- Apply Logarithm Properties for Sample B

Using the quotient property: $pH_B = -(\log_{10}(10^{-5}) - \log_{10}(10^2))$

$$pH_B = -\log_{10}(10^{-5}) + \log_{10}(10^2) \text{ Using the power property: } pH_B = 5 + 2 = 7$$

(Sample B has a neutral pH of 7, making it safe).

This question reinforces health and environment standards by demonstrating how mathematical modeling directly ensures the safety of shared community resources.

Application activity 1.13



- Without using calculator, compare the numbers a and b.
 - $a = 3\log 2$ and $b = \log 7$
 - $a = \log 2 + \log 40$ and $b = 4\log 2 + \log 5$
 - $a = 2\log 2$ and $b = \log 16 - \log 3$
- Given that $\log 2 = 0.30$, $\log 3 = 0.48$ and $\log 5 = 0.69$. Calculate
 - $\log 150$
 - $\log\left(\frac{9}{2}\right)$
 - $\log(0.2) + \log 10$
- Find co-logarithm of
 - 100
 - 42
 - 15

1.14 Simple equations involving decimal logarithms

Activity 1.14



Consider the logarithmic equations below: a) $\log x = 3$ b) $\log 2x = 4$ c) $\log x = \log(4x - 9)$

- For each equation:
 - Rewrite it in exponential form
 - Identify which logarithmic property is being applied.
- Solve each equation step by step by justifying each step using logarithmic laws.
- Determine the conditions on x for each equation to be defined and check whether your solutions satisfy these conditions.
- Compare how equations (a) and (b) differ from (c) in structure. What special property allows you to solve equation (c) directly?

Content Summary

Logarithmic equations can be solved using the laws of logarithms. These laws allow us to rewrite logarithms and form more convenient expressions. The goal is to reduce the logarithmic equation until you get a single logarithm on each side or a single logarithm on one side. Based on this, we can distinguish two types of logarithmic equations. We must recognize these two types to facilitate solving the equations.

Remarks

When solving logarithmic equations, you cannot simply manipulate symbols and accept all algebraic solutions. Logarithms are only defined for positive arguments (and a positive base, though we use base 10 here).

The correct two-step process is:

1. Find the domain of solution: Write down all conditions that make every logarithm in the equation valid.
For decimal logarithms (base 10), the rule is: "The input (argument) of every log must be strictly greater than 0. So if you see $\log(f(x))$, you require $f(x) > 0$."
2. Solve algebraically: Solve the equation as usual (by considering logarithmic properties aligned with the given equation).
3. Check each solution: Compare every algebraic solution against the domain of solution.
 - Keep only those values that satisfy all domain conditions.
 - Reject any value that falls outside the domain; even if it makes the algebra work, it is not a valid solution to the original equation.

Example 1:

Solve for x in the set of real number: $\log(x-3) = 2$

Solution

- Domain of solution: The argument must be positive: $x-3 > 0 \rightarrow x > 3$
Domain of solution: $(3, \infty)$
- Algebraic solution: Convert to exponential form (here we have base 10) :
 $x-3 = 10 \rightarrow x-3 = 100 \rightarrow x = 103$
- Check against domain: $103 \in (3, \infty) \rightarrow \text{valid}$.
- Therefore, $S = \{103\}$

Example 2:

Solve for x in the set of real number: $\log x + \log(x-2) = 1000$

Solution

- Domain of solution: Both arguments must be positive: $x > 0$ and $x-2 > 0 \rightarrow x > 2$
Domain: $(2, \infty)$
- Algebraic solution: Use $\log A + \log B = \log AB$ and $\log A = \log B \rightarrow A = B$

Note that $\log x + \log(x-2) = 1000 \Leftrightarrow \log x + \log(x-2) = \log 3$ and

$$\log x + \log(x-2) = \log 3 \Leftrightarrow \log x(x-2) = \log 3$$

Since logs are one-to-one:

$$x(x-2)=3 \rightarrow x^2-2x-3=0$$

$$(x-3)(x+1)=0 \rightarrow x=3 \text{ or } x=-1$$

- Check obtained value against domain of solution $(2, \infty)$:
 - $x=3 \rightarrow$ valid (because $3 > 2$).
 - $x=-1 \rightarrow$ reject (because $-1 < 2$).

Therefore, the solution set is $S = \{3\}$

Types of logarithmic equations

Generally, after applying the laws of logarithms to reduce the equation, we can end up with one of two types of logarithmic equations:

- The first type looks like this:

$$\log_b P = \log_b Q \Leftrightarrow P = Q$$

In cases where we end up with only one logarithm on each side of the equation, we can eliminate the logarithms if they have the same base, and we can form an equation with the arguments. For example, in the expression above, the arguments are the algebraic expressions represented by **P** and **Q**.

- The second type looks like this:

$$\log_b P = Q \Leftrightarrow P = b^Q$$

In cases where we end up with a single logarithm on only one side of the equation, we can write the logarithm as an exponential expression and solve it that way.

Example 1:

Solve the following equation

$$\log(2x+2) + \log 2 = \log(x+1) + \log 3$$

Solution

$$\log(2x+2) + \log 2 = \log(x+1) + \log 3 \quad (\text{All logarithms are decimal, base 10.})$$

Find the domain of solution

For every logarithm, the argument must be strictly positive:

1. From $\log(2x+2)$: $2x+2 > 0 \rightarrow x > -1$
2. From $\log(x+1)$: $x+1 > 0 \rightarrow x > -1$

The constants $\log 2$ and $\log 3$ have no domain restriction.

So the **domain** is the intersection (for 1 and 2): $x > -1$ or $x \in (-1, \infty)$

Solve Algebraically

Use logarithm properties to combine terms.

$$\text{Left side: } \log(2x+2) + \log 2 = \log[(2x+2) \times 2] = \log(4x+4)$$

$$\text{Right side: } \log(x+1) + \log 3 = \log[(x+1) \times 3] = \log(3x+3)$$

$$\text{Now the equation becomes: } \log(4x+4) = \log(3x+3)$$

Since the log function is one-to-one (same base), we equate the arguments:
 $(4x+4) = (3x+3) \rightarrow 4x-3x = 3-4 \rightarrow x = -1$

Check against the domain of solution

Domain of solution requires: $x > -1$ or $x \in (-1, \infty)$

But our algebraic solution is: $x = -1$ and $x \notin (-1, \infty)$

Therefore, $x = -1$ is outside of the domain of solution. Since the only algebraic solution is rejected, the equation has no valid solution.

$$S = \{ \} \text{ or } S = \emptyset$$

Note: If you had blindly cancelled or simplified without checking, you might have thought $x = -1$ works. But plugging $x = -1$ into the original equation gives:

- $\log(2(-1)+2) = \log(-2+2) = \log 0 \rightarrow \text{undefined}$
- $\log(-1+1) = \log 0 \rightarrow \text{undefined}$

So -1 is **extraneous** and must be rejected. Always find the **domain first**, then reject values outside it.

Example 2:

What is the value of x in $\log(x+3) - \log 2 = \log(x-1) - \log 7$

Solution

- We start with the equation: $\log(x+3) - \log 2 = \log(x-1) - \log 7$
- Use the logarithm property $\log a - \log b = \log \frac{a}{b}$ which implies that
 $\log(x+3) - \log 2 = \log(x-1) - \log 7$ gives $\log\left(\frac{x+3}{2}\right) = \log\left(\frac{x-1}{7}\right)$
- Since the log function is one-to-one, we can equate the arguments: $\frac{x+3}{2} = \frac{x-1}{7}$
- Cross-multiply: $\frac{x+3}{2} = \frac{x-1}{7} \rightarrow 7(x+3) = 2(x-1)$ and $7(x+3) = 2(x-1) \rightarrow 7x+21 = 2x-2$
- Solve for x : $7x+21 = 2x-2 \rightarrow 7x-2x = -2-21$ and $x = -\frac{23}{5}$
- Check the domain of solution: $x = -\frac{23}{5} = -4.6 \notin (1, \infty)$
- The solution set is $S = \{ \}$

Example 3:

What is the value of x in $\log(4x + 60) = 2$?

Solution

We start with the equation: $\log(4x + 60) = 2$

- Assume the logarithm is base 10 (common logarithm) unless specified otherwise.
 $\log_{10}(4x + 60) = 2$
- Rewrite in exponential form: $4x + 60 = 10^2 \rightarrow 4x + 60 = 100$
- Solve for x : $4x + 60 = 100 \rightarrow 4x = 100 - 60 \rightarrow 4x = 40 \rightarrow x = 10$

Check the domain of solution. For the log to be defined: $4x + 60 > 0 \rightarrow x > -15$ or $x \in (-15, \infty)$

This value $x = 10$ valid because $10 \in (-15, \infty)$. Therefore, $S = \{10\}$ or the value of x that satisfies $\log(4x + 60) = 2$ is 10.

Application activity 1.14



Solve the following logarithmic equations:

- | | | |
|--------------------------|--------------------------|---------------------------------|
| 1. $\log x = 5$ | 2. $\log(3x) = -3$ | 3. $\log(3x) = \log(x + 2)$ |
| 4. $\log x + \log 2 = 2$ | 5. $3 = \log x - \log 3$ | 6. $\log 2x = \log 3x - \log x$ |

1.15 Simple interest and related problems

Activity 1.15



Use any available resource material(s) to find the meaning of principal and simple interest in investment and discover how to calculate a simple interest from a given principal after a given time.

Suppose you invest 10,000Frw at 10% in a saving cooperative at simple interest.

- a) How much money will you have after first year?
- b) If you keep all your 10,000Frw with its interest in the same bank, how much money will you have after two years?
- c) How much money will you have after three years of simple interest?

Content Summary

Simple interest is the amount charged when one borrows money or loan from a financial

institution which accrues yearly. This interest is a fixed percentage charged on money/loan that is not yet paid. This interest is calculated based on the original principal or loan and is paid at regular intervals. From, above example, we see that when the principal (P), rate in percentage (R) and time in year (T) are given, then simple interest (I) for the given period is

$$\text{given by } I = P \cdot \frac{R}{100} \cdot T = \frac{PRT}{100}$$

The total amount (A) paid back by the borrower or the financial institution on the expiry of the interest period is the sum of the principal and the interest earned (I). Thus, $A = P + I$.

Example 1:

Find the simple interest earned from 3400Frw borrowed for 3years at the rate of 10%p.a.

Solution

$$\text{Interest, } I = \frac{PRT}{100} = \frac{3400 \times 10 \times 3 \text{Frw}}{100} = 1020 \text{ Frw}$$

Example 2:

Gatete borrowed 32 000Frw from a lending institution to start a business. If the institution charged interest at a rate of 8%p.a., calculate the simple interest and the total amount eventually paid back after 4years.

Solution

$$\text{Interest, } I = \frac{PRT}{100} = \frac{32000 \times 8 \times 4 \text{Frw}}{100} = 10240 \text{Frw}$$

$$\text{Amount, } A = (32000 + 10240) \text{ Frw} = 42240 \text{ Frw}$$

Application activity 1.15



1. A 2-years loan of \$500 is made with 4% simple interest. Find the interest earned.
2. You put 100,000Frw into your savings account which pays you by simple interest at 7% per annum and you leave it in the account for 5 years. Calculate your interest from your investment after five years. How much money will you have after five years?
3. In how many years will a sum of 4,000Frw yield a simple interest of 1,440Frw at 12% per annum?

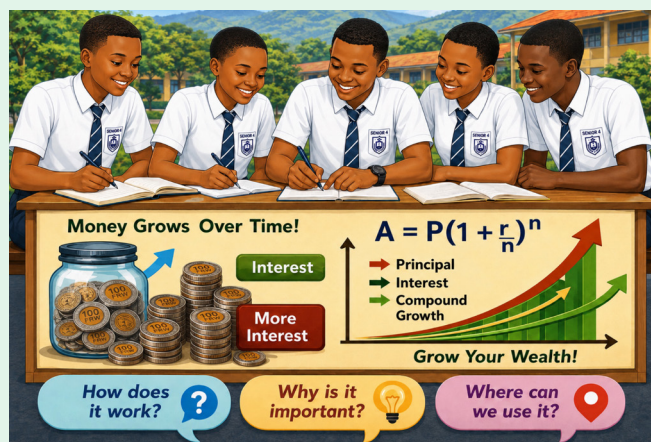
1.16 Compound interest

Activity 1.16



Two friends, Aline and Eric, each save 10,000Frw in two different saving cooperatives that both advertise 10% interest per year.

- Aline's cooperative uses simple interest, where interest is always calculated on the original 10,000Frw.
- Eric's cooperative adds interest to the money each year before calculating the next interest. Investigate how their savings grow over time.



- Calculate the amount of money that each person has after the first year. What do you observe?
- Calculate how much money Aline has after the second year if the interest is still calculated only on the original 10,000Frw.
- Calculate how much money Eric has after the second year if the interest is calculated on the new total amount after the first year.
- After three years, which person has more money? By how much?
- Based on your results, explain why Eric's money grows faster and suggest what this way of calculating interest might be called.

Content Summary

Compound interest is the interest calculated not only on the original money invested (called the principal) but also on the interest that has already been added in previous periods.

The accumulated amount A after the time t (number of years P is invested) is given by:

$A = P\left(1 + \frac{r}{n}\right)^{n \times t}$ where n is the number of interest periods per year, r is the interest rate expressed as decimal, A is the amount after t years.

Note: When R = Annual interest rate (%), the compound interest formula when interest is compounded n times per year: $A = P\left(1 + \frac{R}{n \times 100}\right)^{n \times t}$

Where: A = Total amount after t years; P = Principal amount; n = Number of compounding periods per year; t = Time in years. This formula is the same as the previous one because $r = \frac{R}{100}$.

A general formula for calculating the interest rate can be derived. Starting with the familiar final sum formula. Compound interest = Accumulated amount (A) – Principal (P).

Compound interest is $I = A - P = P\left(1 + \frac{r}{n}\right)^{nt} - P = P\left(\left(1 + \frac{r}{n}\right)^{nt} - 1\right)$, where A is accumulated amount and P principal.

Example 1:

A trader deposited 63000Frw in a fixed deposit account with a local bank which attracted an interest of 8% per annual (p.a) compound interest. Find:

- a) The total amount after 4 years. b) Compound interest.

Solution

$$P = 63000 \text{ Frw}; n = 1; t = 4; r = 8\% \text{ p.a. and } A = P\left(1 + \frac{r}{100}\right)^t$$

$$\text{a) } A = 63000\left(1 + \frac{8}{100}\right)^4 = 63000(1.08)^4 = 63000 \times 1.36048896 = 85710.80 \text{ Frw}$$

$$\text{b) } I = (85710.80 - 63000) \text{ Frw} = 22710.80 \text{ Frw}.$$

Application activity 1.16



1. You put 100,000Frw into your savings account which pays you by compound interest at 7% per annum and you leave it in the account for 5 years. Calculate your interest from your investment after five years.
2. If you deposit 6,500Frw into an account paying 8% annual interest compounded monthly, how much money will be in the account after 7 years?
3. If you deposit 4,000ü into an account paying 6% annual interest compounded quarterly, how much money will be in the account after 5 years?

1.17 Problems related to compound interest

Activity 1.17



Suppose you invest 10,000Frw at 100% in a bank

- a) How much money will you have after first year?
- b) If you keep all your 10,000Frw with its interest in the same bank, how much money will you have after two years? How much money will you have after three years of compound interest?

Content Summary

When tackling problems related to compound interest, the key is to understand that your money or any principal amount grows not just on the initial sum, but also on the interest that accumulates over time. This means you'll be working with the compound interest formula

$A = P\left(1 + \frac{r}{n}\right)^{nt}$, where A is the final amount, P is the principal, n is the annual interest rate (in decimal form), n is the number of times interest is compounded per year, and t is the time in years. Your main tasks will include identifying the given values, correctly substituting them into the formula, solving for the unknown variable (which could be the final amount, principal, rate, time, or compounding frequency), and interpreting the result in a real-world context such as savings accounts, loans, or investments. Always pay careful attention to whether the interest is compounded annually, semiannually, quarterly, monthly, or daily, because that changes n and directly affects your final answer. With consistent practice, you'll learn to set up these problems logically and avoid common mistakes like forgetting to convert percentages to decimals or misplacing the time period.

Example 1:

Find the compound interest earned on 15000Frw invested for 3 years, at 20%p.a. compounded quarterly.

Solution

Here, each year has 4 interest period (quarterly) *i.e* in 3 years, there are 12 interest period

$(3 \times 4 = 12)$. The rate, $r\% = 20 \div 4 = 5\% p.a.$

$$A = 15000\left(1 + \frac{5}{100}\right)^{12}$$

$$A = 15000(1.05)^{12} = 26937.84\text{Frw}$$

$$I = 26937.84 - 15000 = 11937.84\text{Frw}$$

Example 2:

How long will it take 10,000Frw to double it on account earning 2% compounded quarterly?

Solution

For this problem, we'll use the compound Interest formula, $F = P(1+i)^n$ where F is the final value and P the initial value of investment. Since we want to know how long it will take, let t represent the time in years. The number of compounding periods is four times the time or $n = 4t$. The original amount is $P = 10,000$ and the future value is the double

or $F = 20,000$. The interest rate per period is $i = \frac{0.02}{4} = 0.005$. When these values are substituted into the compound interest formula, we get the exponential equation

$20000 = 10000(1.005)^{4t}$. To solve this equation for t , isolate the exponential factor by dividing both sides by 10,000 to give $2 = (1.005)^{4t}$

Convert this exponential form to logarithm form and divide by 4

$$t = \frac{\frac{\log 2}{\log 1.005}}{4} = \frac{\log 2}{4 \log 1.005} \text{ or } t = 34.7$$

It will take 34.7 years to double 10,000 Frw on account earning 2%.

Note:

It is interesting to note that the starting amount is irrelevant when doubling. If we started with P dollars and wanted to accumulate $2P$ at the same interest rate and compounding periods, we would need to solve $2P = P(1.005)^{4t}$.

This reduces to the same equation as above $\log 2 = 4t \log 1.005$ when both sides are divided by P . This means it takes about 37.4 years to double any amount of money at an interest rate of 2% compounded quarterly.

Considering the case in which P is invested in a bank for n interest periods per year at the rate r expressed as a decimal. The accumulated amount A after the time t (number of years

P is invested) is given by: $A = P \left(1 + \frac{r}{n} \right)^{nt}$

Where n is the number of interest periods per year, r is the interest rate expressed as decimal, A is the amount after two years.

Application activity 1.17



1. How much money would you need to deposit today at 9% annual interest compounded monthly to have 12,000 Frw in the account after 6 years?
2. If you deposit \$8000 into an account paying 7% annual interest compounded quarterly, how long until there is \$12,400 in the account?

1.18 Final value of investment

Activity 1.18



Kwizera takes out a simple interest loan of P Frw at rate r .

- a) Find the total amount Kwizera will repay if the loan is for n years and establish the formula giving final value F for simple interest.
- b) Determine the formula giving final value F for compound interest at rate r of principal P after t years. Illustrate your ideas using

$$P = 10,000 \text{ Frw} ; r = 10\% = 0.1; t = 3$$

Content Summary

Consider an investment at compound interest where: P is the initial sum invested; A is the final value of the investment, r is the interest rate per period (as a decimal fraction) and n is the number of time periods. The value of the investment at the end of each year will be $1 + r$ times the sum invested at the start of the year. Thus, for any investment,

- The value after one year $P(1 + r)$
- Value after 2 years $P(1 + r)(1 + r) = P(1 + r)^2$
- Value after 3 years $= P(1 + r)(1 + r)(1 + r) = P(1 + r)^3$ etc

We can see that each value is multiplied by $1 + r$ to the power of number of years that the sum is invested. Thus, after n years the initial sum P is multiplied by $(1 + r)^n$. The formula for the final value A . Thus, after an investment of P money, for n time periods at interest rate r is therefore $A = P(1 + r)^n$.

Example 1:

If \$600 is invested at compound interest of 8% per year, determine the final sum after 3 years.

Solution

$$P = \$600; \quad n = 3; \quad r = 8\% = 0.08$$

Thus, the final sum will be

$$A = P(1 + r)^n = 600(1.08)^3 = \$755.83$$

Example 2:

If \$4,000 is invested for 10 years at an interest rate of 11% per annum what will be the final value of the investment?

Solution

$$P = \$4,000, \quad n = 10 \quad r = 11\% = 0.11$$

$$F = P(1 + r)^n = 4,000(1 + 0.11)^{10} = 11,357.68$$

The final value of the investment be \$11,357.68

So far, we have only calculated F for given values of P , r and n . However, if the values of any three of the variables in this equation are given then one can usually calculate the fourth.

Initial amount

A formula to calculate P , when values for F , r and n are given, can be derived as follows.

Since the final sum formula is $F = P(1 + r)^n$,

Then, dividing through by $(1 + r)^n$, we get the initial sum formula $P = \frac{F}{(1 + r)^n}$ or $P = F(1 + r)^{-n}$

Example 3:

How much money needs to be invested now to accumulate a final sum of \$12,000 in 4 years' time at an annual rate of interest of 10%?

Solution

Using the formula derived above, the initial amount is

$$P = F(1 + r)^{-n} = 12,000(1.1)^{-4} = \$8,196.16$$

What we have done in the above example is to find the sum of money that is equivalent to \$12,000 in 4 years' time if interest rates are 10%.

An investor would therefore be in different between (a) \$8,196.16 now and (b) \$12,000 in 4 years' time.

\$8,196.16 is therefore known as the 'present value' (PV) of \$12,000 in 4 years' time. We shall come back to this concept in the next few sections when methods of appraising different types of investment project are explained.

Time period

Calculating the time is rather trickier than the calculation of the initial amount.

From the final Sum formula.

$$A = P(1 + r)^n, \text{ then } \frac{A}{P} = (1 + r)^n$$

If the values of A , P and r are given and one is trying to find n this means that one has to work out to what power $(1 + r)$ must be raised to equal $\frac{A}{P}$. One way of doing this is via logarithms.

Example 4:

For how many years' must \$1,000 be invested at 10% to accumulate \$1,600?

Solution

$$P = \$1,000, A = \$1,600, r = 10\% = 0.1$$

Substituting these values into the formula

$$\frac{A}{P} = (1 + r)^n \text{ then, } \frac{1600}{1000} = (1 + 0.1)^n$$

$$\text{We get } 1.6 = (1.1)^n$$

Since to find the n^{th} power of a number its logarithm must be multiplied by n . Finding logs, this means that our equation becomes $\log 1.6 = n \log(1.1)$

And $n = \frac{\log(1.6)}{\log(1.1)} = 4.93$. Given that 4.93 years is approximately 5 years,

If investments must be made for whole years, then the answer is 5 years. This answer can be checked using the final sum formula $F = P(1+r)^n = 1000(1.1)^5 = 1,610.51 \approx 1600$

If \$1,000 is invested for a full 5 years, then it accumulates just over \$1,600, which checks out with the answer above. A general formula to solve for n can be derived as follows from the final sum formula: $A = P(1+r)^n$. Therefore, $\frac{A}{P} = (1+r)^n$ $A = P(1+r)^n$, and $n = \frac{\log(A/P)}{\log(1+r)}$.

An alternative approach is to use the iterative method and plot different values on a spreadsheet. To find the value of n for which $1.6 = (1.1)^n$.

This entails setting up a formula to calculate the function $y = (1.1)^n$ and then computing it for different values of n until the answer 1.6 is reached. Although some students who find it difficult to use logarithms will prefer to use a spreadsheet, logarithms are used in the other examples in this section. Logarithms are needed to analyze other concepts related to investment and so you really need to understand how to use them.

Example 5:

How many years will \$2,000 invested at 5% take to accumulate to \$3,000?

Solution

$$P = 2,000; \quad A = 3,000; \quad r = 5\% = 0.05$$

Using these given values in the time period formula derived above gives

$$n = \frac{\log(A/P)}{\log(1+r)} = \frac{\log(1.5)}{\log(1.05)} = 8.34$$

This money will need to be invested in 8.34 years.

Example 6:

How long will any sum of money take to double its value if it is invested at 12.5%?

Solution

Let the initial sum be A . Therefore, the final sum is $A = 2P$ and $r = 12.5\%$.

Substituting this value for A and r into the final sum formula $A = P(1+r)^n$, we find $2A = A(1.125)^n$ or $2 = (1.125)^n$ which gives $n = \frac{\log 2}{\log 1.125} = 5.9$.

For any sum of money, it takes 5.9 years to double its value if it is invested at 12.5%.

Interest rates

When studying interest rates, your main goal is to understand how the cost of borrowing money or the return on saving money is quantified over time. Interest rates are typically expressed as a percentage of the principal (the initial amount) and can be applied in two fundamental ways: simple interest, calculated only on the original principal using $I = Prt$, and compound interest, where interest is earned on both the principal and previously accumulated interest, using $A = P\left(1 + \frac{r}{n}\right)^{n \times t}$. As a student, you'll need to distinguish between nominal (stated) rates and effective (actual) rates, especially when compounding occurs more than once a year, because that affects the true cost or yield. Your tasks will involve converting percentages to decimals, identifying the compounding period, solving for unknowns like time, rate, or principal, and comparing different interest rate offers to make informed financial decisions. Beyond formulas, you should also develop an intuition for how even small changes in the interest rate can significantly impact long-term growth or debt, which is crucial for real-life applications like mortgages, credit cards, savings accounts, and investments. Ultimately, mastering interest rates means not just computing numbers, but also interpreting

Example 7:

If \$4,000 invested for 10 years is projected to accumulate to \$6,000, what interest rate is used to derive this forecast?

Solution

$$P = 4,000 \quad A = 6,000 \quad \text{and} \quad n = 10$$

Substituting these values into the final sum formula

$$A = P(1+r)^n \text{ gives } 6000 = 4000(1+r)^{10} \text{ from which } 1.5 = (1+r)^{10} \text{ or } 1+r = \sqrt[10]{1.5}.$$

Therefore, $r = 4.14\%$

Application activity 1.18



You have 90,000Frw to deposit. Bank A offers 12 percent per year compounded monthly, while Bank B offers 12 percent but will only compound annually. How much will your investment be worth in 10 years at each bank?

1.19 Problems related to final value of investment

Activity 1.19



An investor deposits \$750 at a compound interest rate of 6% per year for 5 years.

- What does the expression $(1+r)^n$ represent in this investment context?
- Calculate $(1.06)^n$. What does this value represent?
- Using the formula $A = P(1+r)^n$, compute the final accumulated value.
- If the investor wanted to reach \$1,000 instead, would 5 years be enough at 6%? Explain briefly.

Content Summary

Example 1:

You want to have \$5,000 in 6 years, and the interest rate is 7% per year.

- Starting from $A = P(1+r)^n$, rearrange the formula to solve for P.
- Substitute the known values into your new formula.
- Computing the present value, how much must be invested today?
- Explain in your own words why the present value is less than \$5,000.

Solution

- Rearranging the formula $A = P(1+r)^n$, we get $P = A(1+r)^{-n}$
- Substituting the known values into the new formula, we get
$$P = 5,000(1+0.07)^{-6} = 5,000(1.07)^{-6} = 5,000 \times 0.666 = 3,331 \text{ or } \$3,331$$
- The present value is \$3,331. This is the value after substitution above.
- Because interest will grow the money over 6 years, you do not need to deposit the full \$5,000 today. The initial amount is discounted using the compound factor $(1+r)^{-n}$.

Example 2:

An investor wants to know how long it will take \$2,500 to grow to \$4,000 at 6% compound interest.

- Starting with the formula $A = P(1+r)^n$, write an equation involving 4000, 2500, and 1.06.
- Rearrange the formula to isolate the exponent n
- Use logarithms to compute the value of n
- Interpret the meaning of your answer in years (round to two decimals)

Solution

- The equation involving 4000, 2500, and 1.06 is $4,000 = 2,500(1.06)^n$
- Rearranging the formula $A = P(1+r)^n$ to isolate n , we divide both sides by 2,500 to get $1.6 = (1.06)^n$.
- Taking the logarithm of both sides of $1.6 = (1.06)^n$ gives $\log(1.6) = n \log(1.06)$
from which $n = \frac{\log(1.6)}{\log(1.06)} = \frac{0.20412}{0.02531} \approx 8.06$
- It takes a little over 8 years for \$2,500 to grow to \$4,000 at 6% annual compound interest.
Rounded to two decimals, the time is 8.06 years.

Application activity 1.19



An account grows from \$5,000 to \$7,200 in 4 years. Determine the annual compound interest rate by going through the following steps:

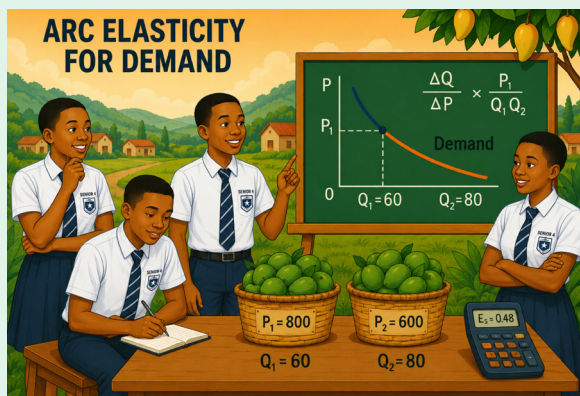
- Starting with $A = P(1+r)^n$ substitute the values into the formula.
- Rearrange the equation to isolate $(1+r)$
- Compute the value of $1+r$, then find r
- Explain whether the rate you found is realistic for common savings or investment accounts.

1.20 Arc of elasticity for demand

Activity 1.20



Using any resource material, explain in your own words what is meant by arc elasticity for demand.



Illustrate your answer using a fully explained formula.

Content Summary

Price elasticity of demand is a measure of the responsiveness of demand to changes in price.

It is usually defined as $e = (-1) \frac{\% \text{ change in quantity demand}}{\% \text{ change in price}}$

The (-1) in this definition ensures a positive value for elasticity as either the change in price or the change in quantity will be negative. When there are relatively large changes in price and quantity it is best to use the concept of ‘arc elasticity’ to measure elasticity along a section of a demand schedule. This takes the changes in quantity and price as percentages of the averages of their values before and after the change. Thus, arc elasticity is usually defined as

$$\text{Arc } e = (-1) \frac{\frac{\text{change in quantity}}{0.5 \times (1^{\text{st}} \text{ quantity} + 2^{\text{nd}} \text{ quantity})} \times 100}{\frac{\text{change in price}}{0.5 \times (1^{\text{st}} \text{ price} + 2^{\text{nd}} \text{ price})} \times 100}$$

Although a positive price change usually corresponds to a negative quantity change, and vice versa, it is easier to treat the changes in both price and quantity as positive quantities. This allows the (-1) to be dropped from the formula. The 0.5 and the 100 will always cancel top and bottom in arc elasticity calculations.

$$\text{Thus, we are left with arc } e = (-1) \frac{\frac{\text{change in quantity}}{0.5 \times (1^{\text{st}} \text{ quantity} + 2^{\text{nd}} \text{ quantity})}}{\frac{\text{change in price}}{0.5 \times (1^{\text{st}} \text{ price} + 2^{\text{nd}} \text{ price})}}$$

as the formula used for calculating price arc elasticity of demand

Example 1:

Calculate the arc elasticity of demand between points A and B on the demand schedule shown in the following figure:



Solution

Between points A and B price falls by 5 from 20 to 15 and quantity rises by 20 from 40 to

$$60. \text{ Using the formula defined above: } \text{arc } e = \frac{\frac{20}{40+60}}{\frac{5}{20+15}} = \frac{7}{5}$$

Example 2:

A local transport cooperative manages a bus route connecting a rural trading center to a nearby city. To cover rising fuel costs, the cooperative decides to adjust the one-way ticket price.

Initial State (Point A): At a price (P_1) of 1,000 FRW, the daily number of passengers (Q_1) is 400.

New State (Point B): When the price (P_2) is increased to 1,200 FRW, the number of passengers (Q_2) drops to 300.

Solution

The arc elasticity of demand measures the responsiveness of the quantity demanded of a good to a change in its price over a specific, distinct segment (or “arc”) of a demand curve.

Instead of measuring elasticity at a single specific point, arc elasticity calculates the average elasticity between two separate points.

The mathematical formula for the coefficient of price elasticity of demand (E_d) is:

$$E_d = \frac{\frac{Q_2 - Q_1}{(Q_1 + Q_2) / 2}}{\frac{P_2 - P_1}{(P_1 + P_2) / 2}}$$

By simplifying the denominators, the operational formula becomes: $E_d = \frac{Q_2 - Q_1}{Q_1 + Q_2} \times \frac{P_1 + P_2}{P_2 - P_1}$

Where:

- P_1 and Q_1 are the initial price and quantity.
- P_2 and Q_2 are the new price and quantity.

Calculate the change and average for Quantity (Q):

- Change in Quantity ($Q_2 - Q_1$) = 300 - 400 = -100
- Sum of Quantities ($Q_1 + Q_2$) = 400 + 300 = 700.

Calculate the change and average for Price (P):

- Change in Price ($P_2 - P_1$) = 1,200 - 1,000 = 200
- Sum of Prices ($P_1 + P_2$) = 1,000 + 1,200 = 2,200

Substitute the values into the simplified formula:

$$E_d = \frac{-100}{700} \times \frac{2,200}{200}$$

$$E_d = -\frac{1}{7} \times 11 = -\frac{11}{7} \approx -1.57$$

Note: In economics, we typically ignore the negative sign and look at the absolute value, so $|E_d| = 1.57$.

Interpreting the Result

Since the absolute coefficient (1.57) is greater than 1, the demand for this bus route over this price range is elastic.

Cross-Cutting Reflection (Financial Education)

This calculation provides a vital lesson in financial literacy and revenue management for the transport cooperative:

Because demand is elastic, the percentage drop in passengers (28.6%) is larger than the percentage increase in ticket price (18.2%).

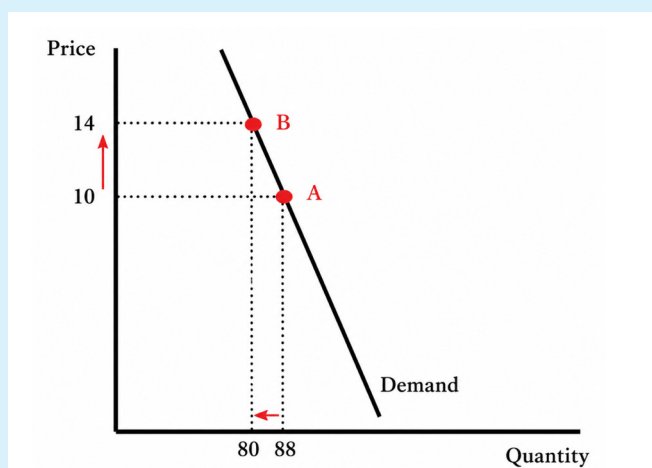
Initial Revenue: $1,000 \text{ RWF} \times 400 = 400,000 \text{ RWF}$

New Revenue: $1,200 \text{ RWF} \times 300 = 360,000 \text{ RWF}$

Application activity 1.20



1. There are 100 people who need to get home on Christmas day and the average cost per ride is 2000 Frw. At this rate, all people are willing to pay that cost to get home. Transport agencies raise the average cost to 3000 per ride (a price increase of 50%). However, most people don't live too far from home, so 75 of them decide to walk. This leaves only 25 people who take buses (a 75% decrease in demand). Calculate price elasticity of demand. Interpret your result.
2. Once again, there are 100 people who need to get home on Christmas day. The average cost per ride is 2000 Frw. At this rate, all people are willing to pay this cost. Assume that transport agencies raise the average cost to 3000 Frw per ride (a price increase of 50%). 25 of them end up calling friends or family instead, leaving only 75 people to join transport agencies (a decrease in demand of 25%). Calculate price elasticity of demand, and interpret your result.
3. Calculate arc elasticity of demand between points A and B on the demand schedule shown in the following figure:



End of unit assessment



Assessing knowledge:

1. Explain the meaning of absolute value and describe one real-life situation where it is used.
2. State the difference between simple interest and compound interest in your own words.
3. Given the expression 3^{-2} , explain what the negative exponent means and rewrite the expression in positive-index form.
4. Define a radical and give two examples of expressions that contain radicals.
5. Explain why percentages are useful when comparing quantities in different contexts such as class performance or price discounts.

Assessing skills:

1. Simplify the expression: $\sqrt{75} - \sqrt{12}$ by rationalizing and using properties of radicals.
2. Solve the logarithmic equation: $\log_{10}(x) + \log_{10}(5) = 2$.
3. The price increases from RWF 12,000 to RWF 15,000. Calculate the percentage increase.
4. A student invests RWF 50,000 at 6% compound interest per year. Calculate the value after 2 years.
5. Given the ratio 4:7, divide RWF 33,000 into two parts in that ratio.

Assessing attitudes and values:

1. Discuss how learning to compare prices using percentages and ratios helps someone make wise and fair consumer decisions.
2. Give an example of a situation where showing patience and persistence is necessary when solving complex problems involving logarithms or radicals.
3. Explain how understanding investment growth encourages young people to develop good saving habits and avoid irresponsible spending.

UNIT 2

EQUATIONS AND INEQUALITIES



Key Unit Competence:

Use linear and quadratic equations and inequalities in solving related problems from daily life situations.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Identify a linear equation, a quadratic equation, and an inequality.
- Describe the concept of an interval and its properties.
- Explain the meaning of the discriminant and its role in determining the nature of the roots of a quadratic equation.
- Identify the supply and demand equations in a market model.
- State the different methods for solving simultaneous equations.

Skills

2

- Solve linear equations and inequalities in one unknown.
- Represent solutions of inequalities on a number line using interval notation.
- Solve systems of two linear equations algebraically (substitution, elimination) and graphically.
- Solve quadratic equations by factorization and by using the quadratic formula/discriminant.
- Analyze and solve inequalities involving products and quotients.
- Apply linear and quadratic models to solve real-life problems.

Attitudes and values

3

- Value the accuracy and logical process required to solve equations.
- Appreciate the power of algebraic methods in making informed decisions in economics and daily life.
- Demonstrate critical thinking when interpreting the feasibility of solutions in real-world contexts (e.g., a negative price is not feasible).

Introduction

Equations and inequalities form a fundamental part of mathematics because they help us describe and solve real-life situations using symbols and logical reasoning. In this unit, you will explore different types of equations, such as linear and quadratic equations, and learn various methods of solving them, including substitution, elimination, factorization, and the use of the discriminant. These skills will enable you to express mathematical relationships, analyze graphs, and justify solutions with confidence. You will also work with intervals, number lines, and algebraic techniques to understand how quantities can vary within specific boundaries.

Beyond classroom exercises, equations and inequalities appear naturally in everyday decision-making and in many fields relevant to society. You will examine real-life problems involving supply and demand, and comparisons between two or more quantities. Understanding these relationships helps you develop critical-thinking skills and prepares you to interpret information, make predictions, and analyze situations logically. By the end of this unit, you will see how mathematical expressions can model real situations, guide reasoning, and support informed choices in daily life.

Introductory activity



1. Using any available resource, do the research and explain what is meant by linear equation.
2. If x is the number of pens for a learner, the teacher decides to give him/her two or more pens. How many pens will a learner with one pen have?
 - a) Complete the following table called table of value to indicate the number $y = f(x) = x + 2$ of pens for a learner who had x pens for $x \geq 0$.

X	-2	-1	0	1	2	3	4
$y = f(x) = x + 2$			2				
(x, y)			(0, 2)				

- b) Use the coordinates of points obtained in the table and complete them in the Cartesian plan.
 - c) Join all points obtained. What is the form of the graph obtained?
 - d) Suppose that instead of writing $f(x) = x + 2$ you write the equation $y = x + 2$. Is this equation a linear equation or a quadratic equation? What is the type of the inequality " $x + 2 \geq 0$ "?
3. Find out an example of problem from the real-life situation that can be solved using linear equation in one unknown.

2.1 Linear equations in one unknown

Activity 2.1



1. Assume that in a competitive market, the supply schedule is $p = 60 + 0.4q$ where p is a price function and q is the quantity supplied. Is the price increasing or decreasing?
2. a) Find the price for $q = 600$ units.
b) What is the value of P for $q = 0$. What does it mean for an industry which is producing a certain good and has just fixed a price P_0 at the beginning ($q = 0$). Can you justify this price P_0 .
3. Solve in the set of real numbers the following equations: a) $3 - x = 3$ b) $x = 16 - x$

Content Summary

An equation is a mathematical statement expressing the equality of two quantities or expressions. Equations are used in every field that uses real numbers. A linear equation is an equation of a straight line.

Consider the statement $x - 1 = 0$, this statement is equation as there are two quantities to be equal and is true for the value $x = 1$. The value $x = 1$ is called the solution of the statement $x - 1 = 0$ the number 1 is called the root of the equation. Thus, to find a solution to the given equation is to find the value that satisfies that equation.

To do this, rearrange the given equation such that the variables will be on the same side and constant in the other side and then find the value of the variable.

Example:

Solve in set of real numbers the following equations

a) $x + 2 = 10$

b) $3 + x = 18 - 2x$

c) $x = 36 - 3x$

Solution

a)
 $x + 2 = 10$
 $x = 10 - 2$
 $x = 8$
 $S = \{8\}$

b)
 $3 + x = 18 - 2x$
 $x + 2x = 18 - 3$
 $3x = 15$
 $x = \frac{15}{3}$
 $x = 5$
 $S = \{5\}$

c)
 $x = 36 - 3x$
 $x + 3x = 36$
 $4x = 36$
 $x = \frac{36}{4}$
 $x = 9$
 $S = \{9\}$

Application activity 2.1



Solve in the set of real numbers:

a) $4x + 5 = 20 + x$

b) $x - 31 = 50 - 8x$

c) $3 + x = 9 - 2x$

d) $x + 2 \geq 0$

e) $\frac{2x + 5}{x - 6} = 4$

2.2 Real life problems involving linear equations

Activity 2.2



A school expects to enrol students into the next academic year. The school leaders expect four more than twice the number of girls to enrol into the school than boys. The school buildings have the capacity of enrolling 316 students. How many of each gender would the school have to enrol to meet this expectation?

Content Summary

Problems which are expressed in words are known as word problems or applied problems. A word or applied problem involving unknown number or quantity can be translated into linear equation consisting of one unknown number or quantity. The equation is formed by using conditions of the problem. By solving the resulting equation, the unknown quantity can be found.

In solving problem by using linear equation in one unknown the following steps can be used:

- Read the statement of the word problems
- Represent the unknown quantity by a variable
- Use conditions given in the problem to form an equation in the unknown variable
- Verify if the value of the unknown variable satisfies the conditions of the problem.

Example 1:

The sum of two numbers is 80. The greater number exceeds the smaller number by twice the smaller number. Find the numbers.

Solution

Let x be the smaller number

Therefore, the greater number be $80 - x$

According to the problem,

$$(80 - x) - x = 2x$$

$$80 - x - x = 2x$$

$$80 - 2x = 2x$$

$$4x = 80$$

$$x = \frac{80}{4}$$

$$x = 20$$

Now substitute the value of $x = 20$ in $80 -$ we get $80 - 20 = 60$

Therefore, the smaller number is 20 and the greater number is 60.

Example 2:

A boat covers a certain distance downstream in 2 hours and it covers the same distance upstream in 3 hours. If the speed of the stream is 2 km/hr . Find the speed of the boat.

Solution

Let the speed of the boat be $x \text{ km/hr}$

Speed of the stream = 2 km/hr

Speed of the boat downstream = $(x + 2) \text{ km/hr}$

Speed of the boat upstream = $(x - 2) \text{ km/hr}$

Distance covered in both the cases is same.

$$2(x + 2) = 3(x - 2)$$

$$2x + 4 = 3x - 6$$

$$2x - 3x = -6 - 4$$

$$-x = -10$$

$$x = 10$$

Therefore, the speed of the boat is 10 km/hr .

Product equation

The equation in the form $(ax + b)(cx + d) = 0$ is product equation since the product of factors is null (zero) either one of them is zero. To solve this, we proceed as follows:

$$(ax + b)(cx + d) = 0$$

$$ax + b = 0 \text{ or } cx + d = 0$$

$$x = \frac{-b}{a} \text{ or } x = \frac{-d}{c}$$

Example:

Solve the equation $(2x + 4)(x - 1) = 0$

Solution

$$2x + 4 = 0, \quad x - 1 = 0$$

$$x = -2, \quad x = 1$$

Fractional equation of the first degree

The general form $\frac{ax+b}{cx+d} = 0$ to find solution of this equation we need to have the condition of

existence $cx + d \neq 0$ and $\frac{ax+b}{cx+d} = 0 \Rightarrow ax + b = 0$

We solve $ax + b$ and we take the value(s) which verify the condition of existence

Example:

Solve $\frac{2x-6}{x+1} = 0$

Solution

The existence condition is $x + 1 \neq 0$, $x \neq -1$

$$2x - 6 = 0 \quad 2x = 6, \quad x = \frac{6}{2}, \quad x = 3$$

Application activity 2.2

1. Mr. Peter wants to fence his garden of rectangular form where the length of that rectangle is twice its breadth. If the perimeter is 72 meters, help Peter to find the length and breadth of the rectangle.
2. The sum of two numbers is 25. One of the numbers exceeds the other by 9. Explain how you can determine these numbers.
3. Find the number whose one fifth is less than the one fourth by 3.

2.3 Meaning of inequalities**Activity 2.3**

1. Find the value s of x such that the following statements are true
 - a) $a)x < 5$
 - b) $x > 0$
 - c) $-4 < x < 12$
2. Using any available resource materials, do the following:
 - a) Discuss the difference between linear equations and linear inequalities

- b) Find out examples of linear inequalities and try to solve them.
- c) Provide examples on how linear inequalities are applied when solving real word problems.

Content Summary

Just as we use the symbol $=$ to represent “is **equal to**”, we also use the symbols $<$ and $>$ to represent “is less than” and “is greater than”, respectively.

If a and b are real numbers, then various statements of inequality can be made:

- $a \neq b$ says that a is not equal to b
- $a < b$ says that a is less than b
- $a > b$ says that a is greater than b
- $a \leq b$ means that a is less than or equal to b
- $a \geq b$ means that a is greater than or equal to b

The table below shows the inequality symbols:

Symbol	Meaning
$\neq$	Not equal
$<$	Less than
$\leq$	Less than or equal to
$>$	Greater than
$\geq$	Greater than or equal to

Inequalities are used to compare two values and determine the range or ranges of values that satisfy the conditions of a given variable.

Note that:

- When the same real number is added or subtracted from each side of inequality the direction of inequality is not changed.

It means that for all real numbers a , b and c ,

1. $a > b$ if and only if $a + c > b + c$

2. $a > b$ if and only if $a - c > b - c$

- The direction of the inequality is not changed if both sides are multiplied or divided by the same positive real number.

It means that for all real numbers a , b and c , with $c > 0$

$$a > b \text{ if and only if } ac > bc$$

- The direction of the inequality is reversed if both sides are multiplied or divided by the same negative real number.

It means that for all real numbers a , b and c , with $c < 0$

$a > b$ if and only if $ac < bc$

Example:

The statement $x + 3 = 10$ is true only when $x = 7$. If x is replaced by 5, we have a statement $5 + 3 = 10$ which is false. To be true we may say that $5 + 3$ is less than 10 or in symbols $5 + 3 < 10$. If x is replaced by 8, the statement $8 + 3 = 10$ is also false. In those two cases we no longer have equality but inequality.

Suppose that we have the inequality $x + 3 < 10$, in this case we have an inequality with one unknown. Here the real value of x satisfies this inequality is not unique. For example, 1 is a solution but 3 is also a solution. In general, all real numbers less than 7 are solutions. In this case we will have many solutions combined in an interval.

Now, the solution set of $x + 3 < 10$ is an open interval containing all real numbers less than 7 whereby 7 is excluded. How?

We solve this inequality as follow

$$x + 3 < 10$$

$$\Leftrightarrow x < 10 - 3$$

$$\Leftrightarrow x < 7$$

And then $S =]-\infty, 7[$

Application activity 2.3



Solve the following linear inequalities in one variable:

a) $7x + 3 < 5x + 9$

b) $3x - 5 \leq 3 - x$

c) $4x + 5 < 6x + 9$

2.4 Concept and properties of intervals

Activity 2.4



Using any available resources, do the following:

Suppose a and b be two real numbers such that $a < b$, how to denote

- a) the open interval from a to b ?
- b) the closed interval from a to b ?
- c) The half-open interval?

Content Summary

A subset of real line is called an interval if it contains at least two numbers and also contains all real numbers between any two of its elements. For example, the set of real numbers x such that $x > 6$ is an interval, but the set of real numbers y such that $y \neq 0$ is not an interval.

If a and b are real numbers and $a < b$, we often refer to:

- The open interval from a to b , denoted by (a, b) or $]a, b[$, consisting of all real numbers x satisfying $a < x < b$
- The closed interval from a to b , denoted by $[a, b]$, consisting of all real numbers x satisfying $a \leq x \leq b$
- The half-open interval $[a, b[$, consisting of all real numbers x satisfying the inequalities $a \leq x < b$
- Half-open interval $]a, b]$, consisting of all real numbers x satisfying the inequalities $a < x \leq b$

Example:

Solve the following inequalities and express the solution in terms of intervals.

a) $2x - 1 > x + 3$ b) $-\frac{x}{3} \geq 2x - 1$ c) $2(x + 5) > 2x - 8$ d) $2x + 5 \leq 2x + 4$

Solution

a)

$$3x - 1 > 4x + 3$$

$$3x - 4x > 3 + 1$$

$$-x > 4$$

$$x < -4$$

The solution set is the interval $] -\infty, -4[$

c)

$$3(x + 5) > 2x + 3$$

$$3x + 15 > 2x + 3$$

$$3x - 2x > 3 - 15$$

$$x > -12$$

The solution set is the interval $] -12, +\infty[$

b)

$$\frac{-2x}{3} \geq 6x - 3$$

$$-2x \geq 3(-6x + 3)$$

$$-2x \geq -18x + 9$$

$$-2x + 18x \geq 9$$

$$16x \geq 9$$

$$x \geq \frac{9}{16}$$

The solution set is the interval $\left[\frac{9}{16}, +\infty \right[$

d)

$$2x + 5 \leq 2x + 4$$

$$2x - 2x \leq 4 - 5$$

$$0x \leq -1$$

Since any real number times zero is zero and zero is not less or equal to -1 then the solution set is the empty set; $S = \emptyset$.

Application activity 2.4



1. Complete the following table by converting the given inequalities into interval notation.

Inequality	Interval notation
$x \leq 3$	
$x < 5$	
$x \geq 2$	

2. Solve the following inequalities and express the solution in terms of intervals.

a) $2x - 1 > x + 3$

b) $-\frac{x}{3} \geq 2x - 1$

c) $2(x + 5) > 2x - 8$

d) $2x + 5 \leq 2x + 4$

2.5 Problems involving intervals

Activity 2.5



1. In a certain country, for a person to be elected as president, he/she should be a minimum of 35 years old. Give an interval representing this information.
2. For a school to participate in an Olympiad exam, the number of students from the school be a minimum of 10. Represent this information using an interval notation.

Content Summary

Introducing intervals, which are bounded sets of numbers and are very useful when describing domain and range. We can use interval notation to show that a value falls between two endpoints. For example, the learners need to score above 65 marks and up to 100 marks. The interval for this information will be written as $65 < x \leq 100$ or $[65, 100]$. The intervals can also be used to determine the amount of time between two given points in time. For example, "The time interval between three o'clock and four o'clock is one hour."

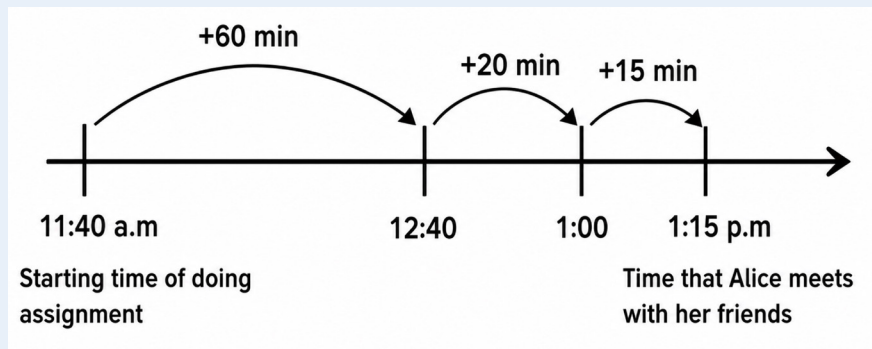
Example 1:

Alice wants to meet her friends in downtown. Before leaving home, she does assignment for 60 minutes and eats lunch for 20 minutes. The walk downtown takes 15 minutes. Alice starts her assignment at 10:45 a.m. At what time does she meet her friends?

Solution

We need to find what time Alice meet her friends in downtown.

The information given is the time Alice starts her assignment before leaving home, the time used for eating lunch, the time it takes to do assignment, and the time it takes to walk downtown. We can start with the time Alice starts her assignment before leaving home and draw a number line to find the time Alice meets her friends.



From the number line we see that Alice meets with her friends at 1:15 p.m.

Example 2:

Find possible values for x if $|x - 4| < 10$.

Solution

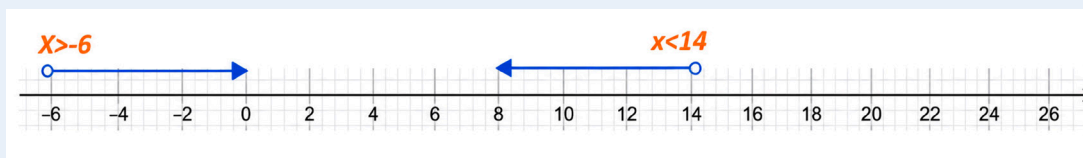
We noted that $|x - 4| = x - 4$ for $x - 4 > 0$ or $|x - 4| = -(x - 4)$ for $x - 4 < 0$

Therefore, $|x - 4| < 10$ implies that $-10 < x - 4 < 10$

- $-10 < x - 4 \Rightarrow -10 + 4 < x - 4 + 4 \Rightarrow -6 < x$
- $x - 4 < 10 \Rightarrow x - 4 + 4 < 10 + 4 \Rightarrow x < 14$

$x > -6$ and $x < 14$ means that we consider all numbers x between -6 and 14 .

$$x \in [-6, 14]$$



Application activity 2.5



1. Peter needs at least 1300 calories a day, but the calorie intake should not exceed 1700 calories. Represent the possible number of calories she could eat using interval notation.
2. A football coach wants players to train for at least 2 hours but no more than 4 hours per day.
 - a) What is the minimum training time?
 - b) What is the maximum training time?
 - c) Let t be training time. How can you represent this situation mathematically?
 - d) Write the inequality.
 - e) Express the solution in interval notation.
3. Karake gets out of school at 2:45 P.M. It takes him 10 minutes to walk home. Then he spends 10 minutes eating a snack. He spends 8 minutes putting on his football uniform. It takes 20 minutes for Karake's father to drive him to football ground for practice.
 - a) By using number line, determine the time for which Karake will arrive at football ground.
 - b) How do you know that the answer is reasonable?

2.6 Inequalities involving products

Activity 2.6



Consider each of the following inequalities:

1. $(x+1)(x-1) < 0$
2. $(x-3)(-2x+4)(x+1) \geq 0$
 - a) Find the zeros of each factor (critical points)
 - b) Place them on a number line
 - c) Split the number line into intervals
 - d) Test the sign (+ or -) of the expression in each interval
 - e) Choose intervals that satisfy the inequality and hence write the resulting solution.

Content Summary

Suppose that we need to solve the inequality of the form $(ax+b)(cx+d) < 0$. For this inequality we need the set of all real numbers that make the left hand side to be negative. Suppose also that we need to solve the inequality of the form $(ax+b)(cx+d) \geq 0$. For this inequality we need the set of all real numbers that make the left hand side to be positive.

We follow the following steps:

- First, we solve for $(ax + b)(cx + d) = 0$
- We construct the table called sign table, find the sign of each factor and then the sign of the product.
- Write the interval considering the given inequality sign.

Example:

Solve in set of real numbers the following inequalities

a) $(x + 2)(x - 5) < 0$

b) $(-2x + 2)(x - 5)(3x - 9) \geq 0$

Solution

a) $(x + 2)(x - 5) < 0$

Start by solving $(x + 2)(x - 5) = 0$

$$x + 2 = 0 \text{ or } x - 5 = 0$$

$$x = -2 \text{ or } x = 5$$

Then, we find the sign table.

x	$-\infty$	-2	5	$+\infty$
$x + 2$	-	0	+	+
$x - 5$	-	-	- 0 +	+
$(x + 2)(x - 5)$	+	0	- 0 +	+

Since the inequality is $(x + 2)(x - 5) < 0$ we will take the interval where the product is negative. Thus, $S =]-2, 5[$

b) $(-2x + 2)(x - 5)(3x - 9) \geq 0$ neighbourhood

$$-2x + 2 = 0 \Rightarrow x = 1, x - 5 = 0 \Rightarrow x = 5, 3x - 9 = 0 \Rightarrow x = 3$$

Then, we find the sign table.

x	$-\infty$	2	3	5	$+\infty$
$-2x + 4$	+	0	-	-	-
$x - 5$	-	-	0	+	+
$3x - 9$	-	-	0	+	+
$(-2x + 4)(x - 5)(3x - 9)$	+	0	- 0 +	0 - -	-

Since the inequality is $(-2x + 5)(x - 5)(3x - 9) \geq 0$ we will take the interval where the product is positive. Thus, $S =]-\infty, 2] \cup [3, 5]$

Application activity 2.6



Solve in set of real numbers the following inequalities

1. $(3x + 7)(x - 2) < 0$

2. $(2x + 7)(3x - 2) \geq 0$

2.7 Inequalities involving quotients

Activity 2.7



Consider each of the following inequalities:

1. $\frac{2x-3}{x} > 0$

2) $\frac{x+2}{x-1} \leq 0$

- determine all values of x that make each numerator and denominator equal to zero and explain why these values are important.
- represent these values on a number line and identify which ones must be excluded from the solution, giving reasons.
- use these points to divide the number line into intervals and state how many intervals are formed in each case.
- select a test value from each interval and determine whether the expression is positive or negative, organizing your results clearly.
- based on the inequality signs (< 0 and ≤ 0), identify the intervals that satisfy each inequality, decide whether the endpoints should be included or excluded;
- Write your final answers in interval notation, explaining your reasoning.

Content Summary

Suppose that we need to solve the inequality of the form $\frac{ax+b}{cx+d} < 0$. For this inequality we need the set of all real numbers that make the left hand side to be negative. Suppose also that we need to solve the inequality of the form $\frac{ax+b}{cx+d} > 0$. For this inequality we need the set of all real numbers that make the left hand side to be positive.

We follow the following summarized steps:

- The first thing we need to do is get a zero on one side of the given inequality. That means we solve for $ax + b = 0$ and $cx + d = 0$
- We construct the table called **sign table**. Find the sign of each factor and then the sign of the quotient.

For the quotient the value that makes the denominator to be zero is always excluded in the solution. For that value we use the symbol $||$ in the row of quotient sign.

- Write the interval considering the given inequality sign.

Example:

1) $\frac{4-x}{x+3} \geq 0$

2) $\frac{2x+3}{4-x} < 0$

Solution

1) $\frac{4-x}{x+3} \geq 0$

$4-x=0 \Rightarrow x=4$

$x+3=0 \Rightarrow x=-3$

Then, we find the sign table

x	$-\infty$	-3	4	$+\infty$		
$4-x$		+	+	0	-	-
$x+3$		-	0	+	+	+
$\frac{x-4}{x+3}$		-		+	0	- -

$S =]-3, 4[$

2) $\frac{2x+3}{4-x} < 0$

$2x+3=0 \Rightarrow x = -\frac{2}{3}$

$4-x=0 \Rightarrow x=4$

Then, we find the sign table.

x	$-\infty$	$-\frac{3}{2}$	4	$+\infty$	
$2x+3$	-	0	+	+	
$4-x$	+	+	0	-	
$\frac{2x+3}{4-x}$	-	0	+		-

$S =]-\infty, -\frac{2}{3}[\cup]4, +\infty[$

Application activity 2.7



Solve in set of real numbers the following inequalities

$$1) \frac{x+4}{2x-1} \geq 0 \quad 2) \frac{x-2}{x+3} < 0 \quad 3) \frac{x+2}{x+4} \leq 0$$

2.8 Real life problems involving linear inequalities

Activity 2.8



Sam and Alex play in the same team at their school. Last Saturday their team played with another team from other school in the same district, Alex scored 3 more goals than Sam. But together they scored less than 9 goals. What is the set of the possible number of goals Alex scored?

Content Summary

Inequalities can be used to model several real-life situations. When converting such word problems into inequalities, begin by identifying how the quantities are related to each other, and then pick the inequality symbol that is appropriate for that situation. When solving these problems, the solution will be a range of possibilities. Absolute value inequalities can be used to model situations where margin of error is a concern.

Examples

1. The width of a rectangle is 20 meters. What must the length be if the perimeter is at least 180 meters?

Solution

Let x be length of rectangle

$$\text{perimeter} = 2\text{length} + 2\text{width}$$

$$2x + 2(20) \geq 180 \rightarrow 2x \geq 180 - 40 \rightarrow x \geq 70$$

The length must be at least 70 meters.

2. John has 1 260 000 Rwandan Francs in an account with his bank. If he deposits 30 000 Rwanda Francs each week into the account, how many weeks will he need to have more than 1 820 000 Rwandan Francs on his account?

Solution

Let x be the number of weeks

We have total amount of deposits to be made plus the current balance is greater than the total amount wanted.

That is $30000x + 1260000 > 1820000$

$$30000x > 1820000 - 1260000$$

$$30000x > 560000 \text{ implies that } x > \frac{560000}{30000} \approx 19$$

Thus, John needs at least 19 weeks to have more than 1 820 000 Rwandan Francs on his account.

Example:

The width of a rectangle is 20 meters. What must the length be if the perimeter is at least 180 meters?

Solution

$$P = 2(L + W), 2(L + 20) \geq 180, L + 20 \geq 90, L \geq 70$$

The length must be at least 70 meter.

Application activity 2.8



1. John enters a race where he must cycle and run. He cycles a distance of 25 km, and then runs for 20 km. His average running speed is half of his average cycling speed. Joe completes the race in less than $2\frac{1}{2}$ hours, what can we say about his average speeds?
2. Explain to your colleague whether or not a solution set for an inequality can have one element.

2.9 Solving algebraically simultaneous linear equations in two unknowns (Solving by equating two same variables)

Activity 2.9



In each of the following systems find the value of one variable from one equation and equate it with the same value of another variable from second equation. Calculate the values of those variables.

$$1. \begin{cases} 2x - 2y = -6 \\ x + y = -2 \end{cases} \quad 2. \begin{cases} x + 2y = 10 \\ -3x + 2y = 12 \end{cases} \quad 3. \begin{cases} x + 4y = 8 \\ y - x = 2 \end{cases}$$

Content Summary

To find the value of unknown from simultaneous equation by equating the same variable in terms of another, we do the following steps:

- a) Find out the value of one variable in first equation
- b) Find out the value of that variable in the second equation
- c) Equate the same values obtained
- d) Solve the equation obtained to find out the unknown variables.

Example 1:

Algebraically, solve the simultaneous linear equation by equating the same variables.

$$\begin{cases} 4x + 5y = 2 \\ x + 2y = -1 \end{cases}$$

Solution

$$\begin{cases} 4x + 5y = 2 \\ x + 2y = -1 \end{cases}$$

From equation (1) $4x + 5y = 2 \Rightarrow x = \frac{2-5y}{4}$, from equation (2) $x + 2y = -1 \Rightarrow x = -1 - 2y$

Equate the values of x from equations (1) and (2)

$$\frac{2-5y}{4} = -1 - 2y$$

$$2 - 5y = 4(-1 - 2y)$$

$$2 - 5y = -4 - 8y$$

$$-5y + 8y = -4 - 2$$

$$y = -2$$

$$x = \frac{2-5y}{4}, x = \frac{2-5(-2)}{4} = 3 \text{ then, } S = \{(3, -2)\}$$

Example 2:

Solve algebraically the following system by equating two same variables

$$\begin{cases} x + y = 1 \\ 2x + 3y = 2 \end{cases}$$

Solution

From equation $x + y = 1 \Rightarrow y = 1 - x$

From equation $2x + 3y = 2 \Rightarrow y = \frac{2-2x}{3}$

By equating those values of y , $1 - x = \frac{2-2x}{3}$

$$3(1 - x) = 2 - 2x$$

$$3 - 3x = 2 - 2x$$

$$-3x + 2x = 2 - 3$$

$$x = 1, y = 1 - x \Rightarrow y = 1 - 1, y = 0$$

$$S = \{(1, 0)\}$$

Example 3:

Solve the following simultaneous equation

$$\begin{cases} x + 4y = 8 \\ y - x = 2 \end{cases}$$

Solution

From equation (1), $x + 4y = 8 \Rightarrow x = 8 - 4y$

From equation (2), $y - x = 2 \Rightarrow x = -2 + y$

By equating values from (1) and (2)

$$8 - 4y = -2 + y$$

$$-4y - y = -2 - 8$$

$$-5y = -10$$

$$y = 2, x = -2 + 2 = 0 \text{ then, } S = \{(0, 2)\}$$

Application activity 2.9



Solve the simultaneous linear equations by equating two same variables

1. $\begin{cases} x - y = 3 \\ 2x - 2y = 6 \end{cases}$

2. $\begin{cases} x + y = 2 \\ 2x + 2y = 4 \end{cases}$

3. $\begin{cases} 3x - 5y = 10 \\ 2x + y = 12 \end{cases}$

4. $\begin{cases} x - 4y = 1 \\ x - y = 2 \end{cases}$

5. $\begin{cases} -x + y = 0 \\ x + 2y = 3 \end{cases}$

6. $\begin{cases} 5y + 3x = 2 \\ 10x + 6y = 0 \end{cases}$

2.10 Solving algebraically simultaneous linear equations in two unknowns (solving by row operations or elimination method)

Activity 2.10



For each of the following, find two numbers to be multiplied to the equations such that one variable will be eliminated when making the addition or subtraction of the two equations.

$$\begin{cases} -2x + 5y = -7 \\ 7x - 3y = -19 \end{cases}$$

Content Summary

To eliminate one of the variables from either of equations to obtain an equation in just one unknown, make one pair of coefficients of the same variable in both equations negatives of one another by multiplying both sides of an equation by the same number. Upon adding the equations, that unknown will be eliminated.

Example 1:

Solve the system of equations using elimination methods.

$$\begin{cases} x + y = 1 \\ 2x + 3y = 2 \end{cases}$$

Solution

$$\begin{cases} x + y = 1 & \times (-2) \\ 2x + 3y = 2 & \times (1) \end{cases} \Leftrightarrow \begin{cases} -2x - 2y = -2 \\ 2x + 3y = 2 \end{cases}$$

$$\underbrace{}_{y=0}$$

$$x + y = 1 \Leftrightarrow x = 1 - y = 1$$

$$S = \{(1, 0)\}$$

Example 2:

Solve the system of equation by using elimination method

$$\begin{cases} -2x + 4y = -7 \\ 7x - 3y = -19 \end{cases}$$

Solution

$$\begin{cases} -2x + 5y = -7 \\ 7x - 3y = -19 \end{cases} \Leftrightarrow \begin{cases} 7(-2x + 5y) = 7 \times (-7) \\ 2(7x - 3y) = 2 \times (-19) \end{cases}$$

$$\begin{cases} -14x + 35y = -49 \\ 14x - 6y = -38 \end{cases} \Leftrightarrow \begin{cases} -14x + 35y = -49 \\ -14x - 6y = -38 \end{cases}$$

$$\underbrace{}_{29y = -87}$$

From $29y = -87$, we have $y = -3$.

$$x = \frac{-7 - 5y}{-2} = \frac{-7 - 5(-3)}{-2} = -4$$

Therefore, $S = \{(-4, -3)\}$

Application activity 2.10

1. Solve the following system of equation by using elimination method

a) $\begin{cases} 3x - 4y = 1 \\ x - 3y = 2 \end{cases}$

b) $\begin{cases} x - 4y = 1 \\ x - y = 2 \end{cases}$

c) $\begin{cases} 3x - 2y = -6 \\ x + y = -2 \end{cases}$

d) $\begin{cases} 2x + 3y = 8 \\ x - y = 2 \end{cases}$

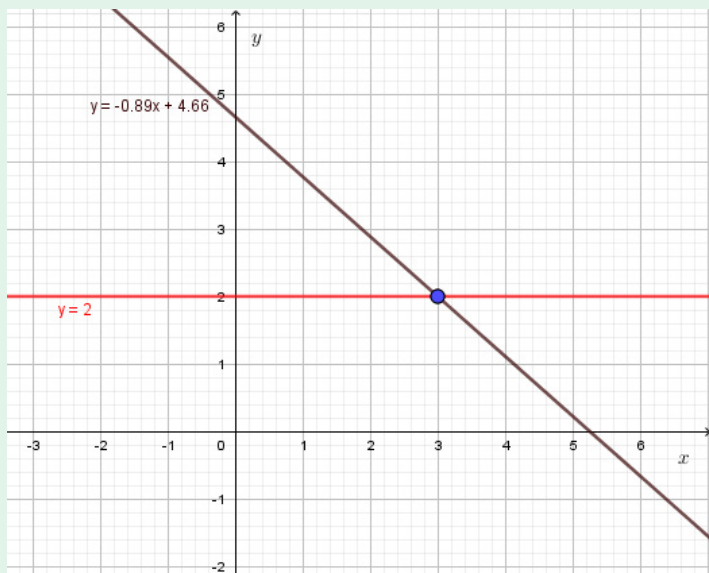
2. Use your own words to explain how to solve algebraically simultaneous linear equations.

2.11 Solving graphically simultaneous linear equations in two unknowns

Activity 2.11



1. Discuss how you can find the coordinate of the point intersection of two lines whose equations are known.



2. Given the system of equations

$$\begin{cases} 3x - 2y = -6 \\ x + y = -2 \end{cases}$$

- a) For each equation from the system, choose any two values of x and use them to find values of y .
- b) Plot the obtained points in XY plane and join these points to obtain the lines.
- c) What is the point of intersection for two lines?

Content Summary

One way to solve a system of linear equations is by graphing. The intersection of the graphs represents the point at which the equations have the same x -value and the same y -value. Thus, this ordered pair represents the solution common to both equations. This ordered pair is called the solution to the system of equations.

The following steps can be applied in solving system of linear equation graphically:

1. Find at least two points for each equation.
2. Plot the obtained points in XY plane and join these points to obtain the lines. Two points for each equation give one line.
3. The point of intersection for two lines is the solution for the given system

Example 1:

Solve the following system by graphical method

$$\begin{cases} x + y = 4 \\ x - y = 2 \end{cases}$$

Solution

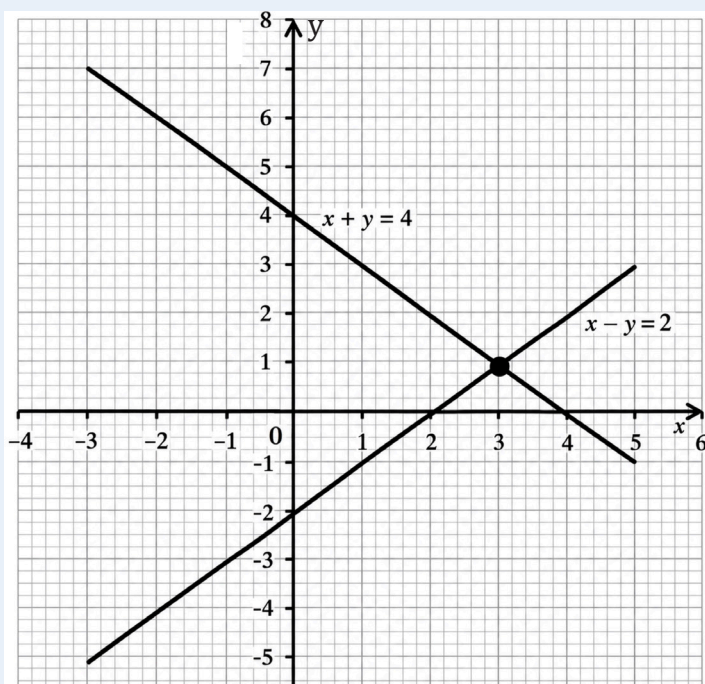
For $x + y = 4$

x	-3	5
y	7	-1

For $x - y = 2$

x	-3	5
y	-5	3

Graph



The two lines intersect at point $(3, 1)$. Therefore, the solution set is $S = \{(3, 1)\}$.

Example 2:

Solve graphically the following system of linear equations

Solve the following equations graphically

$$\begin{cases} x + y = 2 \\ 2y = 4 - 2x \end{cases}$$

Solution

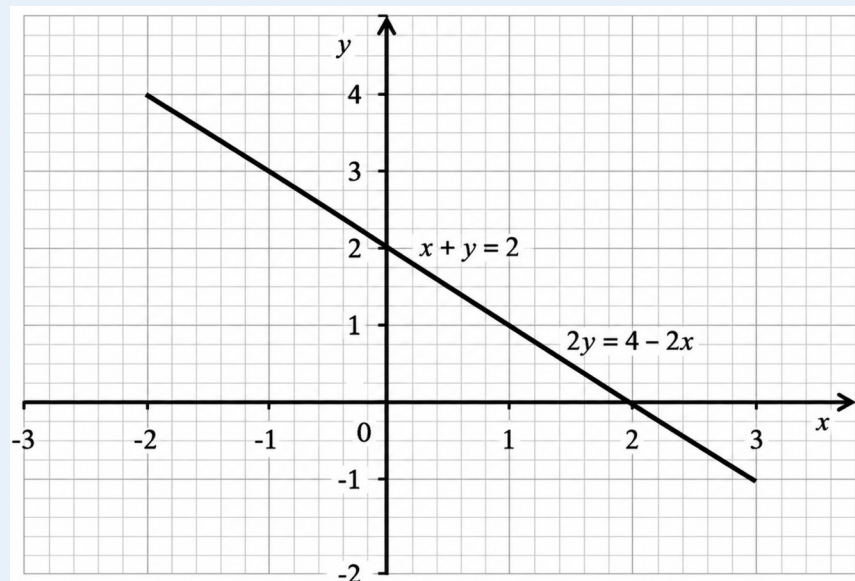
For $x + y = 2$

x	-2	3
y	4	-1

For, $2y = 4 - 2x$

x	-2	3
y	4	-1

Graph



We see that the two lines coincide as a single line. In such case there are infinite number of solutions.

Application activity 2.11



Solve the graphically the following systems of linear equations

1.
$$\begin{cases} 4y + x = 8 \\ -x + y = 2 \end{cases}$$

2.
$$\begin{cases} 3x + y = 1 \\ x - y = -1 \end{cases}$$

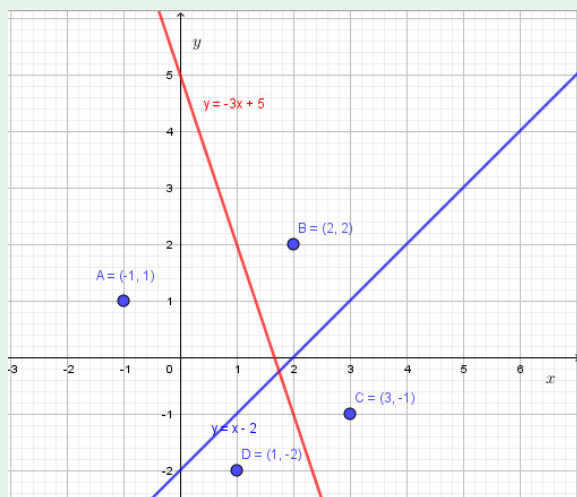
3.
$$\begin{cases} 2x + y = 3 \\ 6x + 3y = 9 \end{cases}$$

2.12 Solving algebraically and graphically simultaneous linear inequalities in two unknowns

Activity 2.12



The following graph illustrates two lines and their equations



For each point A, B, C, and D, replace its coordinate in the two inequalities to verify which one satisfies the following system:

$$\begin{cases} y < x - 2 \\ y > -3x + 5 \end{cases}$$

Content Summary

A system of inequalities consists of a set of two inequalities with the same variables. The inequalities define the conditions that are to be considered simultaneously. Each inequality in the set contains infinitely many ordered pair solutions defined by a region in rectangular coordinate plane. When considering two of these inequalities together, the intersection of these sets will define the set of simultaneous ordered pair solutions.

Linear inequalities with two unknowns are solved to find a range of values of the two unknowns which make the inequalities true at the same time. The solution is represented graphically by a region.

In finding solution, first, graph the “equals” line, then shade in the correct area.

The following steps are used to find the solution of simultaneous inequalities graphically:

1. Rearrange the equation so “ y ” is on the left and everything else on the right.
2. Plot the y line (make it a solid line for $y \leq$ or $y \geq$ and a dashed line for $y <$ or $y >$)
3. Shade above the line for a greater than $y >$ or $y \geq$ or below the line for a less than $y <$ or $y \leq$
4. The intersection will define the set of simultaneous ordered pair solutions.

Example:

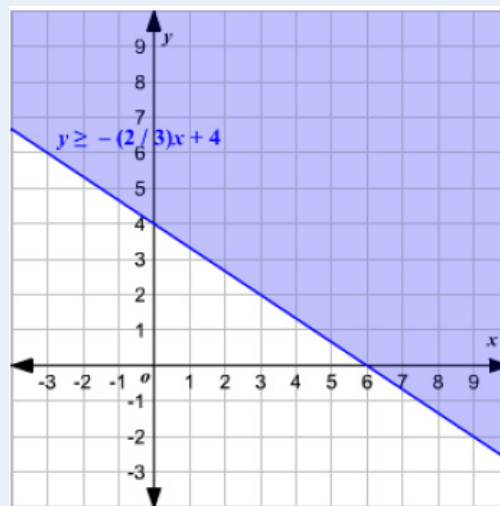
Solve the system of inequalities by graphing:
$$\begin{cases} 2x + 3y \geq 12 \\ 8x - 4y > 1 \\ x < 4 \end{cases}$$

Solution

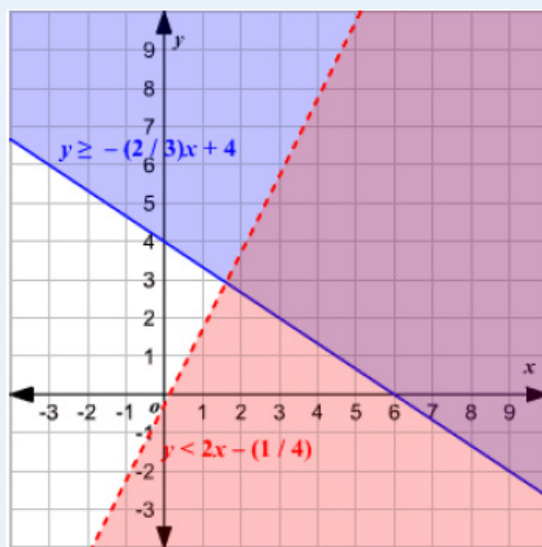
From Each inequality from the system we have $2x + 3y \geq 12$ which implies $y \geq -\frac{2}{3}x + 4$, the related equation to this is

$y = -\frac{2}{3}x + 4$ since the inequality is $\geq$, not a strict one, the border line is solid.

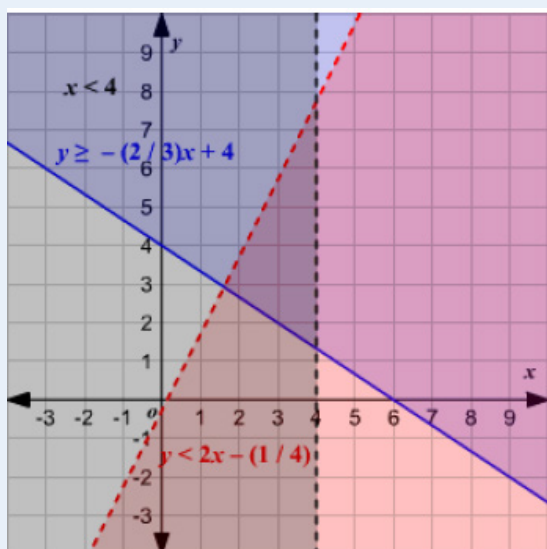
Graph the line $y = -\frac{2}{3}x + 4$



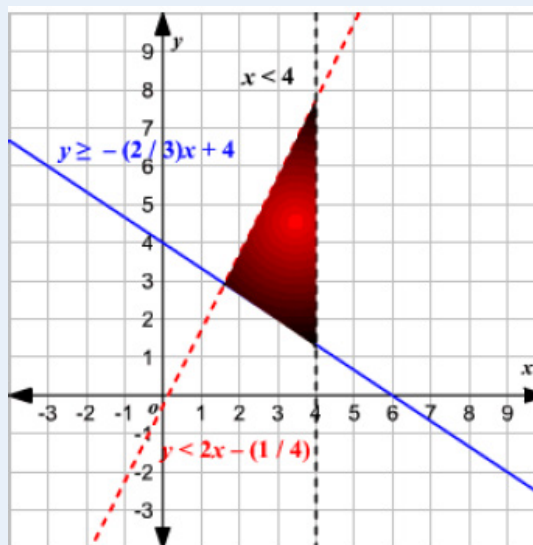
Similarly, draw a dashed line of related equation of the second inequality $y < 2x - \frac{1}{4}$ which has a strict inequality.



Draw the dashed vertical line $x = 4$ which is related to the equation of the third inequality.



The solution of the system of inequalities is the intersection region of the solutions of the three inequalities as it is done in the following figure.



Application activity 2.12



1. Algebraically and graphically, solve the following simultaneous inequalities

a)
$$\begin{cases} y - 2x \leq 1 \\ x + y \leq 10 \\ x \geq 0 \end{cases}$$

b)
$$\begin{cases} -3x + 2y > 6 \\ 6x - 4y > 8 \end{cases}$$

c)
$$\begin{cases} x \geq 0 \\ y \geq 0 \\ 2x + y \leq 4 \end{cases}$$

2. Use your own words to explain how to solve graphically the simultaneous inequalities.

2.13 Solving quadratic equations using factorization

Activity 2.13



1. The sum of the square of a number and 15 is the same as eight times the number.
 - a) Write the equation in the form $ax^2 + bx + c = 0$
 - b) Hence, factorize the equation and find all such numbers.
2. John threw a rock off a bridge into the river. The distance from the rock to the river is modelled by the equation $y = t^2 - 7t - 30$, where y is the height in meter and t is the time in seconds. Find how long it took the rock to hit the surface of the water.

Hint: $-7t = -10t + 3t$

Content Summary

Equations which are written in the form of $ax^2 + bx + c = 0 (a \neq 0)$ are called quadratic equations. To find solution of this equation the two main ways can be used in solving such equation

Use of factorization or finding square roots

Grouping terms or decomposition can be used to factorize the quadratic equations and later help us to find the solution of equation. By having the product of ac and the sum of those two integers which gives b , it helps you to decompose into a product of factors.

Example 1:

Solve in $\mathbb{R}$: $6y^2 + 5y - 25 = 0$

Solution

To obtain a common factor, terms that have a common factor are grouped together, and the common factor of each group is divided as follows:

Guess two integers whose sum is 5 and product is -150. The given equation can be written

as $6y^2 + 15y - 10y - 25 = 0$, or $(6y^2 + 15y) - (10y + 25) = 0$ or $3y(6y + 5) - 5(2y + 5) = 0$

$(2y + 5)(3y - 5) = 0$ from which $(2y + 5) = 0 \Rightarrow y = -\frac{5}{2}$ or $(3y - 5) = 0 \Rightarrow y = \frac{5}{3}$

Thus, the solution set is: $\left\{-\frac{5}{2}, \frac{5}{3}\right\}$.

Example 2:

Solve the equation $x^2 - 4x - 45 = 0$

Solution

$x^2 - 4x - 45 = 0$ is equivalent to $(x - 9)(x + 5) = 0$ from which $(x - 9) = 0 \Rightarrow x = 9$ or $(x + 5) = 0 \Rightarrow x = -5$

Then, the solution set is $S = \{-5, 9\}$.

Example 3:

Find the solution set of $(3x - 1)(2x + 3) = -5$

Solution

We expand the product as the equation is different from zero

$$6x^2 + 7x - 3 = -5$$

$$6x^2 + 7x - 3 + 5 = 0$$

$$(3x + 2)(2x + 1) = 0$$

$$(2x + 1) = 0 \text{ or } (3x + 2) = 0$$

$$\text{Therefore, } x_1 = -\frac{1}{2} \text{ or } x_2 = -\frac{2}{3}$$

$$\text{Thus, the solution set is } \left\{-\frac{2}{3}, -\frac{1}{2}\right\}.$$

Application activity 2.13



a) Use factorization to solve the following equations

1. $3x^2 = 10 - x$

2. $x^2 - 3x = -11$

3. $x^2 - 12x - 11 = 0$

4. $x^2 + 16 = 8x$

b) The area of a rectangular garden is 30 meters square. If the length is 7 meters longer than the width, find the dimensions of the rectangle.

c) Does a quadratic equation have more than one solution? Explain your answer.

2.14 Concept of discriminant

Activity 2.14



Use any available resource materials and answer the following questions:

- What is the discriminant of the quadratic equation of the form $ax^2 + bx + c = 0$?
- How to find the discriminant of a quadratic equation?
- Identify the different discriminant rules.

Content Summary

Equations which are written in the form of $ax^2 + bx + c = 0$, ($a \neq 0$) are called quadratic equations.

- The discriminant of a quadratic equation of the form $ax^2 + bx + c = 0$ is $\Delta = b^2 - 4ac$
- To find the discriminant given the quadratic equation $ax^2 + bx + c = 0$, simply record the values of a , b , and c and then substitute them into the discriminant formula: $\Delta = b^2 - 4ac$. This will give the value of the discriminant. It tells if the solutions to the quadratic equation real number or not as well as how many solutions to the equation exist.
- Here are the rules for the discriminant: Suppose that $f(x) = 0$ is a quadratic equation, then:
 - If $\Delta = b^2 - 4ac > 0$, there are two real roots
 - If $\Delta = b^2 - 4ac = 0$, there is one double real root
 - If $\Delta = b^2 - 4ac < 0$, there are no real roots.

Example:

Equation	Discriminant	Nature of solutions
$4x^2 - 7x - 1 = 0$	$\Delta = b^2 - 4ac = (-7)^2 - 4(4)(-1) = 49 + 16 = 65$	Two real roots
$4x^2 + 12x + 9 = 0$	$\Delta = b^2 - 4ac = (12)^2 - 4(4)(9) = 144 - 144 = 0$	One double real root
$5x^2 + 2x + 1 = 0$	$\Delta = b^2 - 4ac = (2)^2 - 4(5)(1) = 4 - 20 = -16$	There are no real roots

Application activity 2.14



Find the value of the discriminant for each quadratic equation. Then describe the number and type of roots for the equation:

- $9x^2 - 12x + 4 = 0$
- $2x^2 + 16x + 33 = 0$
- $2x^2 - 5x + 1 = 0$

2.15 Solving quadratic equations using discriminant

Activity 2.15



1. For what value of k will the equation $x^2 + 2x + k = 0$ have one double root. Find that root using $x_1 = x_2 = \frac{-b}{2a}$ for $ax^2 + bx + c = 0$ ($a \neq 0$).
2. Use discriminant to solve in $\mathbb{R}$ the following equation: $6x^2 + 5x - 25 = 0$

Content Summary

The quadratic equations which are written in the form of $ax^2 + bx + c = 0$ ($a \neq 0$) can be solved by using discriminant.

In quadratic equation $ax^2 + bx + c = 0$, Let $\Delta = b^2 - 4ac$ be discriminant,

- The two values of unknown x are generated as follows:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

- Note that, when the $\Delta = 0$ two equal roots are generated as $x_1 = x_2 = \frac{-b}{2a}$.
- When $\Delta < 0$ the equation has no root in the set of real numbers.

These two values of x can help us to write down factor form of a quadratic expression.

Example 1:

Solve in $\mathbb{R}$: $6y^2 + 5y - 25 = 0$

Solution

$$a = 6; \quad b = 5; \quad c = -25$$

$$\Delta = b^2 - 4ac = (5)^2 - 4(6)(-25) = 625$$

$$\sqrt{\Delta} = \sqrt{625} = 25$$

$$x_1 = \frac{-5 + 25}{12} = \frac{20}{12} = \frac{5}{3} \quad \text{and} \quad x_2 = \frac{-5 - 25}{12} = \frac{-30}{12} = \frac{-5}{2}$$

$$\text{Then, the solution set is } S = \left\{ -\frac{5}{2}, \frac{5}{3} \right\}.$$

Example 2:

For what value of k will the equation $4x^2 - 4x + k = 0$ have one double root? Find that root.

Solution

For one double root $\Delta = 0$.

$$\Delta = b^2 - 4ac$$

$$\Delta = (-4)^2 - 4 \times 4 \times k$$

$$\Delta = 16 - 16k$$

$$16 - 16k = 0$$

$$\Rightarrow -16k = -16 \Rightarrow k = 1$$

Thus, the value of k is 1.

That root is $x = \frac{-b}{2a} = \frac{-(-4)}{2 \times 4} = \frac{4}{8} = \frac{1}{2}$. Then, the solution set is $S = \left\{ \frac{1}{2} \right\}$.

Application activity 2.15



a) Use discriminant to solve the following equations:

1. $3x^2 = 10 - x$

2) $x^2 - 3x = -11$

3. $x^2 - 12x - 11 = 0$

4) $x^2 + 16 = 8x$

b) A farmer in Rwanda owns a rectangular garden with an area of 96 m^2 . The length is 10 meters longer than the width.

c) Does a quadratic equation have more than one solution? Explain your answer

2.16 Applications of linear equations: Problems about supply, demand and finance

Activity 2.16



Assume that a firm can sell as many units of its product as it can manufacture in a month at 180 Frw each. It must pay out 2400 Frw fixed costs plus a marginal cost of 140 Frw for each unit produced. How much does it need to produce to breakeven point (where total revenue equals to total cost)?

Content summary

When only two or single variables and equations are involved, a simultaneous equation system can be related to familiar graphical solutions, such as supply and demand analysis.

For example, assume that in a competitive market the demand schedule is given by

$$p = 420 - 0.2q \quad (1) \text{ and the supply schedule is given by } p = 60 + 0.4q \quad (2).$$

If this market is in equilibrium, the equilibrium price and quantity will be where the demand

and supply schedules intersect. This requires you to solve the system formed by the two simultaneous equations. Its solution will correspond to a point which is on both the demand schedule and the supply schedule. Therefore, the equilibrium values of p and q will be such that both equations (1) and (2) hold.

Example 1:

In a competitive market the demand schedule is given by $p = 420 - 0.2q$ and the supply schedule is given by $p = 60 + 0.4q$, solve for p and q the simultaneous equation and determine the point at which the market is in equilibrium.

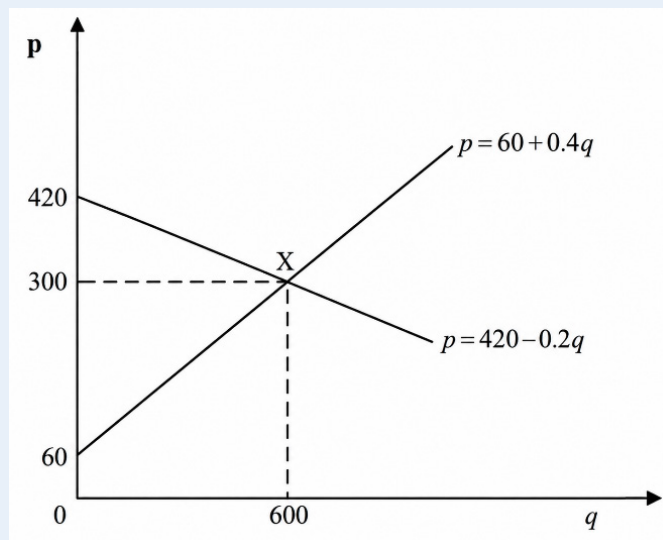
Solution

Let us solve the system

$$\begin{cases} p = 420 - 0.2q \\ p = 60 + 0.4q \end{cases}$$

Equating for the value of p , we find $420 - 0.2q = 60 + 0.4q$ which gives $q = 600$. Replacing this value in the given two equations, we find $p = 300$. The market is in equilibrium at the point $q = 600$.

These two functional relationships are plotted in the figure, and both hold at the intersection point $X(600, 300)$.



Example 2:

Calculate the equilibrium values of p and q in a competitive market where the demand schedule is $p = 200q^{-1}$ and the supply is $p = 30 + 2q$.

Solution

In equilibrium, demand price equals supply price. Therefore, $200q^{-1} = 30 + 2q$

Multiplying through by q , the equation above becomes $200 = 30q + 2q^2$ or $0 = 30q + 2q^2 - 200$ or $0 = (2q - 10)(q + 20)$ from which $(2q - 10) = 0$ or $(q + 20) = 0$.

Therefore, $q_1 = 5$ or $q_2 = -20$

We can ignore the second solution as negative quantities do not exist. Thus, the equilibrium quantity is 5. Substituting this value into the supply function gives equilibrium price $p = 30 + 2(5) = 40$.

Example 3:

A firm makes two goods, A and B which require two inputs K and L. One unit of A requires 6 units of K plus 3 units of L and one unit of B requires 4 units of K plus 5 units of L. The firm has 420 units of K and 300 units of L at its disposal. How much of A and B should it produce if it wishes to exhaust its supplies of K and L totally?

Solution

This question requires you to use the economic information given to set up a mathematical problem in a format that can be used to derive the desired solution.

The total requirements of input K are 6 for every unit of A and 4 for each unit of B, which can be written as $K = 6A + 4B$

Similarly, the total requirements of input L can be specified as $L = 3A + 5B$

As we know that $K = 420$ and $L = 300$ because all resources are used up

Then, $420 = 6A + 4B$ and $300 = 3A + 5B$

Solve the system of linear equations to find values of A and B

$$\begin{cases} 420 = 6A + 4B \\ 300 = 3A + 5B \end{cases}$$

From the first equation, we have $420 = 6A + 4B \Rightarrow A = \frac{420 - 4B}{6}$

From the second equation, we have $300 = 3A + 5B \Rightarrow A = \frac{300 - 5B}{3}$

From the two equations above, we have the following equality:

$$\frac{420 - 4B}{6} = \frac{300 - 5B}{3} \text{ from which } 1260 - 12B = 1800 - 30B \text{ or } 30B - 12B = 1800 - 1260 \text{ or } 18B = 540 \text{ or } B = 30.$$

Using $B = 30$ in $A = \frac{420 - 4B}{6}$ gives $A = \frac{420 - 4(30)}{6} = \frac{420 - 120}{6} = \frac{300}{6} = 50$ or $A = 50$.

The firm should therefore produce 50 units of A and 30 units of B.

Example 4:

Two student teachers were driving at constant speeds in the same direction during their holidays. The first travelling at 40 km/h left Kigali at 8:30 a.m. The second travelling at 60 km/h followed him after 1 hour.

- What is the distance between the two drivers one hour after the departure of the first driver?
- If it is the same time is used by the two drivers after the departure of the second driver and the entire distance covered,
 - Establish the functions $y = f(t)$ of the first.
 - Establish the functions $y = g(t)$ of the second driver
 - When did the second bus overtake the first bus?
 - Illustrate the two functions on the Cartesian plan from $t = 0$ to $t = 5h$ and show the point where the second met the first driver.

Solution

The speed of the first driver is 40 km/hour or $v = 40 \text{ km} / h$

The second driver has a speed of 60km/hour and left after the departure of the first.
 $v_2 = 60 \text{ km} / h$.

a) Distance covered by the first driver is $y_0 = v_1 t = (40 \text{ km} / h) \times 1 h = 40 \text{ km}$

b) i) The distance covered by the first driver is the initial distance plus the distance covered after the departure of the second driver.

This is $y = y_0 + v_1 t = 40 + 40t$.

Then $y = f(t) = 40 + 40t$

ii) The distance y covered by the second driver is $y = v_2 t = 60t$.

Therefore, $y = g(t) = 60t$

iii). After the departure of the second driver, the two drivers will be using the same time t and when this second driver will meet the first, the two will be covered the same distance

$d \text{ } \forall y \quad g(t) \quad f(t)$

This means: $40 + 40t = 60t$ or

$$40 + 40t = 60t$$

$$60t - 40t = 40$$

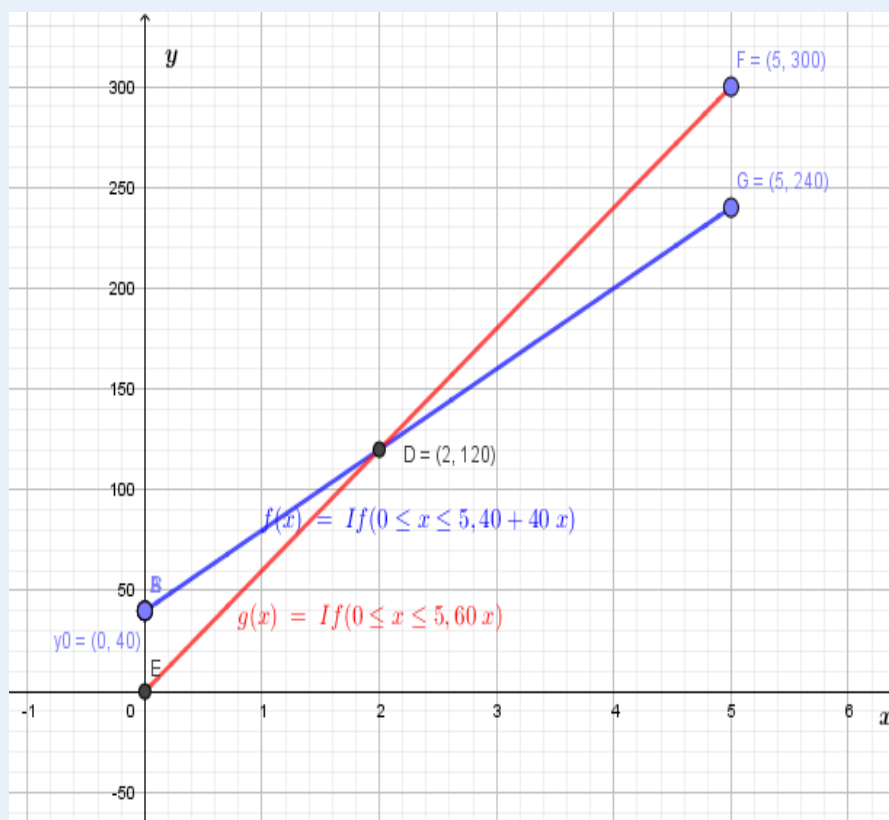
$$20t = 40$$

$$t = 2$$

The second driver will meet the first after 2 hours. It will be at 11h30 because $9\text{h}30 + 2\text{hs} = 11\text{h}30$.

iv) The table of values can help to draw the graphs of the function f and g .

T	0	1	2	3	4	5
$y = f(t) = 40 + 40t$ (in km)	40	80	120	160	200	240
$y = g(t) = 60t$ (in km)	0	60	120	180	240	300



Example 5:

Market equilibrium for rice cooperatives

In a local agricultural market, the supply and demand for high-quality organic rice fluctuate based on the market price per kilogram (P , expressed in Rwandan Francs, Frw).

An economist studying the market models the monthly dynamics using the following linear functions:

- Demand function: $Q_d = 12,000 - 4P$
- Supply function: $Q_s = -3,000 + 6P$

Where Q represents the quantity of rice in kilograms.

- c) Find the equilibrium price: Calculate the market equilibrium price (P) where the quantity supplied equals the number of kilograms demanded ($Q_s = Q_d$).

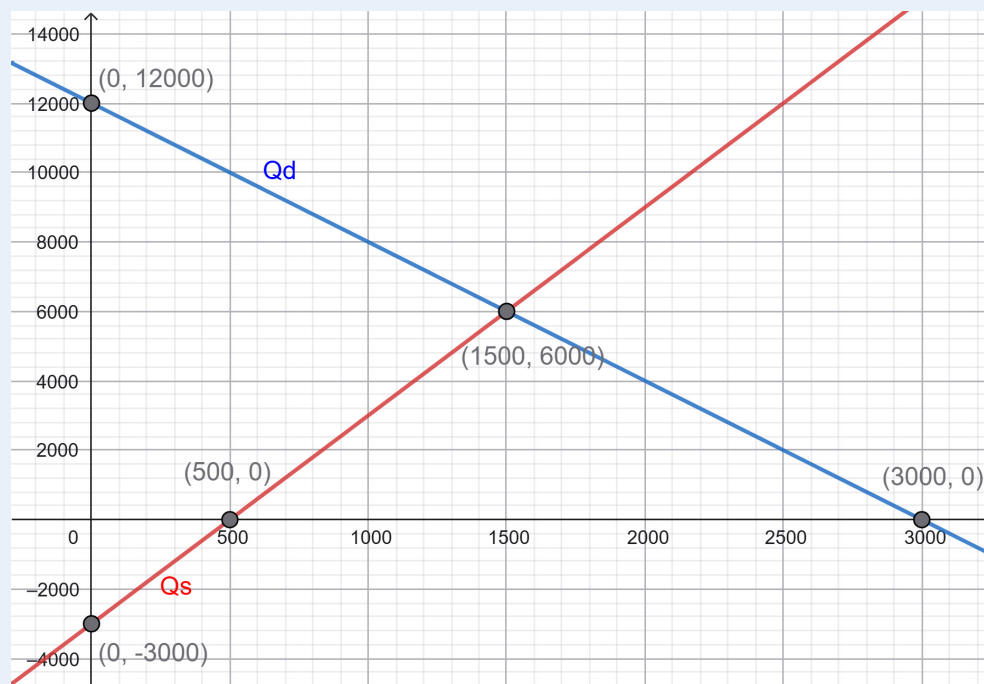
- d) Find the equilibrium quantity: Substitute your equilibrium price back into either function to find the exact quantity of rice that will be traded monthly.
- e) Financial literacy reflection: If the government sets a price ceiling of 1,200 Frw per kilogram to protect consumers, calculate the new Q_d and Q_s . Will this policy result in a market surplus or a shortage? Explain the financial risk this poses to farming cooperatives.

Solution

- a) Equilibrium price (P): Set the equations equal to each other:

$$12,000 - 4P = -3,000 + 6P$$

$$15,000 = 10P \Rightarrow P = 1,500 \text{ Frw}$$



- b) Equilibrium quantity (Q): $Q = 12,000 - 4(1,500) = 6,000 \text{ kg}$
- c) Price ceiling analysis: At 1,200 Frw, $Q_d = 12,000 - 4(1,200) = 7,200 \text{ kg}$ and $Q_s = -3,000 + 6(1,200) = 4,200 \text{ kg}$.

This creates a shortage of 3,000 kg $(7,200 - 4,200)$, which could lead to black markets and reduced income for cooperative farmers.

Example 6:

Maximizing total revenue

A youth entrepreneurship group operates a community solar-powered charging kiosk. Currently, they charge a flat rate of 150Frw per phone charge, serving an average of 400 customers per week.

A market survey reveals that for every 10 Frw increase in the charging price, they will lose 20 customers per week to competitors.

The total weekly revenue (R) can be modeled as a quadratic function of the number of price increases (x): $R(x) = (\text{Price}) \times (\text{Quantity}) = (150 + 10x)(400 - 20x)$.

- Expand the Revenue Function: Multiply the terms to write the total revenue function in standard quadratic form: $R(x) = ax^2 + bx + c$.
- Find the optimal strategy: Determine the value of x that maximizes the weekly revenue by using the vertex formula $x = -\frac{b}{2a}$.
- Determine the best price and max revenue: What is the ideal price per charge the group should set, and what will their maximum weekly revenue be?

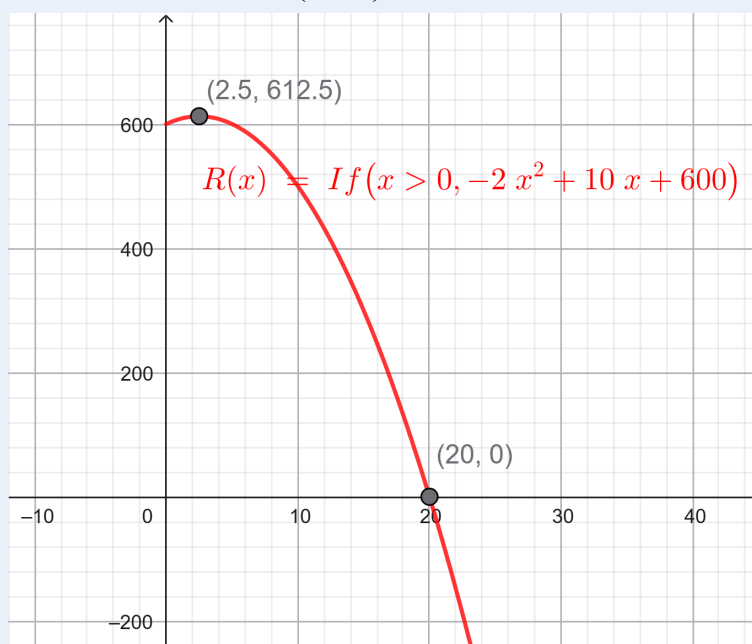
Solution

- a) Standard quadratic form:

$$R(x) = 60,000 - 3,000x + 4,000x - 200x^2$$

$$R(x) = -200x^2 + 1,000x + 60,000$$

- b) Finding the vertex (x): $x = -\frac{1,000}{2(-200)} = \frac{-1,000}{-400} = 2.5$ price increases



- c) Calculating ideal price and revenue:

$$\text{New price: } 150 + 10(2.5) = 175 \text{ Frw}$$

$$\text{New quantity: } 400 - 20(2.5) = 350 \text{ customers}$$

$$\text{Max revenue: } 175 \times 350 = 61,250 \text{ Frw (Compared to 60,000 Frw originally).}$$

Example 7:

Break-even analysis for a startup

A small manufacturing company produces eco-friendly, school exercise books made from recycled paper. The company's financial planning team establishes the following weekly cost and revenue structures:

- Total cost function: $C(x) = 200x + 150,000$ (where 150,000 RWF represents fixed overhead costs like rent, and 200 RWF is the variable production cost per book).
- Total revenue function: $R(x) = 800x - 0.5x^2$ (where revenue is non-linear because they offer bulk-buying discounts to large schools).

Here, x represents the number of exercise books produced and sold per week.

- Formulate the profit function: Write the simplified profit function,
Profit $P(x) = R(x) - C(x)$.
- Calculate the break-even points: Find the minimum and maximum number of books the factory must sell each week to break even (Profit = 0) using the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Financial risk analysis: Why are there two break-even points for this company instead of just one? What operational advice would you give the production manager based on the maximum break-even point?

Solution

a) Profit function: $P(x) = (800x - 0.5x^2) - (200x + 150,000)$

$$P(x) = -0.5x^2 + 600x - 150,000$$

b) Break-even points ($P(x) = 0$): $-0.5x^2 + 600x - 150,000 = 0$

Multiply by -2 to clear the decimal: $x^2 - 1,200x + 300,000 = 0$

Using the quadratic formula where $a = 1, b = -1,200, c = 300,000$:

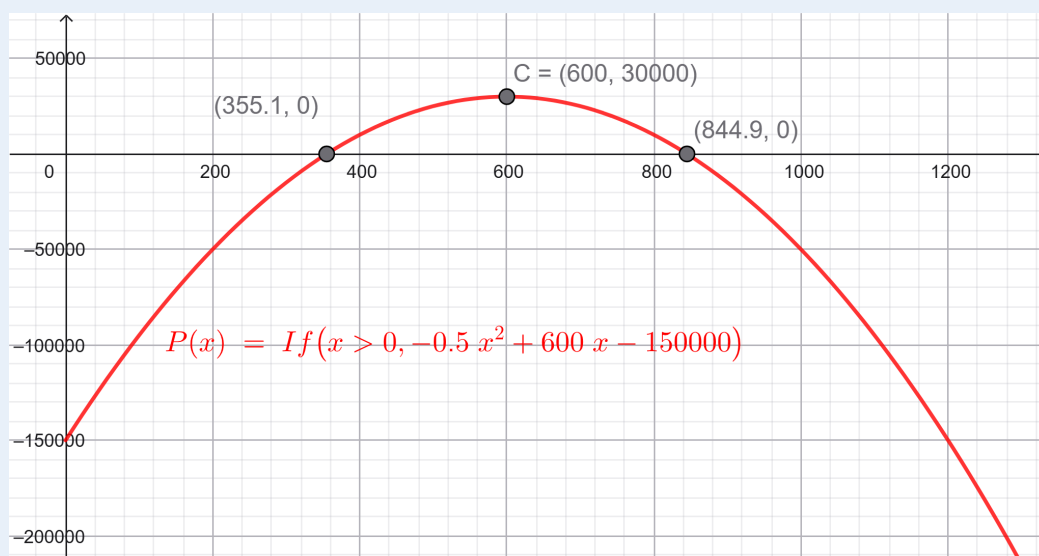
$$x = \frac{1,200 \pm \sqrt{(-1,200)^2 - 4(1)(300,000)}}{2(1)}$$

$$x = \frac{1,200 \pm \sqrt{1,440,000 - 1,200,000}}{2} = \frac{1,200 \pm \sqrt{240,000}}{2}$$

$$x \approx \frac{1,200 \pm 489.9}{2}$$

Lower break-even point: $x_1 \approx 355$ books

Upper break-even point: $x_2 \approx 845$ books



- c) Financial interpretation: The lower point (355) is the classic minimum needed to cover fixed startup costs. The upper point (845) exists because their bulk discounts eventually reduce the marginal price so much that revenue can no longer cover the variable production costs. The manager must strictly keep production between 355 and 845 units to remain profitable.

Application activity 2.16



Read carefully the following word problems and solve the related equations

1. Assume that in a competitive market the demand schedule is $p = 420 - 0.2q$ and the supply schedule is $p = 60 + 0.4q$ ($p = \text{price}, q = \text{quantity}$). If the market is in equilibrium, then the equilibrium price and quantity will be where the demand and supply schedules intersect. As this will correspond to a point which is on both the demand schedule and the supply schedule the equilibrium values of p and q will be such that both equations hold. Find the equilibrium quantity and the equilibrium price by solving the following $420 - 0.2q = 60 + 0.4q$.
2. Today a house-worker has 3000Frw. His boss pays him 400Frw per day and he saves all the money received.
 - a) What is the money the house-worker will have after 2 days, 5 days and after 10 days?
 - b) Discuss the function $y = f(t)$ for the money saved by the house-worker where t represents the time in days.
 - c) Illustrate the function $y = f(t)$ in the Cartesian plane on a manila paper.
 - d) Discuss the money the house-worker will have after 2 months.
 - e) Is it possible for a house-worker to save the money? Explain your answer.

3. The national income in the basic Keynesian macroeconomic model is given by $Y = C + I$. If C is given by $40 + 0.5y$ and $I = 200$. Calculate the national income in the basic Keynesian macroeconomic model.

2.17 Revision of unit content

Knowledge and understanding:

1. Define a linear equation in one unknown.
2. State the meaning of inequality in mathematics.
3. What is an interval and how is it represented on a number line?
4. List two methods used to solve simultaneous linear equations algebraically.
5. What is the discriminant of a quadratic equation and what information does it provide?

Skills:

1. Solve the linear equation: $3x - 8 = 16$
2. Solve the inequality and represent the solution on a number line: $2x + 5 < 13$
3. Solve the simultaneous linear equations by elimination:
$$\begin{cases} 2x + y = 7 \\ x - y = 1 \end{cases}$$
4. Solve the quadratic equation by factorization: $x^2 - 5x + 6 = 0$
5. A shop sells pens and books. A total of 40 items were sold, including 12 books. Form and solve a linear equation to find the number of pens sold.

Attitudes and values:

1. Explain why it is important to show all steps when solving equations and inequalities in classwork or assessments.
2. Describe how solving real-life problems using inequalities can help students make fair and well-reasoned decisions (e.g. budgeting or time management).

End of unit assessment



Assessing knowledge and understanding

- Which of the following represents a linear equation in one unknown?
A) $x^2 + 3x - 4 = 0$ B) $5x - 7 = 3$ C) $2y + 3 = 9$ D) $xy = 12$
- Which inequality correctly represents the statement “x is greater than or equal to 4”?
A) $x > 4$ B) $x \leq 4$ C) $x \geq 4$ D) $x < 4$
- Which of the following pairs of equations forms a system of simultaneous linear equations?
A) $x^2 - y = 7$ and $2x + 3 = 0$ B) $3x - 2y = 5$ and $x + y = 4$
C) $2x = 3y^2$ and $x + 4 = 0$ D) $xy = 4$ and $y - 3 = 0$
- What is the value of the discriminant of the quadratic equation $x^2 - 6x + 9 = 0$?
A) -6 B) 0 C) 6 D) 36
- Which of the following best describes a real-life situation modeled by linear inequalities?
A) Finding the exact number of days in a year
B) Determining the possible scores a student needs to pass a test
C) Calculating the roots of a quadratic equation
D) Finding the area of a rectangle

Assessing skills:

- A student wants to buy exercise books that cost 300 *Frw* each. The student has 2,400 *Frw* saved and plans to buy as many books as possible.
 - Form a linear equation that represents the total cost of buying x exercise books.
 - Write an inequality showing how many books the student can afford with 2,400 *Frw*.
 - Solve the inequality and determine the maximum number of books the student can buy.
 - Explain how solving this inequality helps develop responsible financial decision-making.

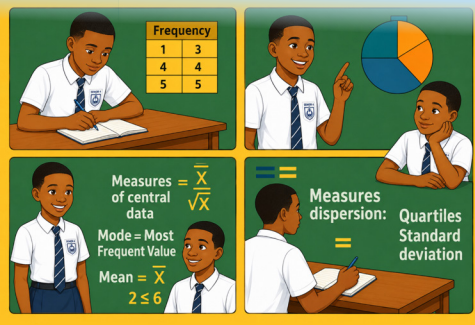
2. Two transport services charge differently for trips to the same destination.
- Company A charges a fixed fee of 500 *Frw* plus 100 *Frw* per kilometer.
 - Company B charges a fixed fee of 300 *Frw* plus 120 *Frw* per kilometer. A group of students wants to compare the two to decide which is cheaper for different distances.
- a) Write two linear equations representing the cost (C) of each company for x kilometers.
 - b) Solve the simultaneous equations to find the distance at which both companies cost the same.
 - c) For distances less than this value, which company is cheaper? Explain briefly.
 - d) Explain how analyzing such equations helps students develop real-life decision-making skills.

Assessing attitudes and values:

1. A student is planning to save money weekly to buy a school bag that costs 15,000 *Frw*. The learner decides to save x *Frw* each week. This situation can be represented using inequalities.
 - a) Explain why it is important for the learner to record savings honestly when forming equations or inequalities.
 - b) If the student commits to saving at least 1,500 *Frw* per week, discuss how discipline and responsibility help achieve the goal.
 - c) Explain how using inequalities in this context helps students appreciate planning and financial literacy.
2. A group of students compares two restaurant meal packages using simultaneous linear equations.
 - Package A: $c = 100 + 200x$
 - Package B: $c = 600 + 250x$.
 - Where c is the cost in *Frw* and x is the number of items.
 - a) Why is it important for the group to listen to each member's reasoning when forming the equations?
 - b) Solve the simultaneous equations to find when both packages cost the same.
 - c) How can students use this result to make a fair decision about which package to choose?
 - d) Explain how analyzing such equations encourages ethical decision-making in daily life.

UNIT 3

DESCRIPTIVE STATISTICS



Key Unit Competence:

Analyze statistical data from daily life situations for drawing meaningful conclusions.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Describe measures of central tendency of data: mode, mean and Median.
- Distinguish between types of data.
- Describe different methods of data presentation such as frequency tables, bar charts, pie charts, histograms, and polygons.
- Explain the concepts and formulas for measures of central tendency and dispersion.
- Interpret graphs and statistical results correctly.

Skills

2

- Represent statistical information using frequency distribution table, bar chart, pie chart, histogram, polygon.
- Interpret graph of presented data.
- Calculate and interpret measures of central tendency and measures of dispersion.
- Display the distribution of data using the box plots.
- Collect, represent, analyze and interpret data to answer a question.

Attitudes and values

3

- Appreciate how data collection, data representation and data interpretation can be used for solving real life situations.
- Appreciate the importance of measures of dispersion in the interpretation of data
- Show concern on how to use the standard deviation as measure of variability of data.

Introduction

Descriptive statistics help us understand how to collect, organize, and summarize information in a meaningful way. In this unit, you explore different types of data and learn how to present them using tools such as frequency distribution tables, bar charts, pie charts, histograms, and polygons. These representations make it easier to identify trends, compare groups, and communicate findings clearly. Through graph interpretation and practical examples, you begin to see how data is used to explain real-world situations in fields like economics, social sciences, etc.

The unit also introduces important numerical measures used to describe data more precisely. Students study measures of central tendency mode, median, and mean for both ungrouped and grouped data, as well as measures of dispersion such as quartiles, variance, standard deviation, and coefficient of variation. These concepts help learners analyze how data values spread out and how typical or unusual certain values may be. By applying these tools to real-life contexts, you develop critical-thinking skills that support informed decision-making and deepen your understanding of patterns and differences in everyday information.

Introductory activity



1. At the market a fruit-seller has the following daily sales in Rwandan francs for five consecutive days: 1000 *Frw*, 1200 *Frw*, 1250 *Frw*, 1000 *Frw*, and 1300 *Frw*. Help her to determine the money she could get if the sales are equally distributed per day to get the same total amount of money.



2. During the welcome test of Mathematics for the first term, 10 students of S4 scored the following marks out of 10: 3, 5, 6, 3, 8, 7, 8, 4, 8 and 6.
 - a) What is the mean mark of the class?
 - b) Choose the mark that was obtained by many students.
 - c) Compare the differences between the mean of the group and the mark for every student.

3.1 Definition of terms related to statistics and type of data

Activity 3.1



Use any available resource materials to determine the meaning of statistics and types of data. Use your findings to select qualitative and quantitative data from this list: male, female, tall, age, 20 sticks, 45 students, 20 meters, 4 pieces of chalk.

Content Summary

- Statistics is the branch of mathematics that deals with data collection, data organization, summarization, analysis and drawing conclusions from data.

The use of graphs, charts, and tables and the calculation of various statistical measures to organize and summarize information is called descriptive statistics. Descriptive statistics help to reduce our information to a manageable size and put it into focus.

Every day, we come across a wide variety of information in form of facts, numerical figures or table groups.

- A variable is a characteristic or attribute that can assume different values.
- Data are the values (measurements or observations) that the variables can assume.

Variables whose values are determined by chance are called random variables. A collection of data values forms a data set. Each value in the data set is called a data value or a datum.

For example, information related to profit/ loss of the school, attendance of students and tutors, used materials, school expenditure in term or year, etc. These facts or figure which is numerical or otherwise, collected with a definite purpose is called data. This is the word derived from Latin word “Datum” which means pieces of information.

- Qualitative variable: The qualitative variables are variables that cannot be expressed using a number. A qualitative data is determined when the description of the characteristic of interest results in a non-numerical value. A qualitative variable may be classified into two or more categories. Data obtained by observing values of a qualitative variable are referred to as **qualitative data**.

Example:

Qualitative variable	Possible categories of data
Marital status	Single, married, divorced
Gender	Male, Female
Pain level	None, moderate, severe
Colour	Red, black, green, yellow

The possible categories for qualitative variables are often coded for the purpose of performing computerized statistical analysis so that we have qualitative data.

- Quantitative variable: Quantitative variables are variables that are expressed in numerical terms, counted or compared on a scale. A quantitative data is determined when the description of the characteristic of interest results in numerical value. When measurement is required to describe the characteristic of interest or when it is necessary to perform a count to describe the characteristic, a quantitative variable is defined.
- Discrete data is quantitative data whose values are countable. Discrete data usually results from counting.
- Continuous data** is a quantitative data that can assume any numerical value over an interval or over several intervals. Continuous data usually results from making a

measurement of some type. Data obtained by observing values of a quantitative variable are referred to as **quantitative data**. Quantitative data obtained from a discrete variable are also referred to as discrete data and quantitative data obtained from a continuous variable are called continuous data.

Example:

Discrete variable	Possible values of data
The number of defective needles in boxes of 100 diabetic syringes	0, 1, 2, 100
The number of individuals in groups of 30 with a type A personality	0, 1, 2, 3, 30
Number of students in classroom	0, 1, 2, 45

Continuous variable	Possible values of data
The household income for households with incomes less than or equal to 200,000 Rwandan francs	All the real numbers between A and 200,000 where A is the smallest household income in the population
Height	All values for the length in length measurements.

Data can be used in different ways. The body of knowledge called statistics is sometimes divided into two main areas, depending on how data are used. The two areas are descriptive statistics and inferential statistics.

- 1. Descriptive statistics** consists of the collection, organization, summarization, and presentation of data.
 - 2. Inferential statistics** consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.
- A **population** consists of all subjects (human or otherwise) that are being studied.
 - A **sample** is a group of subjects selected from a population.

Application activity 3.1



1. Select qualitative and quantitative data from the list below:

Product rating, basketball team classification, number of student-teachers in the classroom, weight, age, number of rooms in a house, number of teachers in school.

2. The table below gives fasting blood sugar reading for five patients at a small medical clinic. What is the variable? Are they continuous? Compare this data and give your observations.

Patient	Fasting blood sugar level
1 st Patient	125
2 nd patient	175
3 rd patient	160
4 th patient	110

3.2 Presentation / organization of data using frequency distribution table

Activity 3.2



At the beginning of the school year, a Mathematics test was administered in Senior four to 50 students to test their level of understanding. Their results out of 20 were recorded as follows:

16	10	11	11	17	12	13	16	15	15
10	9	9	10	11	9	17	10	16	8
18	10	17	9	11	10	14	12	9	8
17	18	15	13	10	10	18	9	10	18
9	10	10	11	11	9	16	11	11	9

- Looking at the list of 50 test scores, what challenges do we face if we try to interpret them directly without organizing?
- How can we decide suitable ranges (class intervals) to group the scores so that the data becomes easier to summarize?
- Once we have intervals, how can we count how many students fall into each group?
- How can we present this grouped information on a simple table that shows the intervals and the number of students in each?
- What does this table tell us about the overall level of understanding of the learners, and how might teachers use it to support them?

Content Summary

After the collection of data, the researcher needs to organize and present them in order to help those who will benefit from the research and lead him or her to the conclusion. When the data are in original form, they are called raw data and are listed next.

Raw data

After the collection of data, one can present them in the raw form presentation.

Example:

The following are numbers of notebooks for 54 student-teachers.

3 4 5 6 6 7 3 2 5 4 5 7 4 3 2 3 6 5

8 5 6 4 2 6 5 3 4 7 9 8 6 7 4 5 4 6

3 4 3 6 7 8 2 7 6 5 4 6 4 7 8 9 9 5

Since little information can be obtained from looking at raw data, the researcher organizes the data into what is called a frequency distribution. A frequency distribution consists of the number of times each value appears in the raw data and sometimes the corresponding percentage vice-versa the total number in the sample.

1. Frequency distribution

Data can be presented in various forms depending on the type of data collected. A **frequency distribution** is the organization of raw data in the table form by using frequencies.

Example:

The following data represent marks obtained by 12 student-teachers out of 20 in mathematics test of a certain TTC.

13 10 15 17 17 18

17 17 11 10 17 10

The set of outcomes is displayed in a frequency table, as illustrated below:

Marks	Tallies	Frequencies f_i
10		3
11		1
13		1
15		1
17		5
18		1
Total		12

The total frequency is the number of items in the population. In the survey about the marks of student-teachers above, the total frequency is the number of all student-teachers. We find this by adding up the numbers from the third column of the table:

$3+1+1+1+5+1$ which is equal to 12.

Grouped data

When the number of data is too large, a simple distribution is not appropriate. In this case we come up with a grouped frequency distribution. A grouped frequency distribution organizes data into groups or classes.

Definitions related to grouped frequency distribution

- a) **Class limits:** The class limits are the lower and upper values of the class
- b) **Lower class limit:** Lower class limit represents the smallest data value that can be included in the class.
- c) **Upper class limit:** Upper class limit represents the largest data value that can be included in the class.
- d) **Class mark or class midpoint:** $\text{class mid point} = \frac{\text{lower class} + \text{upper class}}{2}$

Example:

The following data represent the marks obtained by 40 students in Mathematics test. Organize the data in the frequency table; grouping the values into classes, starting from 41-50:

54 83 67 71 80 65 70 73 45 60 72 82 79 78 65 54 67 64 54 76 45 63 49 52
60 70 81 67 45 58 69 53 65 43 55 68 49 61 75 52

Solution

Classes	Class midpoint x	Frequency f
41-50	45.5	6
51-60	55.5	10
61-70	65.5	13
71-80	75.5	8
81-90	85.5	3

The classes: 41-50, 51-60, 61-70, 71-80, 81-90

Lower class limits: 41, 51, 61, 71, and 81

Upper class limits: 50, 60, 70, 80, and 90

$$\text{Class mid point for the first class} = \frac{41 + 50}{2} = 45.5$$

Class boundaries

Class boundaries are the midpoints between the **upper-class** limit of a class and the **lower-class limit of the next class**.

Therefore, each class has a lower and an upper-class boundary.

Example:

Classes	Class midpoint	Frequency
[5,10[	7.5	2
[10,15[	12.5	6
[15,20[	17.5	4
[20,25[	22.5	3
[25,30[	27.5	4
[30,35[	32.5	1

For [10,15[, the lower-class boundary is 10, the upper-class boundary is 15. Therefore, class width = $15 - 10 = 5$.

We have Categorical **Frequency Distributions and Grouped Frequency Distributions**

The categorical frequency distribution is used for data that can be placed in specific categories, such as nominal- or ordinal-level data.

Example:

Twenty-five army inductees were given a blood test to determine their blood type.

The data set is:

A B B AB O O O B AB B B O
A O A O O O AB AB A O B A

Class	Tally	Frequency	Percent
A		5	20
B		7	28
AB		9	36
O		4	16
Total		25	100

Grouped Frequency Distributions

When the range of the data is large, the data must be grouped into classes that are more than one unit in width, in what is called a grouped **frequency distribution**. For example, a distribution of the number of hours that boat batteries lasted is the following:

Class limits	Class boundaries	Tally	Frequency
24-30	23.5-30.5		3
31-37	30.5-37.5		1
38-44	37.5-44.5		5
45-51	44.5-51.5		9
52-58	51.5-58.5		6
59-65	58.5-65.5		1
Total			25

The procedure for constructing a grouped frequency distribution, i.e., when the classes contain more than one data value. These data represent the record high temperatures in degrees Fahrenheit (*F) for each of the 50 states. Construct a grouped frequency distribution for the data using 7 classes.

112 100 127 120 134 118 105 110 109 112
 110 118 117 116 118 122 114 114 105 109
 107 112 114 115 118 117 118 122 106 110
 116 108 110 121 113 120 119 111 104 111
 120 113 120 117 105 110 118 112 114 114

Source: *The World Almanac and Book of Facts*

1. Determine the classes

- Find the highest value and lowest value: H =134 and L=100.
- Find the range: R = highest value-lowest value H -L, so R=134 -100 = 34
- Select the number of classes desired (usually between 5 and 20). In this case, 7 is arbitrarily chosen.
- Find the class width by dividing the range by the number of classes

$$\text{width} = \frac{R}{\text{number of classes}} = \frac{34}{7} = 4.9$$

- Select a starting point for the lowest class limit. This can be the smallest data value or any convenient number, less than the smallest data value. In this case, 100 is used. Add the width to the lowest score taken as the starting point to get the lower limit of the next class. Keep adding until there are 7 classes, as shown, 100, 105, 110, etc.
- Subtract one unit from the lower limit of the second class to get the upper limit of the first class. Then add the width to each upper limit to get all the upper limits. 105-1=104.
- The first class is 100–104, the second class is 105–109, etc.
- Find the class boundaries by subtracting 0.5 from each lower-class limit and adding 0.5 to each upper-class limit:

99.5–104.5, 104.5–109.5, etc.

2. Tally the data
3. Find the numerical frequencies from the tallies.

The completed frequency distribution is

Class limits	Class boundaries	Tally	Frequency
100-104	99.5-104.5		2
105-109	104.5-109.5		8
110-114	109.5-114.5		18
115-119	114.5-119.5		13
120-124	119.5-124.5		7
125-129	124.5-129.5		1
130-134	129.5-134.5		1
Total			$n = \sum f = 50$

2. Cumulative frequency

The cumulative frequency corresponding to a particular value is the sum of all frequencies up to the last value including the first value. Cumulative frequency can also define as the sum of all previous frequencies up to the current point.

Example:

The set of data below shows marks obtained by student-teachers in Mathematics. Draw a cumulative table for the data.

11	15	18	15	10	16	11	10	17
13	17	11	17	16	17	15	13	16

Solution

The cumulative frequency at a certain point is found by adding the frequency at the present point to the cumulative frequency of the previous point.

Marks	Frequencies (f_i)	Cumulative frequencies (cuf_i)
10	2	2
11	3	$2 + 3 = 5$
13	2	$5 + 2 = 7$
15	3	$7 + 3 = 10$
16	3	$10 + 3 = 13$
17	4	$13 + 4 = 17$
18	1	$17 + 1 = 18$

3. Relative frequency and percentage

The relative frequency is obtained by dividing the frequency for every data by the sum of all the frequencies. The percentage for a category is obtained by multiplying the relative frequency for that category by 100.

Example:

Marks	Frequencies (f_i)	Relative frequency	percentage
10	2	$2/18 = 0.111$	11.1
11	3	$3/18 = 0.167$	16.7
13	2	$2/18 = 0.11$	11.1
15	3	$3/18 = 0.167$	16.7
16	3	$3/18 = 0.167$	16.7
17	4	$4/18 = 0.222$	22.2
18	1	$1/18 = 0.056$	5.5

Application activity 3.2



1. Suppose that a teacher conducted a test for learners and the marks out of 10 were as follows: 3 3 3 5 6 4 6 7 8 3 8 8 8 10 9 10 9 10 8 10 6 Present this data using frequency distribution table.

In a quiz, the marks obtained by 20 learners out of 30 are given as follows:

2. 12, 15, 15, 29, 30, 21, 30, 30, 15, 17, 19, 15, 20, 20, 16, 21, 23, 24, 23, 21

Present this data in the form of a frequency distribution.

3. The mass of 50 tomatoes (measured to the nearest g) was measured and recorded in the table below.

86	101	114	118	87	92	93	116	105
102	92	93	101	111	96	117	100	106
118	101	107	96	101	102	104	92	99
107	98	105	113	100	103	108	92	109
95	100	103	110	113	99	106	116	101
105	86	88	108	92				

Construct a frequency distribution table, using equal class intervals of width 5g and taking the lower-class boundary of the first interval as 84.5g.

3.3 Presentation of data using bar chart

Activity 3.3



At the beginning of the school year students came to school with all requirements materials and others didn't attend the school on first day. The table below shows the number of student-teachers who attended the school in 5 classrooms on first day.

Number of student-teachers	4	5	6	7	8
Classroom	A	B	C	D	E

- Looking at the table of classrooms and the number of student-teachers present, what difficulties do we face if we try to compare the classrooms directly without visual aid?
- What kind of visual method could help us show the differences in attendance more clearly than a table?
- If we decide to use a chart, what should we place on the horizontal axis (x-axis) and what should go on the vertical axis (y-axis)?
- How can we represent each classroom's attendance using bars of different heights, and what rules should we follow to keep the chart accurate?
- Once the chart is complete, what patterns or comparisons can we observe about attendance across the classrooms, and what conclusions can we draw about student readiness on the first day?

Content Summary

A bar chart or bar graph is a chart or graph that presents numerical data with rectangular bars with heights or lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally. A vertical bar chart is sometimes called a column graph.

To draw the bar chart, always choose a suitable scale for the vertical axis.

Example:

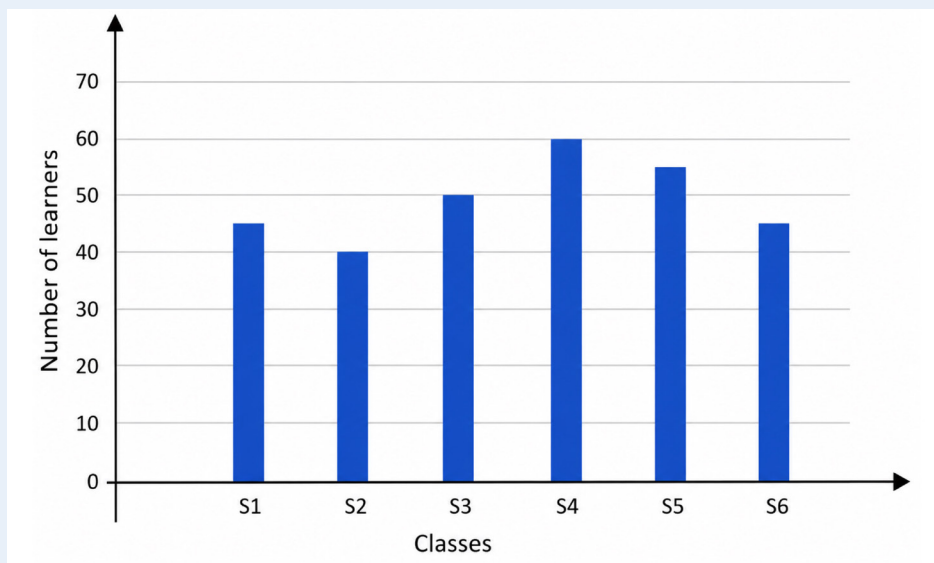
The table below shows number of learners per class in a school in Rwanda.

Class	S1	S2	S3	S4	S5	S6
Number of learners	45	40	50	60	55	45

- Represent the data in a bar chart
- How many learners are in the whole school?

Solution

- a) A bar graph showing number of learners per class in a school



- b) The number of learners that are in the whole school

$$= 45 + 40 + 50 + 60 + 55 + 45 = 295$$

The school has 295 learners.

Application activity 3.3



An insurance company determines vehicle insurance premiums based on known risk factors. If a person is considered a higher risk, their premiums will be higher. One potential factor is the colour of your car. The insurance company believes that people with some colour cars are more likely to get in accidents. To research this, they examine police reports for recent total loss collisions. The data is summarized in the frequency table below:

Colour	Frequency
Blue	25
Green	52
Red	41
White	36
Black	39
Grey	23

- a) From the frequency table above, identify the highest frequency and the lowest frequency.
- b) Present the car data on bar chart indicating frequency against vehicle colour involved in total loss collision.

3.4 Presentation of data using pie chart

Activity 3.4



1. Conduct research in the library or on the internet about the meaning of pie chart and explain steps to be done when statistical data are to be presented using pie chart.
2. One teacher at any secondary school wants to check the level of how his/her learners like different subjects taught in Languages option. The survey is done on 60 students. Here is the number of participants of the survey.

Subject	Number of students
English	12
Mathematics	24
French	6
Entrepreneurship	10
Kinyarwanda	8

Present these data on a pie chart. Using that chart, explain the data.

Content Summary

A pie chart, sometimes called a circle chart, is a way of summarizing a set of data in circular graph. This type of chart is a circle that is divided into sections or wedges according to the percentage of frequencies in each category of the distribution. Each part is represented in degrees.

To present data using pie chart, the following steps are respected:

- Write all the data into a table and add up all the values to get a total.
- To find the values in the form of a percentage divide each value by the total and multiply by 100. That means that each frequency must also be converted to a percentage by using the formula $\% = \frac{f}{n} \times 100\%$
- To find how many degrees for each pie sector we need, we take a full circle of 360° and use the formula: Angle for sector S = $\frac{\text{frequency of S} \times 360^\circ}{\text{total frequency}}$

Since there are 360° in a circle, the frequency for each class must be converted into a proportional part of the circle. This conversion is done by using the formula:

$$\text{Degrees} = \frac{f}{n} \times 360^\circ$$

where f is frequency for each class and n is the sum of the frequencies. Hence, the following conversions are obtained. The degrees should sum to 360° .

Once all the degrees for creating a pie chart are calculated, draw a circle (pie chart) using the calculated measurements with the help of a protractor, and label each section with the name and percentages or degrees.

Example 1:

In the summer, a survey was conducted among 400 people about their favourite beverages: 2% like cold-drinks, 6% like Iced-tea, 12% like Cold-coffee, 24% like Coffee and 56% like Tea.

- How many people like tea?
- How many more people like coffee than cold coffee?
- What is the total central angle for iced tea and cold drinks?
- Draw a pie chart to represent the provided information.

Solution

- a) Total number of people = 400

$$\text{Number of people like tea: } 400 \times \frac{56}{100} = 224$$

- b) Number of people like coffee: $400 \times \frac{24}{100} = 96$

$$\text{Number of people like cold-coffee} = 400 \times \frac{12}{100} = 48$$

$$\text{Number of people like coffee more than cold-coffee} = 96 - 48 = 48$$

- c) Number of people like iced-tea: $400 \times \frac{6}{100} = 24$

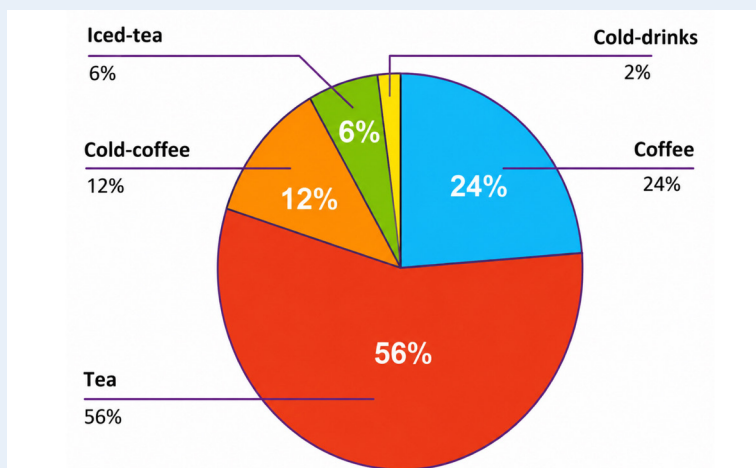
$$\text{Number of people like cold-drinks: } 400 \times \frac{2}{100} = 8$$

$$\text{Central angle for iced-tea: } \frac{24}{400} \times 360^\circ = 21.6^\circ$$

$$\text{Central angle for cold-drinks: } \frac{8}{400} \times 360^\circ = 7.2^\circ$$

$$\text{Total central angle} = 21.6^\circ + 7.2^\circ = 28.8^\circ.$$

- d)



Example 2:

Construct a pie graph showing the blood types of the army inductees described above. The frequency distribution is repeated here.

Class	Frequency	Percent
A	5	20
B	7	28
O	9	36
AB	4	16
Total	25	100

Solution

- Find the number of degrees for each class, using the formula: $\text{Degrees} = \frac{f}{n} \times 360^\circ$

For each class the following results are obtained:

$$A: \frac{5}{25} \times 360^\circ = 72^\circ ;$$

$$B: \frac{7}{25} \times 360^\circ = 100.8^\circ$$

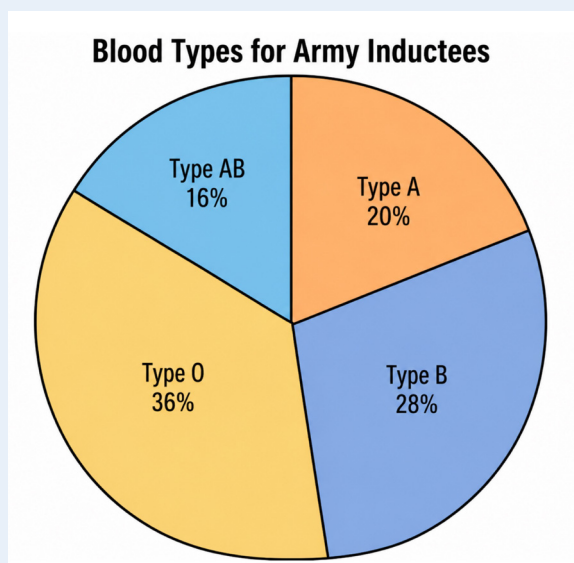
$$O: \frac{9}{25} \times 360^\circ = 129.6^\circ ;$$

$$AB: \frac{4}{25} \times 360^\circ = 57.6^\circ$$

- Find the percentages.

$$A: \frac{5}{25} \times 100\% = 20\% \quad B: \frac{7}{25} \times 100\% = 28\% \quad O: \frac{9}{25} \times 100\% = 36\% \quad AB: \frac{4}{25} \times 100\% = 16\%$$

- Using a protractor, graph each section and write its name and corresponding percentage, as shown in the Figure



Application activity 3.4



A person spends his time on different activities daily (in hours):

Activity	Office work	Exercise	Travelling	Watching shows	Sleeping	Miscellaneous
Number of hours	9	1	2	3	7	2

- Find the central angle and percentage for each activity.
- Draw a pie chart for this information
- Use the pie chart to comment on these findings.

3.5 Presentation of data using histogram and polygon

Activity 3.5



The mass of 50 tomatoes (measured to the nearest g) was measured and recorded in the table below.

Class boundaries (in g)	84.5 – 89.5	89.5 – 94.5	94.5 – 99.5	99.5 – 104.5	104.5 – 109.5	109.5 – 114.5	114.5 – 119.5
Frequency	4	7	6	13	10	5	5

- Determine cumulative frequency
- Try to draw a histogram
- Construct the frequency polygon

Content Summary

After you have organized the data into a frequency distribution, you can present them in graphical form. The purpose of graphs in statistics is to convey the data to the viewers in pictorial form. It is easier for most people to comprehend the meaning of data presented graphically than data presented numerically in tables or frequency distributions.

The three most used graphs in research are

- The histogram.
- The frequency polygon.
- The cumulative frequency graph or ogive (pronounced o-jive).

a) The Histogram

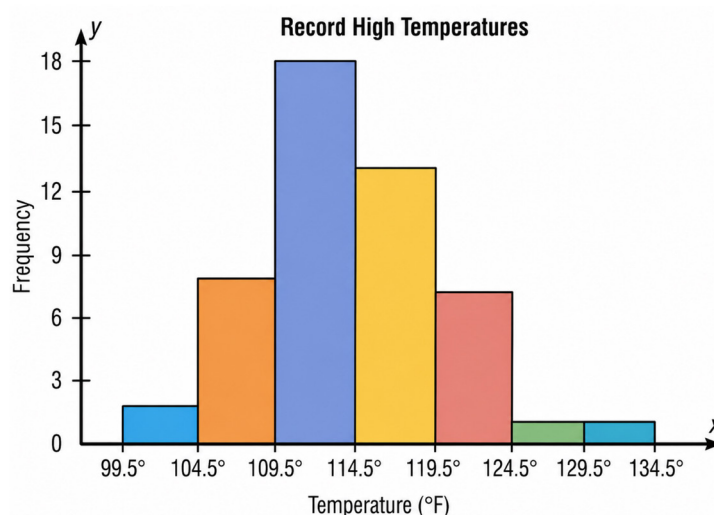
The histogram is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes.

Example:

Construct a histogram to represent the data shown for the record high temperatures for each of the 50 states.

Class boundaries	Frequency
99.5-104.5	2
104.5-109.5	8
109.5-114.5	18
114.5-119.5	13
119.5-124.5	7
124.5-129.5	1
129.5-134.5	1

- Draw and label the x and y axes. The x axis is always the horizontal axis, and the y axis is always the vertical axis.
- Represent the frequency on the y axis and the class boundaries on the x axis.
- Using the frequencies as the heights, draw vertical bars for each class. See Figure below



b) The Frequency Polygon

The frequency polygon is a graph that displays the data by using lines that connect points plotted for the frequencies at the midpoints of the classes. The frequencies are represented by the heights of the points.

Example:

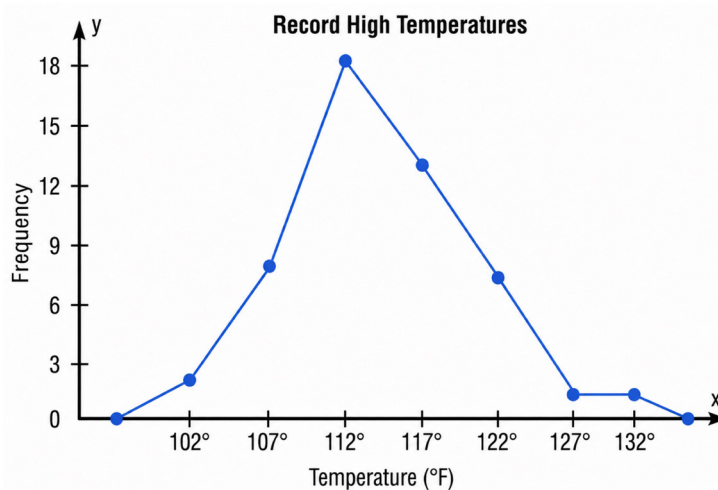
Using the frequency distribution given in Example 2-4, construct a frequency polygon

- Find the midpoints of each class. Recalling that midpoints are found by adding the upper and lower boundaries and dividing by 2: $\frac{99.5 + 104.5}{2} = 102$, $\frac{104.5 + 109.5}{2} = 107$ and so on.

The midpoints are:

Class boundaries	Midpoints	Frequency
99.5-104.5	102	2
104.5-109.5	107	8
109.5-114.5	112	18
114.5-119.5	117	13
119.5-124.5	122	7
124.5-129.5	127	1
129.5-134.5	132	1

- Draw the x and y axes. Label the x axis with the midpoint of each class and then use a suitable scale on the y axis for the frequencies.
- Using the midpoints for the x values and the frequencies as the y values, plot the points.
- Connect adjacent points with line segments. Draw a line back to the x axis at the beginning and end of the graph, at the same distance that the previous and next midpoints would be located, as shown in Figure 2-3.



The frequency polygon and the histogram are two different ways to represent the same data set. The choice of which one to use is left to the discretion of the researcher.

c) The Ogive

The ogive is a graph that represents the cumulative frequencies for the classes in a frequency distribution.

- Find the cumulative frequency for each class.

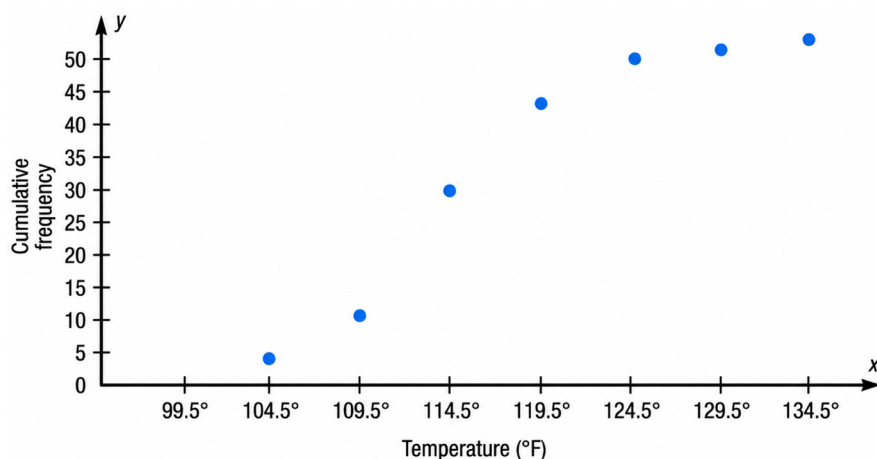
Classes	Cumulative frequency
Less than 99.5	0
Less than 104.5	2
Less than 109.5	10
Less than 114.5	28
Less than 119.5	41
Less than 124.5	48
Less than 129.5	49
Less than 134.5	50

- Draw the x and y axis. Label the x axis with the class boundaries. Use an appropriate scale for the y axis to represent the cumulative frequencies.

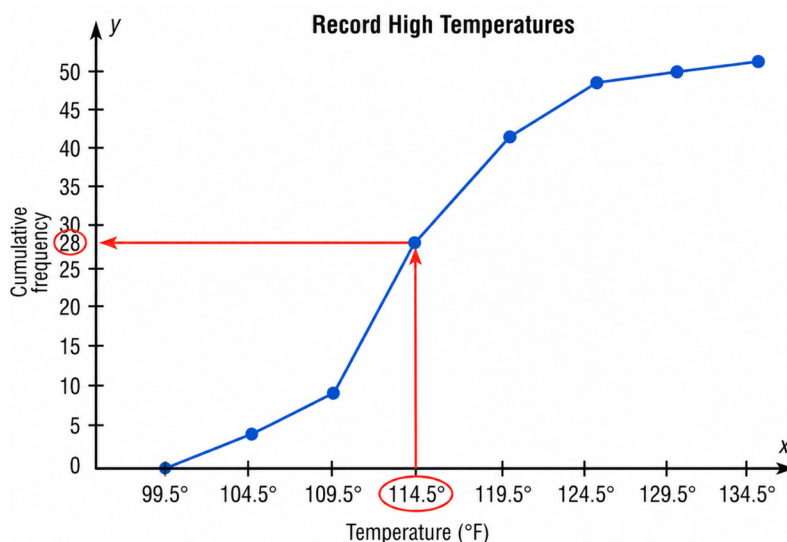
(Depending on the numbers in the cumulative frequency columns, scales such as 0, 1, 2, 3, ..., or 5, 10, 15, 20, ..., or 1000, 2000, 3000, ... can be used.

Do *not* label the y axis with the numbers in the cumulative frequency column.) In this example, a scale of 0, 5, 10, 15, ... will be used.

- Plot the cumulative frequency at each upper-class boundary, as shown in Figure below. Upper boundaries are used since the cumulative frequencies represent the number of data values accumulated up to the upper boundary of each class.



- Starting with the first upper class boundary, 104.5, connect adjacent points with line segments, as shown in the figure. Then extend the graph to the first lower class boundary, 99.5, on the x axis.



Cumulative frequency graphs are used to visually represent how many values are below a certain upper-class boundary. For example, to find out how many record high temperatures are less than 114.5_F, locate 114.5_F on the x axis, draw a vertical line up until it intersects the graph, and then draw a horizontal line at that point to the y axis. The y axis value is 28, as shown in the figure.

d) Relative Frequency Graphs

The histogram, the frequency polygon, and the ogive shown previously were constructed by using frequencies in terms of raw data. These distributions can be converted to distributions using *proportions* instead of raw data as frequencies. These types of graphs are called **relative frequency graphs**.

Example:

Construct a histogram, frequency polygon, and ogive using relative frequencies for the distribution (shown here) of the kilometers that 20 randomly selected runners ran during a given week.

Class boundaries	Frequency
5.5-10.5	1
10.5-15.5	2
15.5-20.5	3
20.5-25.5	5
25.5-30.5	4
30.5-35.5	3
35.5-40.5	2
Total	20

Solution

- Convert each frequency to a proportion or relative frequency by dividing the frequency for each class by the total number of observations.

For class 5.5-10.5, the relative frequency is $\frac{1}{20} = 0.05$;

For class 10.5-15.5, the relative frequency is $\frac{2}{20} = 0.10$;

For class 15.5-20.5, the relative frequency is $\frac{3}{20} = 0.15$; so, on and so forth.

Place these values in the column labeled Relative frequency.

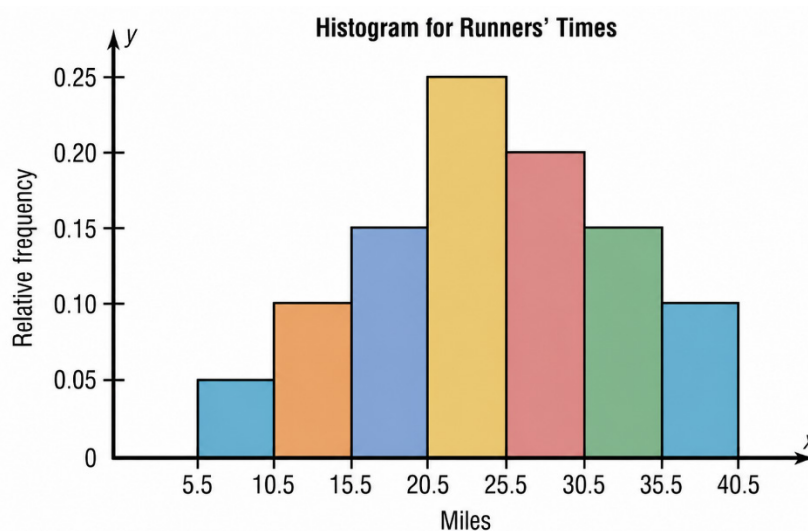
Class boundaries	Midpoints	Relative frequency
5.5-10.5	8	0.05
10.5-15.5	13	0.10
15.5-20.5	18	0.15
20.5-25.5	23	0.25
25.5-30.5	28	0.20
30.5-35.5	33	0.15
35.5-40.5	38	0.10
Total		1.00

- Find the cumulative relative frequencies. To do this, add the frequency in each class to the total frequency of the preceding class. In this case, $0 + 0.05 = 0.05$, $0.05 + 0.10 = 0.15$, $0.15 + 0.15 = 0.30$, $0.30 + 0.25 = 0.55$, etc. Place these values in the column labeled Cumulative relative frequency.

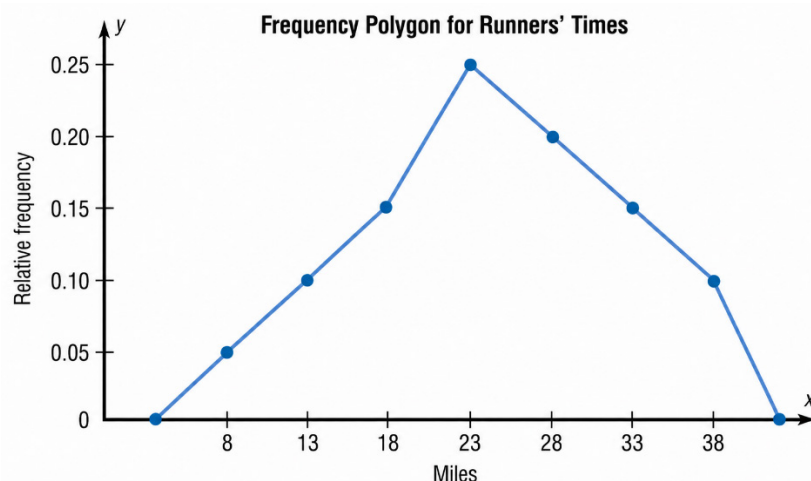
An alternative method would be to find the cumulative frequencies and then convert each one to a relative frequency.

Classes	Cumulative frequency	Cumulative frequency relative
Less than 5.5	0	0.00
Less than 10.5	1	0.05
Less than 15.5	3	0.15
Less than 20.5	6	0.30
Less than 25.5	11	0.55
Less than 30.5	15	0.75
Less than 35.5	18	0.90
Less than 40.5	20	1.00

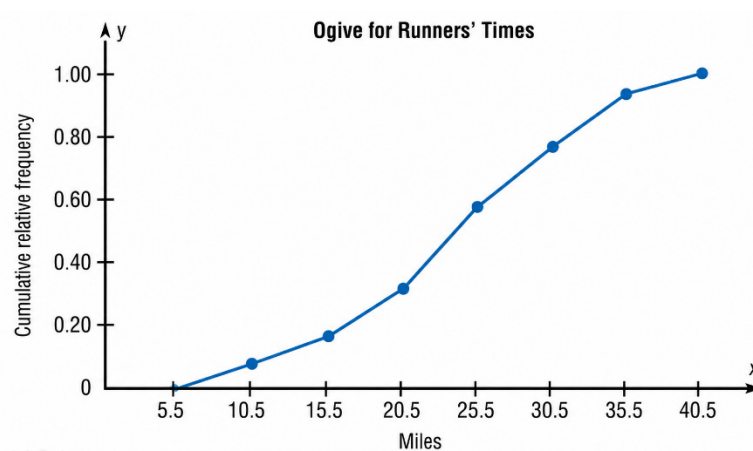
- Draw each graph as shown in figures below. For the histogram and ogive, use the class boundaries along the x axis. For the frequency polygon, use the midpoints on the x axis. The scale on the y axis uses proportions.



(a) Histogram



(b) Frequency polygon



(c) Ogive

Application activity 3.5



The cumulative frequency distribution table below shows distances (in km) covered by twenty runners during the week of competition.

Class boundaries	Frequency	Cumulative frequency
5.5-10.5	1	1
10.5-15.5	2	3
15.5-20.5	3	6
20.5-25.5	5	11
25.5-30.5	4	15
30.5-35.5	3	18
35.5-40.5	2	20

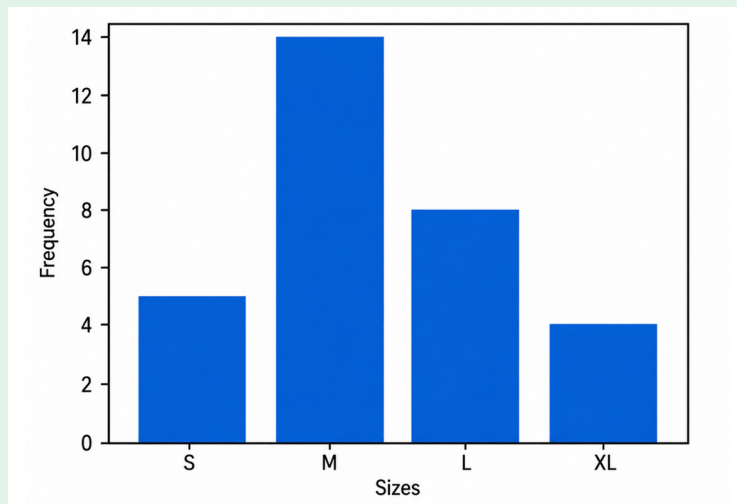
- Construct a histogram.
- Construct the frequency polygon.

3.6 Interpretation of a statistical graph

Activity 3.6



The graph below shows the sizes of sweaters worn by 30 S4 students in a certain school. Observe it and interpret it by answering the questions below:



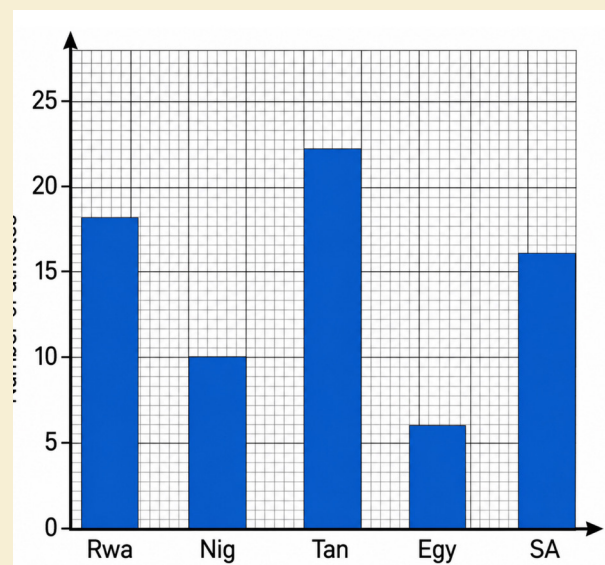
- How many students are with small size (S)?
- How many students with medium size (M), large size (L) and extra-large size (XL) are there?

Content Summary

Once data has been collected, they may be presented or displayed in various ways including graphs. Such displays make it easier to interpret and compare the data.

Example 1:

The bar graph shows the number of athletes who represented five African countries in an international championship.



- What was the total number of athletes representing the five countries?
- What was the smallest number of athletes representing one country?
- What was the greatest number of athletes representing a country?
- Represent the information on the graph on a frequency table.

Solution

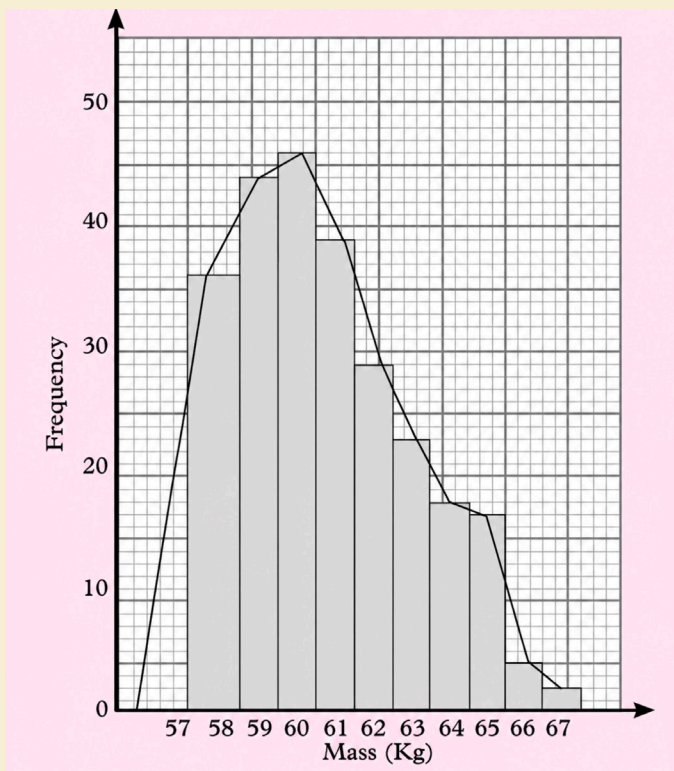
We read the data on the graph:

- Total number of athletes are: $18 + 10 + 22 + 6 + 16 = 72$ athletes
- 6 athletes
- 22 athletes
- Representation of the given information on the graph on a frequency table.

Country	Number of athletes
Rwanda	18
Nigeria	10
Tanzania	22
Egypt	6
South Africa	16
Total	72

Example 2:

Use vertical scale 2cm: 10 students and Horizontal scale 2cm: 10 represented on histogram below to answer the questions that follows



- estimate the mode
- Calculate the range

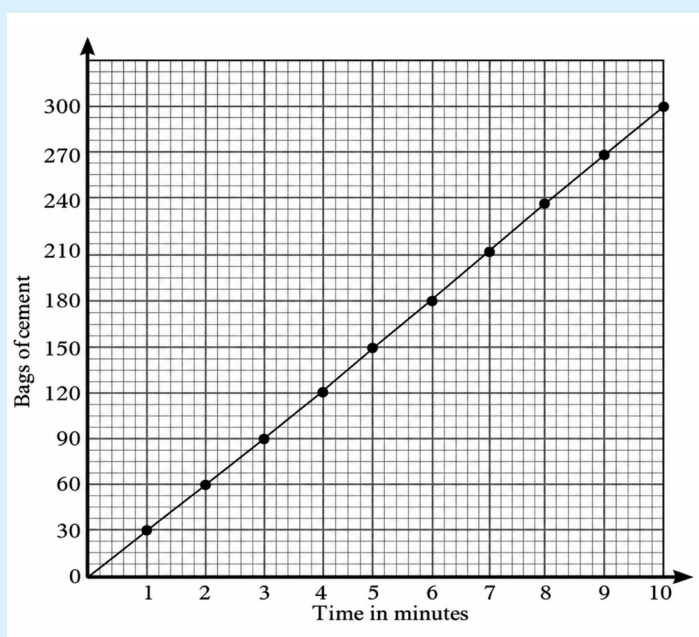
Solution

- a) To estimate the mode graphically, we identify the bar that represents the highest frequency. The mass with the highest frequency is 60 kg. It represents the mode.
- b) The highest mass = 67 kg and the lowest mass = 57 kg
Then, The range = highest mass - lowest mass = $67\text{ kg} - 57\text{ kg} = 10\text{ kg}$

Application activity 3.6



The line graph below shows bags of cement produced by CIMERWA (cement factory) in a minute.



- a) Find how many bags of cement will be produced in 8 minutes, 3 minutes 12 seconds, 5 minutes and 7 minutes.
- b) Calculate how long it will take to produce: 78 bags of cement.
- c) To summarize the information in a), draw a frequency table to show the number of bags produced and the time taken.

3.7 Measures of central tendency for ungrouped data: Mode and mean

Activity 3.7



Conduct research from any available resource materials and explain measures of central tendency, their types and provide examples.

Insist on explaining how to determine the Mean, Mode and their role when interpreting statistical data.

Content Summary

Measures of central tendency were studied in S1 and S2.

1. The Mean

The mean, also known as the arithmetic average, is found by adding the values of the data and dividing by the total number of values.

The following steps are used to calculate the mean:

- Add the numbers
- Count how many numbers there are in the data set
- Find the mean by dividing the sum of the data values by the number of data values.

Suppose that a fruit seller earned the following money from Monday to Friday respectively: 300, 200, 600, 500, and 400 Rwandan francs. The Mean of this money explains the same daily amount of money that she should earn to totalize the same amount in 5 days.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{300 + 200 + 600 + 500 + 400}{5} = 400$$

$$\text{as } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

This is called the mean and is equivalent to sharing out all data evenly.

2. The mode:

The mode is the number that appears the most often from the set of data. It represents the value which appears more frequently in the data. No calculation is needed to find the mode. You may have to sort the data, and then you count to find the value that appears more frequently.

The following steps are used to calculate the mode:

- First sort the data, writing the values in order from smallest to largest.
- Identify the mode, which is the value that appears the most often

Note: Set of data can have no mode if each data repeats once. Data set can have more than one modes if there are more than one data with equal frequencies greater than one.

Example 1:

Calculate the Mean and mode of the following set of numbers: 3, 4, 4, 6, 8, 5, 4, and 8

Solution

$$\bar{X} = \frac{1}{n} \sum xfi$$

x	fi	xfi
3	1	3
4	3	12
5	1	5
6	1	6
8	2	16
	$\sum fi = n = 8$	$\sum xfi = 42$

The Mean is given by $\bar{x} = \frac{1}{n} \sum xfi$, from which $\bar{x} = \frac{42}{8} = 5.25$

Arrange data first 3, 4, 4, 4, 5, 6, 8, 8. Total observation is 8, Mode is 4.

Example 2:

Considering the following set of data: 4; 6; 7; 3; 4; 8; 4; 2; 9.

Find the mean value and mode of the data set.

Solution

To find the mean:

- Add the numbers.
Sum of the data values = $4 + 6 + 7 + 3 + 4 + 8 + 4 + 2 + 9 = 47$
- Count how many numbers there are in the data set.
Number of data values = 9.
- Find the mean by dividing the sum of the data values by the number of data values.

$$\text{The mean} = \frac{\text{sum of data values}}{\text{number of data values}} = \frac{\sum_{i=1}^n x_i}{n} = \frac{47}{9} = 5.2$$

The mean is 5.2.

Arrange the data first: 2, 3, 4, 4, 4, 6, 7, 8, 9. The mode is 4.

Example

Data set: 70, 82, 82, 85, 85, 90, 90, 93, 95, 95

a) $\bar{X} = \frac{867}{10} = 86.7$

b) Mode: 82, 85, 90, 95 (Multimodal)

Application activity 3.7



1. A group of learners from languages option were asked how many books they had read in previous year, the results are shown in the frequency table below.

Number of books	0	1	2	3	4	5	6	7	8
Frequency (number of student teachers)	5	5	6	9	11	7	4	2	1

Calculate the mean and mode of the number books read.

2. During oral presentation of internship report for senior six students, the first 10 students scored the following marks out of 10: 8, 7, 9, 10, 8, 9, 8, 6, 7 and 10. Calculate the mean and the mode of the group.

3.8 Median as a measure of central tendency for ungrouped data

Activity 3.8



Use any available resources and explain measures of central tendency, their types and provide examples.

Insist on explaining how to determine the Median and its role when interpreting statistic data.

Content Summary

The Median:

If the data is well arranged in an order from the smallest to the largest, the median is the middle number or the central number of the range. When total observation ($\sum f_i = n$) is odd the median is given by $\left(\frac{n+1}{2}\right)^{th}$ number which is located on this position. On the other side when n is even, $\frac{1}{2} \left[\left(\frac{n}{2}\right)^{th} + \left(\frac{n+1}{2}\right)^{th} \right]$, then the median is a half of the sum of number located on those two positions.

Example 1:

Calculate the median of the following numbers: 4, 5, 7, 2, 1

Solution

Arrange data from lowest to highest number as 1, 2, 4, 5, 7, the position of the median is given by $\left(\frac{n+1}{2}\right)^{th}$, from which $Me = \left(\frac{5+1}{2}\right)^{th} = 3^{rd} \text{ position}$. Then, $Me = 4$.

Example 2:

Calculate the median of the following numbers: 4, 5, 7, 2, 1 and 8

Solution

Arrange numbers in ascending order: 1, 2, 4, 5, 7, 8. Total observation (n) = 6 since the total observation is even then, the position of the median is $\left[\left(\frac{n}{2}\right) + \left(\frac{n+1}{2}\right) \right]^{th} \text{ element}$

$$\text{and } Me = \frac{1}{2} \left[\left(\frac{n}{2}\right) + \left(\frac{n+1}{2}\right) \right]^{th} \text{ element}$$

Then, the position of the median is $\left[\left(\frac{6}{2} \right) + \left(\frac{6}{2} + 1 \right) \right]^{th}$ element which is the $(3+4)^{th}$ and $Me = \frac{4+5}{2} = 4.5$.

Application activity 3.8



A secondary school has two cricket teams: a junior and a senior team. The junior team consists of 17 players (including reserves), and the senior team consists of 16 players (including reserves). The mass of each team member is given below. Use the data to answer the questions that follow:

Junior team masses (kg) = {56, 60, 67, 45, 51, 53, 64, 49, 56, 48, 42, 51, 64, 52, 64, 49, 50}

Senior team masses (kg) = {88, 81, 53, 62, 83, 68, 70, 62, 91, 78, 64, 74, 73, 54, 62, 62}

1. What is the Mean mass of the senior team?
2. Arrange the masses of the senior team in ascending order.
3. Determine the mode of the senior team.
4. Determine the median of the senior team.
5. Calculate the mean of the masses of the junior team correct to one decimal digit.
6. Arrange the masses of the junior team in ascending order.
7. Calculate the mode of the masses of the junior team.
8. Calculate the median of the masses of the junior team.
9. Look at the answers you found for the junior and senior teams. Which measure do you think gives the best measure of the real 'average' of each data set?

3.9 Mode as the measure of central tendency for grouped data

Activity 3.9



Carry out research from reference books or from internet to discover how to determine the mode for grouped data and then answer the following question. Suppose we have the following frequency distribution that shows the number of points scored per game by 60 basketball players:

Points scored	Frequency
1-10	8
11-20	25
21-30	14
31-40	9
41-50	4

Determine the mode of the above statistical data.

Content Summary

The mode for grouped data is found from the class with the largest frequency which is called **modal class**, and it is determined using the following formula:

$$\text{Mode} = L + w \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right). \text{ Where:}$$

- L : the lower limit of the modal class
- f_m : the modal frequency
- w : modal class width
- f_1 : the frequency of the immediate class below the modal class
- f_2 : the frequency of the immediate class above the modal class

Example:

Find the modal class for the frequency distribution of kilometers that 20 runners ran in one week.

Class boundaries	Frequency
5.5-10.5	1
10.5-15.5	2
15.5-20.5	3
20.5-25.5	5 ← Modal class
25.5-30.5	4
30.5-35.5	3
35.5-40.5	2

The modal class is 20.5–25.5, since it has the largest frequency. Sometimes the midpoint of the class is used rather than the boundaries; hence, the mode could also be given as 23 km per week.

Application activity 3.9



Suppose we have the following frequency distribution that shows the exam scores received by 40 students in a certain class:

Exam score	Frequency
51-60	4
61-70	8
71-80	15
81-90	8
91-100	5

Determine the modal exam score of the above frequency distribution.

3.10 Mean as the measure of central tendency for grouped data

Activity 3.10



1. Conduct research in the library or on the internet and explain measures of central tendency for grouped data and provide examples.

Insist on explaining how to determine the Mean, Mode, Median and their role when interpreting statistic data.

2. The data below represents the number of kilometers (distance) covered by a sample of 20 runners for one week. Using the frequency distribution, find the Mean.

A	B	C	D
Class boundaries	Frequency f	Midpoint x_m	$f \cdot x_m$
5.5-10.5	1		
10.5-15.5	2		
15.5-20.5	3		
20.5-25.5	5		
25.5-30.5	4		
30.5-35.5	3		
35.5-40.5	2		
	$n = 20$		

What does this Mean represent considering the class in which it is in the data?

Content Summary

The process of finding the mean is the same as the one applied in the ungrouped data with the exception that the midpoints x_m of each class in grouped data play the role of x_i used in ungrouped data.

$$\bar{x} = \frac{\sum f \cdot x_m}{n}$$

Weighted mean

It is a Mean of a data set in which not all values are equally represented. Find the weighted mean of a variable X by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

$$\bar{X} = \frac{w_1 X_1 + w_2 X_2 + w_3 X_3 + \dots + w_n X_n}{w_1 + w_2 + w_3 + \dots + w_n} = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}$$

where $w_1, w_2, \dots, w_n$ are the weights and $X_1, X_2, \dots, X_n$ are the values.

Example 2:

A student received an A in English Composition I (3 credits), a C in Introduction to Psychology (3 credits), a B in Biology I (4 credits), and a D in Physical Education (2 credits). Assuming A=4 grade points, B=3 grade points, C =2 grade points, D =1 grade point, and F=0 grade points, find the student's grade point average.

Solution

Course	Credits (w)	Grade (x)
English Composition	3	A (4 points)
Introduction	3	C (2 points)
Biology I	4	B (3 points)
Physical Education	2	D (1 points)

$$\bar{X} = \frac{\sum_{i=1}^n w_i X_i}{\sum w_i} = \frac{3 \times 4 + 3 \times 2 + 4 \times 3 + 2 \times 1}{3 + 3 + 4 + 2} = \frac{32}{12} = 2.7$$

Application activity 3.10



The data below shows the marks scored by a group of students in a mathematics out of 100: 72; 63; 51; 25; 31; 49; 51; 27; 46; 42; 25; 39; 38; 39; 55; 38; 35; 64; 67; 37.

Use a grouped data of 5 intervals and determine the mean mark.

3.11 Median as the measure of central tendency for grouped data

Activity 3.11



Carry out research from reference books or from internet to discover how to determine the mode for grouped data and then answer the following question.

Suppose we have the following frequency distribution that shows the number of points scored per game by 60 basketball players:

Points scored	Frequency
1-10	8
11-20	25
21-30	14
31-40	9
41-50	4

Determine the median of the above statistical data.

Content Summary

In the case of continuous frequency of distribution, we first locate the median by cumulating the frequency until $\left(\frac{n}{2}\right)^{th}$ point is reached. Finally, the median is determined within this class by using formula. The procedures thus involve the following steps:

1. Compute cumulative frequencies
2. Locate the median class in cumulative frequency column where the size of $\left(\frac{n}{2}\right)^{th}$ item falls.
3. Obtain the median value by applying the formula mentioned below

The median represents the value that lies directly in the middle of a dataset, when all of the values are arranged from smallest to largest.

The median of grouped data is given by $Median = L + \frac{w}{f} \left(\frac{N}{2} - C_{<} \right)$ where:

L : Lower limit of median class

W : Width of median class

N : Total Frequency

$C_{<}$: Cumulative frequency up to median class

f : Frequency of median class

Example 2:

1. The following table shows the weekly consumption of electricity of 56 families

Weekly consumption	0-10	10-20	20-30	30-40	40-50
Number of families	16	12	18	6	4

Calculate the median weekly consumption.

Solution

Weekly consumption	Frequency	Cumulative frequency
0-10	16	16
10-20	12	28
20-30	18	46
30-40	6	52
40-50	4	56
Sum	56	

Median is the value of $\left(\frac{n}{2}\right)^{th}$ position $= \left(\frac{56}{2}\right)^{th} = 28^{th}$ position, which lies in the class 10-20.

Thus 10-20 is the median class

$$\text{Median} = L + \frac{w}{f} \left(\frac{N}{2} - C_{<} \right)$$

$$\text{Median} = 10 + \frac{(20-10)}{12} \left(\frac{56}{2} - 16 \right)$$

$$\text{Median} = 10 + 10 = 20$$

Hence the median is 20 units.

Application activity 3.11



1. Suppose we have the following frequency distribution that shows the exam scored receive by 40 students in a certain class:

Exam score	Frequency
51-60	4
61-70	8
71-80	15
81-90	8
91-100	5

Determine the median of the above frequency distribution.

2. Calculate the median for the following distribution

Class interval	130-140	140-150	150-160	160-170	170-180	180-190	190-200
Frequency	5	9	17	28	24	10	7

3.12 Quartiles as Measures of dispersion

Activity 3.12



Carry out research from reference books or from internet to discover how to determine the quartiles for grouped data and then answer the following question.

The following table gives the amount of time spent (in minutes) on the internet each evening by a group of 56 students. Compute the cumulative frequency for the following frequency distribution.

Time spent on internet	9.5–12.5	12.5–15.5	15.5–18.5	18.5–21.5	21.5–24.5
Number of students	3	12	15	24	2

Find quartiles for the given grouped data.

Content Summary

The word dispersion has a technical meaning in statistics. The average measures the center of the data. It is one aspect of observations. Another feature of the observations is how the observations are spread about the center. The observation may be close to the center, or they may be spread away from the center. If the observations are close to the center (usually the arithmetic mean or median), we say that dispersion, scatter or variation is small. If the observations are spread away from the center, we say that dispersion is large.

The study of dispersion is very important in statistical data. If in a certain factory there is consistency in the wages of workers, the workers will be satisfied. But if some workers have high wages and some have low wages, there will be unrest among the low paid workers, and they might go on strikes and arrange demonstrations. If in a certain country some people are very poor and some are very high rich, we say there is economic disparity. It means that dispersion is large.

The extent or degree in which data tend to spread around an average is also called the dispersion or variation. Measures of dispersion help us in studying the extent to which observations are scattered around the average or central value. Such measures are helpful in comparing two or more sets of data regarding their variability.

Quartiles are values that split up a dataset into four equal parts.

The middle term, between the median and first term is known as the first or Lower Quartile and is written as Q_1 . Similarly, the value of mid-term that lies between the last term and the median is known as the third or upper quartile and is denoted as Q_3 . Second Quartile is the median and is written as Q_2 . You can use the following formula to calculate quartiles for grouped data:

$$Q_i = L + \frac{w}{f} \left(\frac{iN}{4} - C_{<} \right), \text{ where:}$$

L is lower limit of i^{th} quartile class

w is width of i^{th} quartile class

N is total frequency

$C_{<}$ is cumulative frequency of the class previous to the i^{th} quartile class

f is frequency of the interval that contains the i^{th} quartile.

Application activity 3.12



The scores of students in a Mathematics test are given in the table below:

Class Interval	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Frequency	6	8	12	10	5	4

Find quartiles for the given grouped data.

3.13 Variance as the measure of dispersion

Activity 3.13



Before starting the third term, a class teacher calculated the mean mark 16 students got in second term in Mathematics and he/she obtained that the mean mark is $\bar{x} = 16.875$.

Use this mean to complete the table below:

x	f	$(x - \bar{x})$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
12	4			
13	2			
15	1			
19	4			
21	5			
	$\sum f =$			$\sum f(x - \bar{x})^2 =$

Explain the expression $f(x - \bar{x})^2$ in your own words.

Content Summary

Variance measures how far a set of numbers is spread out. A variance of zero indicates that all the values are identical. Variance is always non-negative: a small variance indicates that the data points tend to be very close to the mean and hence to each other, while a high variance indicates that the data points are very spread out around the mean and from each other.

The variance is denoted and defined by $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$

Note that for the variance of a sample, we divide by $n-1$. This means that $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$

Developing the formula for the variance of population we have

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + (\bar{x})^2)}{n} \text{ or}$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \frac{1}{n} 2\bar{x} \sum_{i=1}^n x_i + \frac{1}{n} (\bar{x})^2 \sum_{i=1}^n 1 \text{ or}$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - 2x_i\bar{x} + (\bar{x})^2 \text{ or}$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - (\bar{x})^2$$

Thus, the variance is also defined by

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - (\bar{x})^2$$

Recall that the mean of the set of n values $x_1, x_2, x_3, \dots, x_n$ is denoted and defined by

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \sum_{i=1}^n \frac{x_i}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example 2:

Calculate the variance of the following distribution: 9, 3, 8, 8, 9, 8, 9, 18

Solution

$$\bar{x} = \frac{9+3+8+8+9+8+9+18}{8} = 9$$

$$\sigma^2 = \frac{(9-9)^2 + (3-9)^2 + (8-9)^2 + (8-9)^2 + (9-9)^2 + (8-9)^2 + (9-9)^2 + (18-9)^2}{8} = 15$$

Application activity 3.13



- The heights (in meters) of six children are 1.42, 1.35, 1.37, 1.50, 1.38 and 1.30.
Calculate the mean height and the variance of the heights.
- The number of customers served lunch in a restaurant over a period of 60 days is as follows:

Number of customers served lunch	20–29	30–39	40–49	50–59	60–69	70–79
Number of days in the 60-day period	6	12	16	14	8	4

Find the mean and variance of the number of customers served lunch using this grouped data.

3.14 Standard deviation

Activity 3.14



Using an available resource material, explain what is meant by standard deviation as measure of dispersion.

Insist on explaining how standard deviation is calculated and why it is used in real life problems for grouped data and for ungrouped data.

Content Summary

The standard deviation has the same dimension as the data and hence is comparable to deviations from the mean. We define the standard deviation to be the square root of the variance.

Thus, the standard deviation is denoted and defined by:

a) For ungrouped data: $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$ or $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$

The **standard deviation** (S) of the sample is given by $S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$

b) For grouped data

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}}, \text{ where } \sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i} \text{ is the variance.}$$

The following results follow directly from the definitions of mean and standard deviation:

- When all the data values are multiplied by a constant a , the new mean and new standard deviation are equal to $|a|$ times the original mean and standard deviation. That is, the mean of $ax_1, ax_2, ax_3, \dots, ax_n$ is $a(\bar{x})$ and the standard deviation is σ .
- When a constant value, b , is added to all data values, then the new mean is increased by b . However standard deviation does not change. That is, the mean of $x_1+b, x_2+b, x_3+b, \dots, x_n+b$ is $\bar{x}+b$ and the standard deviation is σ .

Example 1:

The six runners in a 200-meter race clocked times (in seconds) of 24.2, 23.7, 25.0, 23.7, 24.0, 24.6. Find the mean and standard deviation of these times.

Solution

$$\bar{x} = \frac{24.2 + 23.7 + 25.0 + 23.7 + 24.0 + 24.6}{6} = 24.2 \text{ seconds}$$

$$\begin{aligned}\sigma &= \sqrt{\frac{(24.2 - 24.2)^2 + (23.7 - 24.2)^2 + (25.0 - 24.2)^2 + (23.7 - 24.2)^2 + (24.0 - 24.2)^2 + (24.6 - 24.2)^2}{6}} \\ &= 0.473 \text{ seconds}\end{aligned}$$

The method which uses the formula for the standard deviation is not necessarily the most efficient. Consider the following: $\frac{\sum (x - \bar{x})^2}{n} = \frac{1}{n} \sum x^2 - (\bar{x})^2$

Example 2:

The heights (in meters) of six children are 1.42, 1.35, 1.37, 1.50, 1.38 and 1.30. Calculate the mean height and the standard deviation of the heights.

Solution

$$\text{Mean} = \frac{1}{6}(1.42 + 1.35 + 1.37 + 1.50 + 1.38 + 1.30) m = 1.39 m$$

$$\text{Variance} = \left[\frac{1}{6}(1.42^2 + 1.35^2 + 1.37^2 + 1.50^2 + 1.38^2 + 1.30^2) - 1.39^2 \right] m^2 = 0.00386 m^2$$

$$\text{Standard deviation} = \sqrt{0.00386 m^2} = 0.0621 m$$

Application activity 3.14



1. In the following statistical series, calculate the standard deviation of the following set of data: 56, 54, 55, 59, 58, 57, 55.
2. In the classroom of language education, the first ten student-teachers scored the following marks out of 10 in a quiz of French: 5, 6, 5, 2, 4, 7, 8, 9, 7, 5.
 - a) Calculate the mean, median and the modal mark
 - b) Calculate the quartiles and inter-quartile range
 - c) Calculate the variance and the standard deviation.

3.15 Coefficient of variation as the measure of dispersion

Activity 3.15



Two plants C and D of a factory show the following results about the number of workers and the wages paid to them.

	Plant C	Plant D
No. of workers	5000	6000
Average monthly wages	\$2500	\$2500
Standard deviation	9	10

- What information do we have about the number of workers, their average monthly wages, and the spread (standard deviation) of wages in each plant?
- Why might it be important to look not only at the average wage but also at how much individual wages differ from the average?
- If two plants have the same average wage but different spreads, how can we create a fair comparison of wage stability?
- What kind of calculation could combine the spread (standard deviation) and the average wage to give us a percentage measure of variability? How is this kind of variability different from the standard deviation?
- Once we have this measure, which plant shows greater wage stability, and how might that influence an investor's choice?

Content Summary

The coefficient of variation measures variability in relation to the mean (or average) and is used to compare the relative dispersion in one type of data with the relative dispersion in another type of data. It allows us to compare the dispersions of two different distributions if their means are positive. The greater coefficient of variation indicates greater relative dispersion.

By determining the coefficient of variation of different securities, an investor identifies the risk-to-reward ratio of each security and develops an investment decision. Generally, an investor seeks a security with a lower coefficient (of variation) because it provides the most optimal risk-to-reward ratio with low volatility but high returns. However, the low coefficient is not favorable when the average expected return is below zero.

Formula for Coefficient of Variation

The coefficient of variation is a calculation built on other calculations: the standard deviation

and the mean as follows: $Cv = \frac{\sigma}{\bar{x}} \times 100$

Where:

- σ is the standard deviation
- $\bar{x}$ is the mean.

In the context of finance, we can re-write the above formula in the following way:

$$\text{Coefficient of Variation} = \frac{\text{Volatility}}{\text{Expected Return}} \times 100\%$$

Example:

One data series A has a mean of 140 and standard deviation 28.28. The second data series B has a mean of 150 and standard deviation 24. Which of the two has a greater dispersion?

Solution

$$CV_1 = \frac{28.28}{140} \times 100 = 20.2\%$$

$$CV_2 = \frac{24}{150} \times 100 = 16\%$$

From the given two data series (A and B), data series A has a greater dispersion.

Application activity 3.15



Fred wants to find a new investment for his portfolio. He is looking for a safe investment that provides stable returns. He considers the following options for investment:

- Stocks: Fred was offered stock of ABC Corp. It is a mature company with strong operational and financial performance. The volatility of the stock is 10%, and the expected return is 14%.
- ETFs: Another option is an Exchange-Traded Fund (ETF) which tracks the performance of the S&P 500 index. The ETF offers an expected return of 13% with a volatility of 7%.
- Bonds: Bonds with excellent credit ratings offer an expected return of 3% with 2% volatility.

To select the most suitable investment opportunity, Fred decided to calculate the coefficient of variation of each option. In which option does Fred want to invest?

3.16 Application: Collection, organization, presentation and interpretation of data and drawing conclusions

Activity 3.16



Students were interested in getting information on the number of hours patients spent in the hospital in a certain week. The table below summarizes the data they recorded where the frequency indicates the number of patients.

Class boundaries (number of hours)	Frequency (number of patients)
7.5-12.5	3
12.5-17.5	5
17.5-22.5	15
22.5-27.5	5
27.5-32.5	2

Determine and interpret the average, the modal class, the median, the variance and the standard deviation related to the number of hours a sick person spends in that hospital.

What is the advice you can provide for the manager of that hospital if he/she has a few numbers of beds?

Content Summary

With observation, data is collected through direct observation. Information could also be collected using an existing table that shows types of data to collect in a certain period of time, this method of collecting data is reliable and accurate.

Once data has been collected, they may be presented or displayed in various ways. Such displays make it easier to interpret and compare the data.

After the collection of data there is need to interpret them and here are some tips:

- Collect your data and make it as clean as possible.
- Choose the type of analysis to perform qualitative or quantitative
- Analyze the data through various statistical methods such as mean, mode, standard deviation or frequency distribution tables
- Reflect on your own thinking and reasoning and analyze your data and then interpret them referring to the reality.

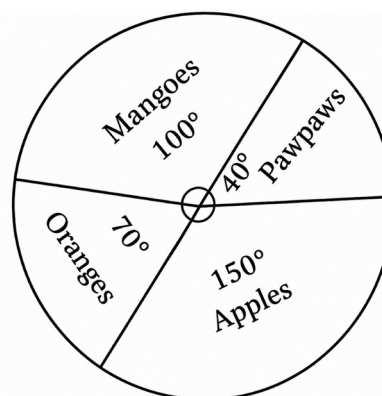
During interpretation, avoid subjective bias, false information and inaccurate decisions.

Application activity 3.8



After selling fruits in a market, Martha had a total of 144 fruits remaining. The pie chart below shows each type of fruit that remains.

- Find the total cost of mangoes and pawpaws remaining if a mango sells at 30Frw and a pawpaw at 160Frw
- Which type of fruit remained the most?



- c) What was the median number of fruits that remained?
 - d) Draw a frequency table to display the information on the pie chart.
- 2) Work in groups and use tailor's meters to correct data on the height for 20 people.
- a) Organize the corrected data with a frequency distribution
 - b) Determine and interpret the mean, the mode and the median for the data.
 - c) Determine and interpret the range, the quartiles and the standard deviation.
 - d) Organize data into a grouped data distribution of 10 groups and determine: the modal class and the standard deviation. Compare this standard deviation with the one found in (c).

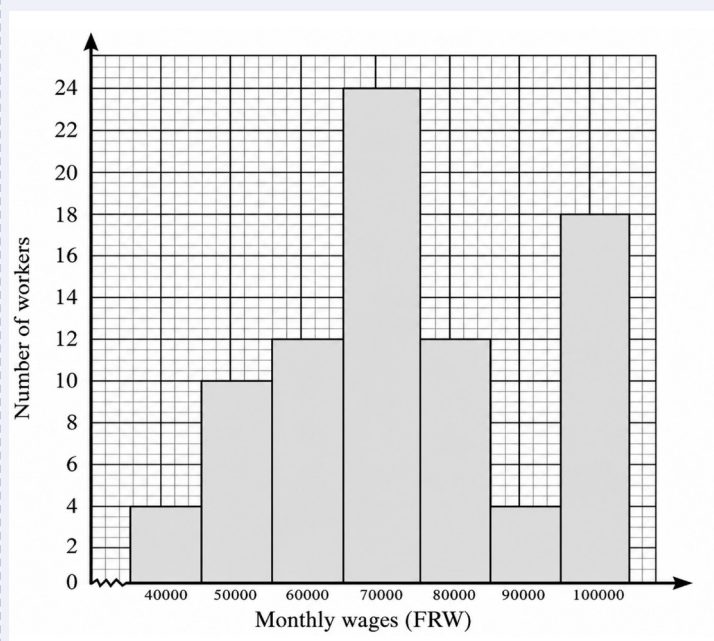
3.17 Questions for revision of unit content

Knowledge and understanding:

1. State two advantages of using a bar chart to present data.
2. What is a pie chart and when is it most appropriate to use?
3. Define the terms *mode*, *median*, and *mean* for ungrouped data.
4. What is the purpose of calculating the *variance* of a dataset?
5. Explain the meaning of *standard deviation* in simple terms.

Skills:

6. Use the graph below to answer the questions that follow



- a) Estimate the mode.
- b) State the range of the distribution.

- c) Draw a frequency distribution table from the graph
 - d) Draw the frequency polygon represented by the histogram
7. In test of mathematics, 10 student-teachers got the following marks: 6, 7, 8, 5, 7, 6, 6, 9, 4, 8. Calculate the mean, mode, quartiles and inter-quartile range

Attitude and values:

- 8. Why is accuracy important when collecting and presenting statistical data?
- 9. Give an example of how misleading graphs can influence people's decisions and discuss why honesty in data presentation matters.
- 10. Explain the value of using statistics when making decisions in daily life (e.g., budgeting, studying habits, community activities).

End of unit assessment



Assessing knowledge:

1. Which of the following best describes a frequency distribution table?
 - A) A diagram showing how data changes over time
 - B) A table that groups data values and shows how often each occurs
 - C) A circle divided into proportional sectors
 - D) A graph with bars drawn side by side
2. Which measure of central tendency represents the most frequently occurring value in a dataset?
 - A) Mean B) Median C) Mode D) Range
3. Which graph is most suitable for showing how a whole is divided into parts?
 - A) Histogram B. Pie chart
 - C) Line graph D. Polygon
4. The standard deviation of a dataset mainly shows:
 - A) The number of values in the dataset
 - B) The middle value when the data is arranged
 - C) How spread out the data values are
 - D) The highest value minus the lowest value
5. Which of the following is a measure of dispersion?
 - A) Mean B) Median
 - C) Quartiles D) Mode

Assessing skills:

1. A class conducted a survey on the number of hours students spent reading per week. The responses (in hours) are:
4, 5, 3, 5, 6, 4, 2, 6, 5, 3, 4, 6
 - a) What type of data is this qualitative or quantitative?
 - b) Construct a simple frequency distribution table for the data.
 - c) Based on the frequencies, which reading-hour value is the *mode*?
 - d) Suggest one appropriate graph to represent this data and explain why.
2. Five households in a community recorded their monthly water consumption (in liters):
750l, 820l, 900l, 880l, 1050l
 - a) Calculate the Mean water consumption.
 - b) Determine the Median of the dataset.
 - c) Calculate the *Range* and explain what it means.
 - d) Explain why knowing measures of dispersion is important for community planning.

Assessing attitudes and values:

A school club wants to present information about students preferred extracurricular activities. They plan to use graphs to share their findings with the school community.

- a) Why should the club avoid manipulating graphs (e.g., altering scales) to make one activity appear more popular?
- b) How does presenting clear and truthful graphs promote trust?
- c) Why is it important for the club to respect privacy when collecting data about students' interests?
- d) How can interpreting the results together help develop cooperation and positive relationships in the school?

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